



Nanyang Technological University

Muchas gracias afición, esto es
para vosotros

Bui Dang Nguyen, Nguyen Trung Dung, Dao Thien Phong

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- 1 Contest
- 2 Mathematics
- 3 Data structures
- 4 Numerical
- 5 Number theory

6 Combinatorial

7 Graph

8 Geometry

9 Strings

10Various

Contest (1)

template.cpp	14 lines
#include <bits/stdc++.h> using namespace std; #define rep(i, a, b) for(int i = a; i < (b); ++i) #define all(x) begin(x), end(x) #define sz(x) (int)(x).size() typedef long long ll; typedef pair<int, int> pii; typedef vector<int> vi; int main() { cin.tie(0)->sync_with_stdio(0); cin.exceptions(cin.failbit); }	

build.sh	5 lines
# Builds a file and executes immediately. # note: run 'chmod +x build.sh' before using filename="\${1%.cpp}" g++ -Wall -Wconversion -Wfatal-errors -g -std=c++17 -fsanitize=undefined, address "\$1" -o "\$filename" ./"\$filename"	

hash.sh	5 lines
# Hashes a file, ignoring all whitespace and comments. Use for # verifying that code was correctly typed. # run 'chmod +x hash.sh' before using # usage: ./hash.sh <filename> cpp -dD -P -fpreprocessed \$1 tr -d '[:space:]' md5sum cut -c-6	

linux-compare	2 lines
system("./code1"); system("./code2"); if (system("diff -w -s task.out task.ans")) exit(0);	

troubleshoot.txt	52 lines
Pre-submit: Write a few simple test cases if sample is not enough. Are time limits close? If so, generate max cases. Is the memory usage fine? Could anything overflow? Make sure to submit the right file.	

1	Wrong answer: Print your solution! Print debug output, as well. Are you clearing all data structures between test cases? Can your algorithm handle the whole range of input? Read the full problem statement again.
1	Do you handle all corner cases correctly? Have you understood the problem correctly? Any uninitialized variables?
2	Any overflows? Confusing N and M, i and j, etc.?
6	Are you sure your algorithm works? What special cases have you not thought of?
9	Are you sure the STL functions you use work as you think? Add some assertions, maybe resubmit.
12	Create some testcases to run your algorithm on. Go through the algorithm for a simple case. Go through this list again.
13	Explain your algorithm to a teammate. Ask the teammate to look at your code.
18	Go for a small walk, e.g. to the toilet. Is your output format correct? (including whitespace) Rewrite your solution from the start or let a teammate do it.
22	Runtime error: Have you tested all corner cases locally? Any uninitialized variables?
23	Are you reading or writing outside the range of any vector? Any assertions that might fail? Any possible division by 0? (mod 0 for example) Any possible infinite recursion? Invalidated pointers or iterators? Are you using too much memory? Debug with resubmits (e.g. remapped signals, see Various).

Time limit exceeded: Do you have any possible infinite loops? What is the complexity of your algorithm? Are you copying a lot of unnecessary data? (References) How big is the input and output? (consider scanf) Avoid vector, map. (use arrays/unordered_map) What do your teammates think about your algorithm?	Memory limit exceeded: What is the max amount of memory your algorithm should need? Are you clearing all data structures between test cases?
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Mathematics (2)

2.1 Equations

$$ax^2+bx+c=0\Rightarrow x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

The extremum is given by $x=-b/2a$.

$$\begin{array}{lcl} ax+by=e & x= & \frac{ed-bf}{ad-bc} \\ cx+dy=f & y= & \frac{af-ec}{ad-bc} \end{array} \Rightarrow$$

In general, given an equation $Ax=b$, the solution to a variable x_i is given by

$$x_i=\frac{\det A'_i}{\det A}$$

where A'_i is A with the i 'th column replaced by b .

2.2 Recurrences

If $a_n=c_1a_{n-1}+\cdots+c_ka_{n-k}$, and $r_1,\ldots,r_k$ are distinct roots of $x^k+c_1x^{k-1}+\cdots+c_k$, there are $d_1,\ldots,d_k$ s.t.

$$a_n=d_1r_1^n+\cdots+d_kr_k^n.$$

Non-distinct roots r become polynomial factors, e.g.

$$a_n=(d_1n+d_2)r^n.$$

2.3 Trigonometry

$$\sin(v+w)=\sin v\cos w+\cos v\sin w$$

$$\cos(v+w)=\cos v\cos w-\sin v\sin w$$

$$\tan(v+w)=\frac{\tan v+\tan w}{1-\tan v\tan w}$$

$$\sin v+\sin w=2\sin\frac{v+w}{2}\cos\frac{v-w}{2}$$

$$\cos v+\cos w=2\cos\frac{v+w}{2}\cos\frac{v-w}{2}$$

$$(V+W)\tan(v-w)/2=(V-W)\tan(v+w)/2$$

where V,W are lengths of sides opposite angles v,w .

$$a\cos x+b\sin x=r\cos(x-\phi)$$

$$a\sin x+b\cos x=r\sin(x+\phi)$$

where $r=\sqrt{a^2+b^2},\phi=\operatorname{atan2}(b,a)$.

2.4 Geometry

2.4.1 Triangles

Side lengths: a,b,c

Semiperimeter: $p=\frac{a+b+c}{2}$

Area: $A=\sqrt{p(p-a)(p-b)(p-c)}$

Circumradius: $R=\frac{abc}{4A}$

Inradius: $r=\frac{A}{p}$

Length of median (divides triangle into two equal-area triangles):

$$m_a=\frac{1}{2}\sqrt{2b^2+2c^2-a^2}$$

Length of bisector (divides angles in two):

$$s_a=\sqrt{bc\left[1-\left(\frac{a}{b+c}\right)^2\right]}$$

Law of sines: $\frac{\sin\alpha}{a}=\frac{\sin\beta}{b}=\frac{\sin\gamma}{c}=\frac{1}{2R}$

Law of cosines: $\frac{a}{a^2}=\frac{b}{b^2+c^2-2bc\cos\alpha}$

Law of tangents: $\frac{a+b}{a-b}=\frac{\tan\frac{\alpha+\beta}{2}}{\tan\frac{\alpha-\beta}{2}}$

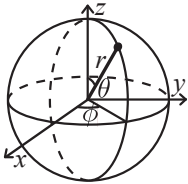
2.4.2 Quadrilaterals

With side lengths a,b,c,d , diagonals e,f , diagonals angle θ , area A and magic flux $F=b^2+d^2-a^2-c^2$:

$$4A=2ef\cdot\sin\theta=F\tan\theta=\sqrt{4e^2f^2-F^2}$$

For cyclic quadrilaterals the sum of opposite angles is 180° , $ef=ac+bd$, and $A=\sqrt{(p-a)(p-b)(p-c)(p-d)}$.

2.4.3 Spherical coordinates



$$\begin{aligned}x &= r \sin \theta \cos \phi \\y &= r \sin \theta \sin \phi \\z &= r \cos \theta\end{aligned}$$

$$\begin{aligned}r &= \sqrt{x^2 + y^2 + z^2} \\ \theta &= \operatorname{acos}(z/\sqrt{x^2 + y^2 + z^2}) \\ \phi &= \operatorname{atan2}(y, x)\end{aligned}$$

2.5 Derivatives/Integrals

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \tan x = 1 + \tan^2 x$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\int \tan ax = -\frac{\ln|\cos ax|}{a}$$

$$\int x \sin ax = \frac{\sin ax - ax \cos ax}{a^2}$$

$$\int e^{-x^2} = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x)$$

$$\int xe^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1)$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

2.6 Sums

$$c^a + c^{a+1} + \dots + c^b = \frac{c^{b+1} - c^a}{c - 1}, c \neq 1$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(2n+1)(n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

2.7 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, (-\infty < x < \infty)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, (-\infty < x < \infty)$$

2.8 Binomial theorem

$$(x+y)^r = \sum_{k=0}^\infty \binom{r}{k} x^{r-k} y^k \tag{2.1}$$

$$= x^r + rx^{r-1}y + \frac{r(r-1)}{2!}x^{r-2}y^2 + \frac{r(r-1)(r-2)}{3!}x^{r-3}y^3 + \dots \tag{2.2}$$

$$(1+x)^{-1} = \frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots$$

$$\frac{1}{(1+x)^s} = \sum_{k=0}^\infty \binom{-s}{k} x^k = \sum_{k=0}^\infty \binom{s+k-1}{k} (-1)^k x^k$$

$$\frac{1}{\sqrt{1+x}} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \frac{35}{128}x^4 - \frac{63}{256}x^5 + \dots$$

$$\frac{1}{(1-x)^s} = \sum_{k=0}^\infty \binom{s+k-1}{k} x^k$$

$$\frac{1}{\sqrt{1-x}} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \frac{35}{128}x^4 + \frac{63}{256}x^5 + \dots$$

2.9 Probability theory

Let X be a discrete random variable with probability $p_X(x)$ of assuming the value x . It will then have an expected value (mean) $\mu = \mathbb{E}(X) = \sum_x xp_X(x)$ and variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$ where σ is the standard deviation. If X is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.

Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent X and Y ,

$$V(aX + bY) = a^2V(X) + b^2V(Y).$$

2.9.1 Discrete distributions

Binomial distribution

The number of successes in n independent yes/no experiments, each which yields success with probability p is $\operatorname{Bin}(n, p)$, $n = 1, 2, \dots$, $0 \leq p \leq 1$.

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np, \sigma^2 = np(1-p)$$

$\operatorname{Bin}(n, p)$ is approximately $\operatorname{Po}(np)$ for small p .

First success distribution

The number of trials needed to get the first success in independent yes/no experiments, each wich yields success with probability p is $\operatorname{Fs}(p)$, $0 \leq p \leq 1$.

$$p(k) = p(1-p)^{k-1}, k = 1, 2, \dots$$

$$\mu = \frac{1}{p}, \sigma^2 = \frac{1-p}{p^2}$$

Poisson distribution

The number of events occurring in a fixed period of time t if these events occur with a known average rate κ and independently of the time since the last event is $\operatorname{Po}(\lambda)$, $\lambda = t\kappa$.

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$$

$$\mu = \lambda, \sigma^2 = \lambda$$

2.9.2 Continuous distributions

Uniform distribution

If the probability density function is constant between a and b and 0 elsewhere it is $\operatorname{U}(a, b)$, $a < b$.

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = \frac{a+b}{2}, \sigma^2 = \frac{(b-a)^2}{12}$$

Exponential distribution

The time between events in a Poisson process is $\operatorname{Exp}(\lambda)$, $\lambda > 0$.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu = \frac{1}{\lambda}, \sigma^2 = \frac{1}{\lambda^2}$$

Normal distribution

Most real random values with mean μ and variance σ^2 are well described by $\mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ then

$$aX_1 + bX_2 + c \sim \mathcal{N}(\mu_1 + \mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$$

2.10 Markov chains

A *Markov chain* is a discrete random process with the property that the next state depends only on the current state. Let $X_1, X_2, \dots$ be a sequence of random variables generated by the Markov process. Then there is a transition matrix $\mathbf{P} = (p_{ij})$, with $p_{ij} = \Pr(X_n = i | X_{n-1} = j)$, and $\mathbf{p}^{(n)} = \mathbf{P}^n \mathbf{p}^{(0)}$ is the probability distribution for X_n (i.e., $p_i^{(n)} = \Pr(X_n = i)$), where $\mathbf{p}^{(0)}$ is the initial distribution.

π is a stationary distribution if $\pi = \pi \mathbf{P}$. If the Markov chain is *irreducible* (it is possible to get to any state from any state), then $\pi_i = \frac{1}{\mathbb{E}(T_i)}$ where $\mathbb{E}(T_i)$ is the expected time between two visits in state i . π_j / π_i is the expected number of visits in state j between two visits in state i .

For a connected, undirected and non-bipartite graph, where the transition probability is uniform among all neighbors, π_i is proportional to node i 's degree.

A Markov chain is *ergodic* if the asymptotic distribution is independent of the initial distribution. A finite Markov chain is ergodic iff it is irreducible and *aperiodic* (i.e., the gcd of cycle lengths is 1). $\lim_{k \rightarrow \infty} \mathbf{P}^k = \mathbf{1}\pi$.

A Markov chain is an A-chain if the states can be partitioned into two sets $\mathbf{A}$ and $\mathbf{G}$, such that all states in $\mathbf{A}$ are absorbing ($p_{ii} = 1$), and all states in $\mathbf{G}$ leads to an absorbing state in $\mathbf{A}$. The probability for absorption in state $i \in \mathbf{A}$, when the initial state is j , is $a_{ij} = p_{ij} + \sum_{k \in \mathbf{G}} a_{ik} p_{kj}$. The expected time until absorption, when the initial state is i , is $t_i = 1 + \sum_{k \in \mathbf{G}} p_{ki} t_k$.

Data structures (3)

OrderStatisticTree.h

Description: A set (not multiset!) with support for finding the n'th element, and finding the index of an element. To get a map, change null.type.
Time: $\mathcal{O}(\log N)$

```
#include <bits/extc++.h>
using namespace __gnu_pbds;

template<class T>
using Tree = tree<T, null_type, less<T>, rb_tree_tag,
tree_order_statistics_node_update>;

void example() {
    Tree<int> t, t2; t.insert(8);
    auto it = t.insert(10).first;
    assert(it == t.lower_bound(9));
    assert(t.order_of_key(10) == 1);
    assert(t.order_of_key(11) == 2);
    assert(*t.find_by_order(0) == 8);
    t.join(t2); // assuming T < T2 or T > T2, merge t2 into t
}
```

HashMap.h

Description: Hash map with mostly the same API as unordered_map, but ~3x faster. Uses 1.5x memory. Initial capacity must be a power of 2 (if provided).

```
#include <bits/extc++.h>
// To use most bits rather than just the lowest ones:
struct chash { // large odd number for C
    const uint64_t C = 1l(4e18 * acos(0)) / 71;
    ll operator()(ll x) const { return __builtin_bswap64(x*C); }
};
__gnu_pbds::gp_hash_table<ll,int,chash> h({}, {}, {}, {}, {1<x<16});
```

PalindromicTree.h

Description: Builds a palindromic tree over a string. 0 is the imaginary string, 1 is the empty string. [2, n) are the palindromes, s.substr(pos[i], len[i]), occurs freq[i]. fail[i] is the longest suffix palindrome of ith palindrome.
Time: $\mathcal{O}(n)$

```
struct PalinTree {
    const int A = 128;
    string str;
    vi fail, len, pos, lz, freq;
    vector<vi> nxt;
    int n, cur;

    PalinTree(const string &s) : str(s) {
        fail = len = pos = lz = freq = vi(sz(s) + 2);
        nxt.resize(sz(s) + 2);
        n = cur = fail[0] = fail[1] = 0;
        addNode(-1, -1), addNode(0, 0);
        for(int i = 0; i < sz(s); ++i) addChar(i, s[i]);
        propagate();
    }

    void addChar(int i, int c) {
        int u = getFailure(cur, i);
        int &ch = nxt[u][c];
        if (~ch) return (void) ++lz[cur = ch];
        int v = cur = ch = addNode(len[u] + 2, i - len[u] - 1);
        fail[v] = len[v] == 1 ? 1 : nxt[getFailure(fail[u], i)][c];
    }

    int addNode(int l, int p) {
        nxt[n].assign(A, -1);
        len[n] = l, pos[n] = p, lz[n] = 1, freq[n] = 0;
        return n++;
    }

    void propagate() {
        for (int i = n - 1; ~i; --i) {
            freq[i] += lz[i];
            lz[fail[i]] += lz[i];
            lz[i] = 0;
        }
    }

    int getFailure(int u, int i) {
        while (i <= len[u] || str[i] != str[i - len[u] - 1]) u = fail[u];
        return u;
    }
};
```

SegmentTreeBeats.h

Description: Segment Tree Beats. Converts range min-max queries into addition queries. Problem statement: Given an array, perform operations of 4 kinds: range max, range min, range add, range sum.
Time: $\mathcal{O}(n * \log_2(n))$

```
const ll inf = 1e18;
ll a[N];
struct Node {
    ll max1, max2, cnt_max;
    ll min1, min2, cnt_min;
    ll sum, lazy;
};
struct SegmentTree {
    Node T[4 * N];
    void merge(int s) {
        T[s].sum = T[s << 1].sum + T[s << 1 | 1].sum;
        if (T[s << 1].max1 == T[s << 1 | 1].max1) {
            T[s].max1 = T[s << 1].max1;
            T[s].cnt_max = T[s << 1].cnt_max + T[s << 1 | 1].cnt_max;
            T[s].max2 = max(T[s << 1].max2, T[s << 1 | 1].max2);
        }
        else if (T[s << 1].max1 > T[s << 1 | 1].max1) {
            T[s].max1 = T[s << 1].max1;
            T[s].cnt_max = T[s << 1].cnt_max;
            T[s].max2 = max(T[s << 1].max2, T[s << 1 | 1].max1);
        }
        else {
            T[s].max1 = T[s << 1 | 1].max1;
```

```
T[s].cnt_max = T[s << 1 | 1].cnt_max;
            T[s].max2 = max(T[s << 1].max1, T[s << 1 | 1].max2);
        }
        if (T[s << 1].min1 == T[s << 1 | 1].min1) {
            T[s].min1 = T[s << 1].min1;
            T[s].cnt_min = T[s << 1].cnt_min + T[s << 1 | 1].cnt_min;
            T[s].min2 = min(T[s << 1].min2, T[s << 1 | 1].min2);
        }
        else if (T[s << 1].min1 < T[s << 1 | 1].min1) {
            T[s].min1 = T[s << 1].min1;
            T[s].cnt_min = T[s << 1].cnt_min;
            T[s].min2 = min(T[s << 1].min2, T[s << 1 | 1].min1);
        }
        else {
            T[s].min1 = T[s << 1 | 1].min1;
            T[s].cnt_min = T[s << 1 | 1].cnt_min;
            T[s].min2 = min(T[s << 1].min1, T[s << 1 | 1].min2);
        }
    }
    void build(int s = 1, int l = 0, int r = n - 1) {
        if (l == r) {
            T[s].max1 = T[s].min1 = T[s].sum = a[l];
            T[s].cnt_max = T[s].cnt_min = 1;
            T[s].max2 = -inf, T[s].min2 = inf;
            return;
        }
        int mid = (l + r) >> 1;
        build(s << 1, l, mid); build(s << 1 | 1, mid + 1, r);
        merge(s);
    }
    void push_add(int s, int l, int r, ll val) {
        if (!val) return;
        T[s].sum += 1LL * (r - l + 1) * val;
        T[s].max1 += val; if (T[s].max2 != -inf) T[s].max2 += val;
        T[s].min1 += val; if (T[s].min2 != inf) T[s].min2 += val;
        T[s].lazy += val;
    }
    void push_max(int s, ll val, bool v) {
        if (val >= T[s].max1) return;
        T[s].sum -= 1LL * T[s].max1 * T[s].cnt_max;
        T[s].max1 = val;
        T[s].sum += 1LL * T[s].max1 * T[s].cnt_max;
        if (v) T[s].min1 = T[s].max1;
        else {
            if (val <= T[s].min1) T[s].min1 = val;
            else if (val < T[s].min2) T[s].min2 = val;
        }
    }
    void push_min(int s, ll val, bool v) {
        if (val <= T[s].min1) return;
        T[s].sum -= 1LL * T[s].min1 * T[s].cnt_min;
        T[s].min1 = val;
        T[s].sum += 1LL * T[s].min1 * T[s].cnt_min;
    }
    if (v) T[s].max1 = T[s].min1;
    else {
        if (val >= T[s].max1) T[s].max1 = val;
        else if (val > T[s].max2) T[s].max2 = val;
    }
}
    void push_down(int s, int l, int r) {
        if (l == r) return;
        int mid = (l + r) >> 1;
        if (T[s].lazy) {
            push_add(s << 1, l, mid, T[s].lazy);
            push_add(s << 1 | 1, mid + 1, r, T[s].lazy);
            T[s].lazy = 0;
        }
        push_max(s << 1, T[s].max1, l == mid);
        push_max(s << 1 | 1, T[s].max1, mid + 1 == r);
        push_min(s << 1, T[s].min1, l == mid);
        push_min(s << 1 | 1, T[s].min1, mid + 1 == r);
    }
    void update_add(int s, int l, int r, int u, int v, ll val) {
        if (l > v || r < u) return;
        if (l >= u && r <= v) {
            push_add(s, l, r, val);
            return;
        }
    }
```

```

    push_down(s, l, r);
    int mid = (l + r) >> 1;
    update_add(s << 1, l, mid, u, v, val);
    update_add(s << 1 | 1, mid + 1, r, u, v, val);
    merge(s);
}

void update_max(int s, int l, int r, int u, int v, ll val) {
    if (l > v || r < u || val <= T[s].min1) return;
    if (l >= u && r <= v && val < T[s].min2) {
        push_min(s, val, l == r);
        return;
    }
    push_down(s, l, r);
    int mid = (l + r) >> 1;
    update_max(s << 1, l, mid, u, v, val);
    update_max(s << 1 | 1, mid + 1, r, u, v, val);
    merge(s);
}

void update_min(int s, int l, int r, int u, int v, ll val) {
    if (l > v || r < u || val >= T[s].max1) return;
    if (l >= u && r <= v && val > T[s].max2) {
        push_max(s, val, l == r);
        return;
    }
    push_down(s, l, r);
    int mid = (l + r) >> 1;
    update_min(s << 1, l, mid, u, v, val);
    update_min(s << 1 | 1, mid + 1, r, u, v, val);
    merge(s);
}

ll get(int s, int l, int r, int u, int v) {
    if (l > v || r < u) return 0;
    if (l >= u && r <= v) return T[s].sum;
    push_down(s, l, r);
    int mid = (l + r) >> 1;
    return get(s << 1, l, mid, u, v) + get(s << 1 | 1, mid + 1, r, u, v);
}

} it;
```

XorSegmentTree.h

Description: Applicable for problems required operations on XOR indexes (e.g: $a[l \sim x], a[(l + 1) \sim x], \dots$) Query update is only available if Merge(id, id + 1) == Merge(id + 1, id) (Commutative)
Time: $\mathcal{O}(\log(n))$ per query

1c95e3, 53 lines

#define pb push_back

```

const int maxN = 1e5 + 1;
int a[maxN];
struct XORSegtree {
    vector<int> st[4 * maxN];
    int lazy[4 * maxN];
    XORSegtree() {
        rep(i, 0, 4 * maxN) {
            st[i].clear();
            lazy[i] = 0;
        }
    }
    void build(int id, int l, int r){
        lazy[id] = 0;
        st[id].clear();
        if(l == r){
            st[id].pb(a[l]);
            return;
        }
        int mid = (l + r) / 2;
        build(id * 2, l, mid); build(id * 2 + 1, mid + 1, r);
        int len = r - l + 1;
        st[id].resize(len);
        rep(i, 0, len) {
            if(i >= len / 2)
                st[id][i] = st[id * 2][i - len / 2] + st[id * 2 + 1][i - len / 2];
            else st[id][i] = st[id * 2 + 1][i] + st[id * 2][i];
        }
    }
    void update(int id, int l, int r, int pos, int val) {
        lazy[id] += val;
        if (l == r) return;
```

```

        int mid = (l + r) / 2;
        if (pos <= mid) update(id * 2, l, mid, pos, val);
        else update(id * 2 + 1, mid + 1, r, pos, val);
    }
    int get(int id, int l, int r, int u, int v, int x, int depth) {
        if(l > v || r < u) return 0;
        if(l >= u && r <= v){
            return lazy[id] + st[id][x & ((l << depth) - 1)];
        }
        int mid = (l + r) / 2;
        if(x >> (depth - 1) & 1){
            return get(id * 2 + 1, mid + 1, r, u - 1 + (mid + 1), v - 1 + (mid + 1), x, depth - 1)
                + get(id * 2, l, mid, l + u - (mid + 1), l + v - (mid + 1), x, depth - 1);
        }
        else {
            return get(id * 2, l, mid, u, v, x, depth - 1)
                + get(id * 2 + 1, mid + 1, r, u, v, x, depth - 1);
        }
    }
}
};
```

UnionFindRollback.h

Description: Disjoint-set data structure with undo. If undo is not needed, skip st.time() and rollback().
Usage: int t = uf.time(); ...; uf.rollback(t);
Time: $\mathcal{O}(\log(N))$

de4ad0, 21 lines

```

struct RollbackUF {
    vi e; vector<pii> st;
    RollbackUF(int n) : e(n, -1) {}
    int size(int x) { return -e[find(x)]; }
    int find(int x) { return e[x] < 0 ? x : find(e[x]); }
    int time() { return sz(st); }
    void rollback(int t) {
        for (int i = time(); i --> t;)
            e[st[i].first] = st[i].second;
        st.resize(t);
    }
    bool join(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) return false;
        if (e[a] > e[b]) swap(a, b);
        st.push_back({a, e[a]});
        st.push_back({b, e[b]});
        e[a] += e[b]; e[b] = a;
        return true;
    }
}
};
```

DynamicWaveletTree.h

Description: Given an array of n elements with q queries [l, r, k]. For each query print the k-th smallest value and the sum of k smallest values in the subsequence [l, r]. Calculate the position of k in the subsequence [l, r].
Time: $\mathcal{O}(n * \log(a_{max}))$

bc992e, 56 lines

```

struct WaveletTree{
    struct TNode{
        int l = 0, r = 0, lp = 0, rp = 0;
        TNode(int _l, int _r) : l(_l), r(_r) {}
    };
    vector <vector <int>> val;
    vector <vector <ll>> sum;
    vector <int> b;
    vector <TNode> wt;
    int wt_sz = 1;
    WaveletTree(int _n = 0) : val(_n * 4), sum(_n * 4), wt(_n * 4) {}
    template <class T> void build(int id, T i, T j, int l, int r){
        wt[id].l = ll; wt[id].r = r;
        if (l == r) return;
        val[id].pb(0); sum[id].pb(0);
        int mid = (l + r - (l + r < 0)) / 2, maxx = l, minn = r;
        for (auto it = i; it != j; ++it){
            val[id].pb(val[id].back() + (*it <= mid));
            sum[id].pb(sum[id].back() + *it * (*it <= mid));
            if (*it <= mid) maxx = max(maxx, *it);
            else minn = min(minn, *it);
        }
    }
};
```

```

        auto p = stable_partition(i, j, [=](const auto &w) {return w <= mid;});
        wt[id].lp = ++wt_sz; wt[id].rp = ++wt_sz;
        build(wt[id].lp, i, p, l, maxx); build(wt[id].rp, p, j, minn, r);
    }
    ll pos(int k, int i, int j) {
        --i;
        ll cnt = 0; int id = l, l = wt[id].l, r = wt[id].r;
        while (l < r){
            int mid = (l + r - (l + r < 0)) / 2;
            if (k <= mid) i = val[id][i], j = val[id][j], id = wt[id].lp;
            else{
                cnt += val[id][j] - val[id][i];
                i -= val[id][i], j -= val[id][j], id = wt[id].rp;
            }
            l = wt[id].l, r = wt[id].r;
        }
        return cnt + (k >= l) * (j - i);
    }
    ll get(int k, int i, int j){
        if (!k) return 0;
        --i;
        ll res = 0; int id = l, l = wt[id].l, r = wt[id].r;
        while (l < r) {
            if (k <= val[id][j] - val[id][i]) i = val[id][i], j = val[id][j], id = wt[id].lp;
            else{
                res += sum[id][j] - sum[id][i];
                k -= val[id][j] - val[id][i];
                i -= val[id][i], j -= val[id][j], id = wt[id].rp;
            }
            l = wt[id].l, r = wt[id].r;
        }
        return res + k * l;
    }
}
};
```

Matrix.h

Description: Basic operations on square matrices.
Usage: Matrix<int, 3> A;
A.d = {{{{1,2,3}}, {{4,5,6}}, {{7,8,9}}}};
vector<int> vec = {1,2,3};
vec = (A^N) * vec;

fee317, 26 lines

```

template<class T, int N> struct Matrix {
    typedef Matrix M;
    array<array<T, N>, N> d{};
    M operator*(const M& m) const {
        M a;
        rep(i,0,N) rep(j,0,N)
            rep(k,0,N) a.d[i][k] += d[i][j]*m.d[j][k];
        return a;
    }
    vector<T> operator*(const vector<T>& vec) const {
        vector<T> ret(N);
        rep(i,0,N) rep(j,0,N) ret[i] += d[i][j] * vec[j];
        return ret;
    }
    M operator^(ll p) const {
        assert(p >= 0);
        M a, b(*this);
        rep(i,0,N) a.d[i][i] = 1;
        while (p) {
            if (p&1) a = a*b;
            b = b*b;
            p >>= 1;
        }
        return a;
    }
}
};
```

LineContainer.h

Description: Container where you can add lines of the form $kx+m$, and query maximum values at points x. Useful for dynamic programming (“convex hull trick”).
Time: $\mathcal{O}(\log N)$

8ec1c7, 30 lines

```

struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
```

```
bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) { // floored division
        return a / b - ((a ^ b) < 0 && a % b); }
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};
```

LiChaoTree.h
Description: Maintains a set of lines $y(x) = ax + b$ and returns the minimum $y(pos)$.
Time: $\mathcal{O}(n \cdot \log_2 n)$

```
struct line {
    int a, b;
};
int get(line m, int x) {
    return 1ll * m.a * x + m.b;
}
struct LiChaoTree {
    line t[4 * N];
    ll query(int s, int l, int r, int pos) {
        if (l > pos || r < pos) return 1e18;
        int res = get(t[s], pos);
        if (l == r) return res;
        int mid = (l + r) / 2;
        res = min(res, query(s * 2, l, mid, pos));
        res = min(res, query(s * 2 + 1, mid + 1, r, pos));
        return res;
    }
    void update(int s, int l, int r, int u, int v, line val) {
        if (l > v || u > r) return;
        int mid = (l + r) / 2;
        if (l >= u && r <= v) {
            if (get(t[s], l) <= get(val, l) && get(t[s], r) <= get(val, r))
                return;
            if (get(t[s], l) >= get(val, l) && get(t[s], r) >= get(val, r)) {
                t[s] = val;
                return;
            }
            if (get(t[s], l) <= get(val, l) && get(t[s], mid) <= get(val, mid))
                update(2 * s + 1, mid + 1, r, u, v, val);
            return;
        }
        if (get(t[s], l) >= get(val, l) && get(t[s], mid + 1) >= get(val, mid + 1)) {
            update(2 * s + 1, mid + 1, r, u, v, t[s]);
            t[s] = val;
            return;
        }
        if (get(t[s], r) <= get(val, r) && get(t[s], mid) <= get(val, mid))
            update(2 * s, l, mid, u, v, val);
        return;
    }
    if (get(t[s], r) >= get(val, r) && get(t[s], mid + 1) >= get(val, mid + 1)) {
        update(2 * s, l, mid, u, v, t[s]);
        update(2 * s + 1, mid + 1, r, u, v, t[s]);
    }
```

```
t[s] = val;
return;
}
}
update(2 * s, l, mid, u, v, val);
update(2 * s + 1, mid + 1, r, u, v, val);
}
}
} it;
```

Treap.h
Description: A short self-balancing tree. It acts as a sequential container with log-time splits/joins, and is easy to augment with additional data. Also supports range updates.
Time: $\mathcal{O}(\log N)$ per split/merge

```
struct Node {
    Node *l = 0, *r = 0;
    int val, y, c = 1, rev = 0;
    Node(int val) : val(val), y(rand()) {}
};

int cnt(Node* n) { return n ? n->c : 0; }
void pull(Node* n) { if (n) n->c = cnt(n->l) + cnt(n->r) + 1; }
void push(Node* n) {
    if (!n) return;
    if (n->rev) {
        swap(n->l, n->r);
        if (n->l) n->l->rev ^= 1;
        if (n->r) n->r->rev ^= 1;
        n->rev = 0;
    }
}

template<class F> void each(Node* n, F f) {
    if (n) { push(n); each(n->l, f); f(n->val); each(n->r, f); }
}

// Put i first nodes into l, the rest into r
void split(Node* x, Node*& l, Node*& r, int i) {
    if (!x) return void(l = r = 0);
    push(x);
    // replace cnt(x->l) with x->val for lower_bound(i)
    if (i <= cnt(x->l)) split(x->l, l, x->l, i), r = x;
    // and just i instead
    else split(x->r, x->r, r, i - cnt(x->l) - 1), l = x;
    pull(x);
}

// Append r to l, store it in x
void merge(Node*& x, Node* l, Node* r) {
    push(l), push(r);
    if (!l || !r) x = l ? l : r;
    else if (l->y < r->y) merge(r->l, l, r->l), x = r;
    else merge(l->r, l->r, r), x = l;
    pull(x);
}

void insert(Node*& t, Node* n, int pos) {
    Node* l, * r;
    split(t, l, r, pos), merge(l, l, n), merge(t, l, r);
}

// Example application: move the range [l, r) to index k
void move(Node*& t, int l, int r, int k) {
    Node* a, * b, * c;
    split(t, a, c, r), split(a, a, b, l), merge(t, a, c);
    if (k <= l) insert(t, b, k);
    else insert(t, b, k - r + l);
}

// Example application: reverse the range [l, r)
void rev(Node*& t, int l, int r) {
    Node* a, * b, * c;
    split(t, a, c, r), split(a, a, b, l);
    b->rev ^= 1;
    merge(a, a, b), merge(t, a, c);
}
```

FenwickTree.h
Description: Computes partial sums $a[0] + a[1] + \dots + a[pos - 1]$, and updates single elements $a[i]$, taking the difference between the old and new value.
Time: Both operations are $\mathcal{O}(\log N)$.

```
e62fac, 22 lines

struct FT {
    vector<ll> s;
    FT(int n) : s(n) {}
    void update(int pos, ll dif) { // a[pos] += dif
        for (; pos < sz(s); pos |= pos + 1) s[pos] += dif;
    }
    ll query(int pos) { // sum of values in [0, pos)
        ll res = 0;
        for (; pos > 0; pos &= pos - 1) res += s[pos-1];
        return res;
    }
    int lower_bound(ll sum) { // min pos st sum of [0, pos] >= sum
        // Returns n if no sum is >= sum, or -1 if empty sum is .
        if (sum <= 0) return -1;
        int pos = 0;
        for (int pw = 1 << 25; pw; pw >= 1) {
            if (pos + pw <= sz(s) && s[pos + pw-1] < sum)
                pos += pw, sum -= s[pos-1];
        }
        return pos;
    }
};

FenwickTree2d.h  
Description: Computes sums  $a[i,j]$  for all  $i < I, j < J$ , and increases single elements  $a[i,j]$ . Requires that the elements to be updated are known in advance (call fakeUpdate() before init()).  
Time:  $\mathcal{O}(\log^2 N)$ . (Use persistent segment trees for  $\mathcal{O}(\log N)$ .)
```

```
"FenwickTree.h"
157f07, 22 lines

struct FT2 {
    vector<vi> ys; vector<FT> ft;
    FT2(int limx) : ys(limx) {}
    void fakeUpdate(int x, int y) {
        for (; x < sz(ys); x |= x + 1) ys[x].push_back(y);
    }
    void init() {
        for (vi& v : ys) sort(all(v)), ft.emplace_back(sz(v));
    }
    int ind(int x, int y) {
        return (int)(lower_bound(all(ys[x]), y) - ys[x].begin()); }
    void update(int x, int y, ll dif) {
        for (; x < sz(ys); x |= x + 1)
            ft[x].update(ind(x, y), dif);
    }
    ll query(int x, int y) {
        ll sum = 0;
        for (; x; x &= x - 1)
            sum += ft[x-1].query(ind(x-1, y));
        return sum;
    }
};

RMQ.h  
Description: Range Minimum Queries on an array. Returns  $\min(V[a], V[a + 1], \dots V[b - 1])$  in constant time.  
Usage: RMQ rmq(values);  
rmq.query(inclusive, exclusive);  
Time:  $\mathcal{O}(|V| \log |V| + Q)$ 
```

```
510c32, 16 lines

template<class T>
struct RMQ {
    vector<vector<T>> jmp;
    RMQ(const vector<T>& V) : jmp(1, V) {
        for (int pw = 1, k = 1; pw * 2 <= sz(V); pw *= 2, ++k) {
            jmp.emplace_back(sz(V) - pw * 2 + 1);
            rep(j, 0, sz(jmp[k]))
                jmp[k][j] = min(jmp[k - 1][j], jmp[k - 1][j + pw]);
        }
    }
    T query(int a, int b) {
        assert(a < b); // or return inf if a == b
        int dep = 31 - __builtin_clz(b - a);
        return min(jmp[dep][a], jmp[dep][b - (1 << dep)]);
    }
};
```

```
}
};
```

2DSparseTable.h

Description: Self-explanatory.

Time: $\mathcal{O}(n \cdot \log_2^2 n)$ db8d7c, 16 lines

```
void st_prepare() {
    repn(i, 1, n) repn(j, 1, m) st[0][0][i][j] = f[i][j];
    for (int i = 1; (1 << i) <= n; i++)
        repn(x, 1 << i, n) repn(y, 1, m)
            st[i][0][x][y] = max(st[i - 1][0][x][y], st[i - 1][0][x - (1 << (i - 1))][y]);
    for (int i = 0; (1 << i) <= n; i++)
        for (int j = 1; (1 << j) <= m; j++)
            repn(x, 1 << i, n) repn(y, 1 << j, m)
                st[i][j][x][y] = max(st[i][j - 1][x][y], st[i][j - 1][x][y - (1 << (j - 1))]);
}

int query_max(int a, int b, int c, int d) {
    int k1 = 31 - __builtin_clz(c - a + 1);
    int k2 = 31 - __builtin_clz(d - b + 1);
    return max(max(st[k1][k2][c][d], st[k1][k2][a + (1 << k1) - 1][d]),
               max(st[k1][k2][c][b + (1 << k2) - 1], st[k1][k2][a + (1 << k1) - 1][b + (1 << k2) - 1]));
}
```

MoQueries.h

Description: Answer interval or tree path queries by finding an approximate TSP through the queries, and moving from one query to the next by adding/removing points at the ends. If values are on tree edges, change step to add/remove the edge (a, c) and remove the initial add call (but keep in).

Time: $\mathcal{O}(N\sqrt{Q})$ a12ef4, 49 lines

```
void add(int ind, int end) { ... } // add a[ind] (end = 0 or 1)
void del(int ind, int end) { ... } // remove a[ind]
int calc() { ... } // compute current answer
```

```
vi mo(vector<pii> Q) {
    int L = 0, R = 0, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s;
    #define K(x) pii(x.first/blk, x.second ^ -(x.first/blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int s, int t){ return K[Q[s]] < K[Q[t]]; });
    for (int qi : s) {
        pii q = Q[qi];
        while (L > q.first) add(--L, 0);
        while (R < q.second) add(R++, 1);
        while (L < q.first) del(L++, 0);
        while (R > q.second) del(--R, 1);
        res[qi] = calc();
    }
    return res;
}
```

```
vi moTree(vector<array<int, 2>> Q, vector<vi>& ed, int root=0) {
    int N = sz(ed), pos[2] = {}, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s, I(N), L(N), R(N), in(N), par(N);
    add(0, 0), in[0] = 1;
    auto dfs = [&](int x, int p, int dep, auto& f) -> void {
        par[x] = p;
        L[x] = N;
        if (dep) I[x] = N++;
        for (int y : ed[x]) if (y != p) f(y, x, !dep, f);
        if (!dep) I[x] = N++;
        R[x] = N;
    };
    dfs(root, -1, 0, dfs);
    #define K(x) pii(I[x[0]] / blk, I[x[1]] ^ -(I[x[0]] / blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int s, int t){ return K[Q[s]] < K[Q[t]]; });
    for (int qi : s) rep(end, 0, 2) {
        int &a = pos[end], b = Q[qi][end], i = 0;
        #define step(c) { if (in[c]) { del(a, end); in[a] = 0; } \
                        else { add(c, end); in[c] = 1; } a = c; }
        while (!L[b] <= L[a] && R[a] <= R[b])
            I[i++] = b, b = par[b];
        while (a != b) step(par[a]);
        while (i--) step(I[i]);
    }
}
```

2DSparseTable MoQueries OrderedSet Bitset

```
    if (end) res[qi] = calc();
}
return res;
}
```

OrderedSet.h

Description: Ordered set for integers in $[0, N)$.

Time: $\mathcal{O}(n \cdot \log_2 n)$ a58a60, 104 lines

```
// If data type is not integer / not compressible into [0, N):
// typedef __gnu_pbds::tree<T, null_type, less<T>, rb_tree_tag,
//      tree_order_statistics_node_update> ordered_set;
// needs to include <ext/pb_ds/assoc_container.hpp> AND <ext/pb_ds/
//      tree_policy.hpp>
```

```
namespace cp_algo::structures {
    template<typename T, typename Container = std::vector<T>>
    struct fenwick {
        size_t n;
        Container data;

        fenwick(auto &&range) {
            assign(range);
        }

        void to_prefix_sums() {
            for (size_t i = 1; i < n; i++) {
                if (i + (i & -i) <= n) {
                    data[i + (i & -i)] += data[i];
                }
            }
        }

        void assign(auto &&range) {
            n = size(range) - 1;
            data = move(range);
            to_prefix_sums();
        }

        void add(size_t x, T const& v) {
            for (++x; x <= n; x += x & -x) {
                data[x] += v;
            }
        }

        // sum of [0, r)
        T prefix_sum(size_t r) const {
            assert(r <= n);
            T res = 0;
            for (; r; r -= r & -r) {
                res += data[r];
            }
            return res;
        }

        // sum of [l, r)
        T range_sum(size_t l, size_t r) const {
            return prefix_sum(r) - prefix_sum(l);
        }

        // First r s.t. prefix_sum(r) >= k
        // Assumes data[x] >= 0 for all x
        size_t prefix_lower_bound(T k) const {
            int x = 0;
            for (size_t i = std::bit_floor(n); i; i /= 2) {
                if (x + i <= n && data[x + i] < k) {
                    k -= data[x + i];
                    x += i;
                }
            }
            return x;
        }
    };
}
```

```
namespace cp_algo::structures {
    // fenwick-based set for [0, maxc)
    template<size_t maxc>
    struct fenwick_set: fenwick<int, std::array<int, maxc+1>> {
        using Base = fenwick<int, std::array<int, maxc+1>>;
        size_t sz = 0;
        std::bitset<maxc> present;
        fenwick_set(): Base(std::array<int, maxc+1>()) {}
        fenwick_set(auto &&range): fenwick_set() {
            for (auto x: range) {
```

```
                Base::data[x + 1] = 1;
                sz += !present[x];
                present[x] = 1;
            }
            Base::to_prefix_sums();
        }

        void insert(size_t x) {
            if (present[x]) return;
            present[x] = 1;
            sz++;
            Base::add(x, 1);
        }

        void erase(size_t x) {
            if (!present[x]) return;
            present[x] = 0;
            sz--;
            Base::add(x, -1);
        }

        size_t order_of_key(size_t x) const {
            return Base::prefix_sum(x);
        }

        size_t find_by_order(size_t order) const {
            return order < sz ? Base::prefix_lower_bound(order + 1) : -1;
        }

        size_t lower_bound(size_t x) const {
            if (present[x]) {return x;}
            auto order = order_of_key(x);
            return order < sz ? find_by_order(order) : -1;
        }

        size_t pre_upper_bound(size_t x) const {
            if (present[x]) {return x;}
            auto order = order_of_key(x);
            return order ? find_by_order(order - 1) : -1;
        }

    };
}
```

Bitset.h

Description: Actual $n / 64$ bitset.

Time: $\mathcal{O}(\frac{n}{64})$ b54005, 40 lines

```
struct bit {
    const static int L = 782;
    ull t[L];

    void set(int x, int y = 1) {
        if (y) t[x >> 6] |= 1ull << (x & 63);
        else t[x >> 6] &= t[x >> 6] ^ (1ull << (x & 63));
    }

    void reset() {
        memset(t, 0, sizeof(t));
    }

    void flip(int x) {
        t[x >> 6] ^= 1ull << (x & 63);
    }

    int test(int x) {
        return t[x >> 6] >> (x & 63) & 1;
    }

    ull& operator [] (int i) {
        return t[i];
    }

    int count() {
        int res=0;
        rep(i, 0, L) res += __builtin_popcountll(t[i]);
        return res;
    }

    friend bit operator ^ (const bit &a, const bit &b) {
        bit tmp;
        rep(i, 0, L) tmp.t[i] = a.t[i] ^ b.t[i];
        return tmp;
    }

    friend bit operator & (const bit &a, const bit &b) {
        bit tmp;
        rep(i, 0, L) tmp.t[i] = a.t[i] & b.t[i];
        return tmp;
    }

    friend bit operator | (const bit &a, const bit &b) {
        bit tmp;
        rep(i, 0, L) tmp.t[i] = a.t[i] | b.t[i];
    }
}
```

```
        return tmp;
    }
};
```

Numerical (4)

4.1 Polynomials and recurrences

Polynomial.h	c9b7b0, 17 lines
--------------	------------------

```
struct Poly {
    vector<double> a;
    double operator()(double x) const {
        double val = 0;
        for (int i = sz(a); i--;) (val *= x) += a[i];
        return val;
    }
    void diff() {
        rep(i,1,sz(a)) a[i-1] = i*a[i];
        a.pop_back();
    }
    void divroot(double x0) {
        double b = a.back(), c; a.back() = 0;
        for(int i=sz(a)-1; i--;) c = a[i], a[i] = a[i+1]*x0+b, b=c;
        a.pop_back();
    }
};
```

PolyRoots.h

Description: Finds the real roots to a polynomial.
Usage: polyRoots({{2,-3,1}},-1e9,1e9) // solve x^2-3x+2 = 0
Time: $\mathcal{O}\left(n^2 \log(1/\epsilon)\right)$
"Polynomial.h" b00bfe, 23 lines

```
vector<double> polyRoots(Poly p, double xmin, double xmax) {
    if (sz(p.a) == 2) { return {p.a[0]/p.a[1]}; }
    vector<double> ret;
    Poly der = p;
    der.diff();
    auto dr = polyRoots(der, xmin, xmax);
    dr.push_back(xmin-1);
    dr.push_back(xmax+1);
    sort(all(dr));
    rep(i,0,sz(dr)-1) {
        double l = dr[i], h = dr[i+1];
        bool sign = p(l) > 0;
        if (sign ^ (p(h) > 0)) {
            rep(it,0,60) { // while (h - l > 1e-8)
                double m = (l + h) / 2, f = p(m);
                if ((f <= 0) ^ sign) l = m;
                else h = m;
            }
            ret.push_back((l + h) / 2);
        }
    }
    return ret;
}
```

PolyInterpolate.h

Description: Given n points $(x[i], y[i])$, computes an $n-1$ -degree polynomial p that passes through them: $p(x) = a[0]*x^0 + \dots + a[n-1]*x^{n-1}$. For numerical precision, pick $x[k] = c * \cos(k/(n-1) * \pi)$, $k = 0 \dots n-1$.

Time: $\mathcal{O}\left(n^2\right)$	08bf48, 13 lines
--	------------------

```
typedef vector<double> vd;
vd interpolate(vd x, vd y, int n) {
    vd res(n), temp(n);
    rep(k,0,n-1) rep(i,k+1,n)
        y[i] = (y[i] - y[k]) / (x[i] - x[k]);
    double last = 0; temp[0] = 1;
    rep(k,0,n) rep(i,0,n) {
        res[i] += y[k] * temp[i];
        swap(last, temp[i]);
        temp[i] -= last * x[k];
    }
    return res;
}
```

BerlekampMassey.h

Description: Recovers any n -order linear recurrence relation from the first $2n$ terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. Should work on any field, but numerical stability for floats is not guaranteed. Output will have size $\leq n$.

Usage: berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}	
Time: $\mathcal{O}\left(N^2\right)$	
"../number-theory/ModPow.h"	96548b, 20 lines

```
vector<ll> berlekampMassey(vector<ll> s) {
    int n = sz(s), L = 0, m = 0;
    vector<ll> C(n), B(n), T;
    C[0] = B[0] = 1;

    ll b = 1;
    rep(i,0,n) { ++m;
        ll d = s[i] % mod;
        rep(j,1,L+1) d = (d + C[j] * s[i - j]) % mod;
        if (!d) continue;
        T = C; ll coef = d * modpow(b, mod-2) % mod;
        rep(j,m,n) C[j] = (C[j] - coef * B[j - m]) % mod;
        if (2 * L > i) continue;
        L = i + 1 - L; B = T; b = d; m = 0;
    }

    C.resize(L + 1); C.erase(C.begin());
    for (ll& x : C) x = (mod - x) % mod;
    return C;
}
```

LinearRecurrence.h

Description: Generates the k 'th term of an n -order linear recurrence $S[i] = \sum_j S[i-j-1]tr[j]$, given $S[0 \dots \geq n-1]$ and $tr[0 \dots n-1]$. Faster than matrix multiplication. Useful together with Berlekamp-Massey.

Usage: linearRec({0, 1}, {1, 1}, k) // k'th Fibonacci number	
Time: $\mathcal{O}\left(n^2 \log k\right)$	f4e444, 26 lines

```
typedef vector<ll> Poly;
ll linearRec(Poly S, Poly tr, ll k) {
    int n = sz(tr);

    auto combine = [&](Poly a, Poly b) {
        Poly res(n * 2 + 1);
        rep(i,0,n+1) rep(j,0,n+1)
            res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
        for (int i = 2 * n; i > n; --i) rep(j,0,n)
            res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
        res.resize(n + 1);
        return res;
    };

    Poly pol(n + 1), e(pol);
    pol[0] = e[1] = 1;

    for (++k; k; k /= 2) {
        if (k % 2) pol = combine(pol, e);
        e = combine(e, e);
    }

    ll res = 0;
    rep(i,0,n) res = (res + pol[i + 1] * S[i]) % mod;
    return res;
}
```

4.2 Optimization

GoldenSectionSearch.h

Description: Finds the argument minimizing the function f in the interval $[a,b]$ assuming f is unimodal on the interval, i.e. has only one local minimum. The maximum error in the result is ϵ ps. Works equally well for maximization with a small change in the code. See TernarySearch.h in the Various chapter for a discrete version.

Usage: double func(double x) { return 4*x+.3*x*x; }	
double xmin = gss(-1000,1000,func);	
Time: $\mathcal{O}(\log((b-a)/\epsilon))$	31d45b, 14 lines

```
double gss(double a, double b, double (*f)(double)) {
    double r = (sqrt(5)-1)/2, eps = 1e-7;
    double x1 = b - r*(b-a), x2 = a + r*(b-a);
```

```
double f1 = f(x1), f2 = f(x2);
while (b-a > eps)
    if (f1 < f2) { //change to > to find maximum
        b = x2; x2 = x1; f2 = f1;
        x1 = b - r*(b-a); f1 = f(x1);
    } else {
        a = x1; x1 = x2; f1 = f2;
        x2 = a + r*(b-a); f2 = f(x2);
    }
    return a;
}
```

HillClimbing.h

Description: Poor man's optimization for unimodal functions.	Seeef, 14 lines
---	-----------------

```
typedef array<double, 2> P;

template<class F> pair<double, P> hillClimb(P start, F f) {
    pair<double, P> cur(f(start), start);
    for (double jmp = 1e9; jmp > 1e-20; jmp /= 2) {
        rep(j,0,100) rep(dx,-1,2) rep(dy,-1,2) {
            P p = cur.second;
            p[0] += dx*jmp;
            p[1] += dy*jmp;
            cur = min(cur, make_pair(f(p), p));
        }
    }
    return cur;
}
```

Integrate.h

Description: Simple integration of a function over an interval using Simpson's rule. The error should be proportional to h^4 , although in practice you will want to verify that the result is stable to desired precision when epsilon changes.

Usage: integrate(f, a, b, 1e-10) // integral of f from a to b	9956fc, 7 lines
Time: $\mathcal{O}(n^3)$	

```
template<class F>
double quad(double a, double b, F f, const int n = 1000) {
    double h = (b - a) / 2 / n, v = f(a) + f(b);
    rep(i,1,n*2)
        v += f(a + i*h) * (i&1 ? 4 : 2);
    return v * h / 3;
}
```

LagrangeInterpolation.h

Description: Lagrange interpolation knowing $f(1), f(2), \dots, f(k)$.	
Time: $\mathcal{O}(n)$	d31fca, 32 lines

```
int interpolate(int x, int k, bool bf = false) {
    if (k == 0) return x;
    // find 1^k + 2^k + ... + x^k.(k+1) degree polynomial -> (k+2) points
    if (x <= k + 1 || bf) {
        int s = 0;
        for (int i = 1; i <= x; i++) {
            s = (s + qpow(i, k)) % mod;
        }
        return s;
    }
    vector<int> pre(k + 2), suf(k + 2), inv(k + 2);
    inv[0] = 1;
    pre[0] = x;
    suf[k + 1] = x - (k + 1);
    for (int i = 1; i <= k; i++) pre[i] = pre[i - 1] * (x - i) % mod; //
        numerator prefix product
    for (int i = k; i >= 1; i--) suf[i] = suf[i + 1] * (x - i) % mod; //
        numerator suffix product
    for (int i = 1; i <= k + 1; i++) inv[i] = inv[i - 1] * rv(i) % mod; //
        denominator factorial

    int ans = 0;
    int yi = 0; // 0^k ~- i^k
    int num, denom;
    for (int i = 0; i <= k + 1; i++) {
        yi = (yi + qpow(i, k)) % mod; // interpolate point: (i, yi)
        if (i == 0) num = suf[1];
        else if (i == k + 1) num = pre[k];
        else num = pre[i - 1] * suf[i + 1] % mod; // numerator
        denom = inv[i] * inv[k + 1 - i] % mod; // denominator
        if ((i + k) & 1) ans += (yi * num % mod) * denom % mod; // (-1)^(i-
            deg) however deg is k+1 here so :)
        else ans -= (yi * num % mod) * denom % mod;
```

```
ans = (ans % mod + mod) % mod;
}
return ans;
}

IntegrateAdaptive.h
Description: Fast integration using an adaptive Simpson’s rule.
Usage: double sphereVolume = quad(-1, 1, [](double x) {
return quad(-1, 1, [&](double y) {
return quad(-1, 1, [&](double z) {
return x*x + y*y + z*z < 1; });});});};
92dd79, 15 lines
```

```
typedef double d;
#define S(a,b) (f(a) + 4*f((a+b) / 2) + f(b)) * (b-a) / 6

template <class F>
d rec(F& f, d a, d b, d eps, d S) {
d c = (a + b) / 2;
d S1 = S(a, c), S2 = S(c, b), T = S1 + S2;
if (abs(T - S) <= 15 * eps || b - a < 1e-10)
return T + (T - S) / 15;
return rec(f, a, c, eps / 2, S1) + rec(f, c, b, eps / 2, S2);
}

template<class F>
d quad(d a, d b, F f, d eps = 1e-8) {
return rec(f, a, b, eps, S(a, b));
}
```

Simplex.h

Description: Solves a general linear maximization problem: maximize $c^T x$ subject to $Ax \leq b, x \geq 0$. Returns -inf if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of $c^T x$ otherwise. The input vector is set to an optimal x (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that $x = 0$ is viable.

Usage: vvd A = {{1,-1}, {-1,1}, {-1,-2}};

vvd b = {1,1,-4}, c = {-1,-1}, x;

T val = LPSolver(A, b, c).solve(x);

Time: $\mathcal{O}(NM \times \#pivots)$, where a pivot may be e.g. an edge relaxation. $\mathcal{O}(2^n)$ in the general case.

```
aa8530, 68 lines
```

```
typedef double T; // long double, Rational, double + modK>...
typedef vector<T> vd;
typedef vector<vd> vvd;

const T eps = 1e-8, inf = 1./0;
#define MP make_pair
#define ltj(X) if(s == -1 || MP(X[j],N[j]) < MP(X[s],N[s])) s=j

struct LPSolver {
int m, n;
vi N, B;
vvd D;

LPSolver(const vvd& A, const vd& b, const vd& c) :
m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
rep(i,0,m) rep(j,0,n) D[i][j] = A[i][j];
rep(i,0,m) { B[i] = n+i; D[i][n] = -1; D[i][n+1] = b[i]; }
rep(j,0,n) { N[j] = j; D[m][j] = -c[j]; }
N[n] = -1; D[m+1][n] = 1;
}

void pivot(int r, int s) {
T *a = D[r].data(), inv = 1 / a[s];
rep(i,0,m+2) if (i != r && abs(D[i][s]) > eps) {
T *b = D[i].data(), inv2 = b[s] * inv;
rep(j,0,n+2) b[j] -= a[j] * inv2;
b[s] = a[s] * inv2;
}
rep(j,0,n+2) if (j != s) D[r][j] *= inv;
rep(i,0,m+2) if (i != r) D[i][s] *= -inv;
D[r][s] = inv;
swap(B[r], N[s]);
}

bool simplex(int phase) {
int x = m + phase - 1;
for (;;) {
int s = -1;
```

```
rep(j,0,n+1) if (N[j] != -phase) ltj(D[x]);
if (D[x][s] >= -eps) return true;
int r = -1;
rep(i,0,m) {
if (D[i][s] <= eps) continue;
if (r == -1 || MP(D[i][n+1] / D[i][s], B[i])
< MP(D[r][n+1] / D[r][s], B[r])) r = i;
}
if (r == -1) return false;
pivot(r, s);
}
}

T solve(vd &x) {
int r = 0;
rep(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
if (D[r][n+1] < -eps) {
pivot(r, n);
if (!simplex(2) || D[m+1][n+1] < -eps) return -inf;
rep(i,0,m) if (B[i] == -1) {
int s = 0;
rep(j,1,n+1) ltj(D[i]);
pivot(i, s);
}
}
bool ok = simplex(1); x = vd(n);
rep(i,0,m) if (B[i] < n) x[B[i]] = D[i][n+1];
return ok ? D[m][n+1] : inf;
}
};
```

4.3 Matrices

Determinant.h

Description: Calculates determinant of a matrix. Destroys the matrix.

Time: $\mathcal{O}(N^3)$

```
bd5cec, 15 lines
```

```
double det(vector<vector<double>>& a) {
int n = sz(a); double res = 1;
rep(i,0,n) {
int b = i;
rep(j,i+1,n) if (fabs(a[j][i]) > fabs(a[b][i])) b = j;
if (i != b) swap(a[i], a[b]), res *= -1;
res *= a[i][i];
if (res == 0) return 0;
rep(j,i+1,n) {
double v = a[j][i] / a[i][i];
if (v != 0) rep(k,i+1,n) a[j][k] -= v * a[i][k];
}
}
return res;
}
```

IntDeterminant.h

Description: Calculates determinant using modular arithmetics. Modulos can also be removed to get a pure-integer version.

Time: $\mathcal{O}(N^3)$

```
3313dc, 18 lines
```

```
const ll mod = 12345;
ll det(vector<vector<ll>>& a) {
int n = sz(a); ll ans = 1;
rep(i,0,n) {
rep(j,i+1,n) {
while (a[j][i] != 0) { // gcd step
ll t = a[i][i] / a[j][i];
if (t) rep(k,i,n)
a[i][k] = (a[i][k] - a[j][k] * t) % mod;
swap(a[i], a[j]);
ans *= -1;
}
}
ans = ans * a[i][i] % mod;
if (!ans) return 0;
}
return (ans + mod) % mod;
}
```

SolveLinear.h

Description: Solves $A * x = b$. If there are multiple solutions, an arbitrary one is returned. Returns rank, or -1 if no solutions. Data in A and b is lost.

Time: $\mathcal{O}(n^2m)$

```
44c9ab, 38 lines
```

```
typedef vector<double> vd;
const double eps = 1e-12;

int solveLinear(vector<vd>& A, vd& b, vd& x) {
int n = sz(A), m = sz(x), rank = 0, br, bc;
if (n) assert(sz(A[0]) == m);
vi col(m); iota(all(col), 0);

rep(i,0,n) {
double v, bv = 0;
rep(r,i,n) rep(c,i,m)
if ((v = fabs(A[r][c])) > bv)
br = r, bc = c, bv = v;
if (bv <= eps) {
rep(j,i,n) if (fabs(b[j]) > eps) return -1;
break;
}
swap(A[i], A[br]);
swap(b[i], b[br]);
swap(col[i], col[bc]);
rep(j,0,n) swap(A[j][i], A[j][bc]);
bv = 1/A[i][i];
rep(j,i+1,n) {
double fac = A[j][i] * bv;
b[j] -= fac * b[i];
rep(k,i+1,m) A[j][k] -= fac*A[i][k];
}
rank++;
}

x.assign(m, 0);
for (int i = rank; i--;) {
b[i] /= A[i][i];
x[col[i]] = b[i];
rep(j,0,i) b[j] -= A[j][i] * b[i];
}
return rank; // (multiple solutions if rank < m)
}
```

SolveLinear2.h

Description: To get all uniquely determined values of x back from SolveLinear, make the following changes:

"SolveLinear.h"

```
08e495, 7 lines
```

```
rep(j,0,n) if (j != i) // instead of rep(j,i+1,n)
// ... then at the end:
x.assign(m, undefined);
rep(i,0,rank) {
rep(j,rank,m) if (fabs(A[i][j]) > eps) goto fail;
x[col[i]] = b[i] / A[i][i];
fail;; }
}
```

SolveLinearBinary.h

Description: Solves $Ax = b$ over F_2 . If there are multiple solutions, one is returned arbitrarily. Returns rank, or -1 if no solutions. Destroys A and b .

Time: $\mathcal{O}(n^2m)$

```
fa2d7a, 34 lines
```

```
typedef bitset<1000> bs;

int solveLinear(vector<bs>& A, vi& b, bs& x, int m) {
int n = sz(A), rank = 0, br;
assert(m <= sz(x));
vi col(m); iota(all(col), 0);
rep(i,0,n) {
for (br=i; br<n; ++br) if (A[br].any()) break;
if (br == n) {
rep(j,i,n) if(b[j]) return -1;
break;
}
int bc = (int)A[br]._Find_next(i-1);
swap(A[i], A[br]);
swap(b[i], b[br]);
swap(col[i], col[bc]);
rep(j,0,n) if (A[j][i] != A[j][bc]) {
```

```
        A[j].flip(i); A[j].flip(bc);
    }
    rep(j,i+1,n) if (A[j][i]) {
        b[j] ^= b[i];
        A[j] ^= A[i];
    }
    rank++;
}

x = bs();
for (int i = rank; i--;) {
    if (!b[i]) continue;
    x[col[i]] = 1;
    rep(j,0,i) b[j] ^= A[j][i];
}
return rank; // (multiple solutions if rank < m)
}
```

MatrixInverse.h

Description: Invert matrix *A*. Returns rank; result is stored in *A* unless singular (rank < *n*). Can easily be extended to prime moduli; for prime powers, repeatedly set $A^{-1} = A^{-1}(2I - AA^{-1}) \pmod{p^k}$ where A^{-1} starts as the inverse of *A* mod *p*, and *k* is doubled in each step.

Time: $\mathcal{O}(n^3)$

```
ebfff6, 35 lines

int matInv(vector<vector<double>>& A) {
    int n = sz(A); vi col(n);
    vector<vector<double>> tmp(n, vector<double>(n));
    rep(i,0,n) tmp[i][i] = 1, col[i] = i;

    rep(i,0,n) {
        int r = i, c = i;
        rep(j,i,n) rep(k,i,n)
            if (fabs(A[j][k]) > fabs(A[r][c]))
                r = j, c = k;
        if (fabs(A[r][c]) < 1e-12) return i;
        A[i].swap(A[r]); tmp[i].swap(tmp[r]);
        rep(j,0,n)
            swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c]);
        swap(col[i], col[c]);
        double v = A[i][i];
        rep(j,i+1,n) {
            double f = A[j][i] / v;
            A[j][i] = 0;
            rep(k,i+1,n) A[j][k] -= f*A[i][k];
            rep(k,0,n) tmp[j][k] -= f*tmp[i][k];
        }
        rep(j,i+1,n) A[i][j] /= v;
        rep(j,0,n) tmp[i][j] /= v;
        A[i][i] = 1;
    }

    for (int i = n-1; i > 0; --i) rep(j,0,i) {
        double v = A[j][i];
        rep(k,0,n) tmp[j][k] -= v*tmp[i][k];
    }

    rep(i,0,n) rep(j,0,n) A[col[i]][col[j]] = tmp[i][j];
    return n;
}
```

Tridiagonal.h

Description: $x = \text{tridiagonal}(d, p, q, b)$ solves the equation system

$$\begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ \vdots \\ \vdots \\ b_{n-1} \end{pmatrix} = \begin{pmatrix} d_0 & p_0 & 0 & 0 & \cdots & 0 \\ q_0 & d_1 & p_1 & 0 & \cdots & 0 \\ 0 & q_1 & d_2 & p_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & q_{n-3} & d_{n-2} & p_{n-2} \\ 0 & 0 & \cdots & 0 & q_{n-2} & d_{n-1} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ \vdots \\ \vdots \\ x_{n-1} \end{pmatrix}.$$

This is useful for solving problems on the type

$$a_i = b_i a_{i-1} + c_i a_{i+1} + d_i, \quad 1 \leq i \leq n,$$

where a_0, a_{n+1}, b_i, c_i and d_i are known. a can then be obtained from

$$\{a_i\} = \text{tridiagonal}(\{1, -1, -1, \dots, -1, 1\}, \{0, c_1, c_2, \dots, c_n\}, \{b_1, b_2, \dots, b_n, 0\}, \{a_0, d_1, d_2, \dots, d_n, a_{n+1}\}).$$

Fails if the solution is not unique.
If $|d_i| > |p_i| + |q_{i-1}|$ for all i , or $|d_i| > |p_{i-1}| + |q_i|$, or the matrix is positive definite, the algorithm is numerically stable and neither tr nor the check for $\text{diag}[i] == 0$ is needed.
Time: $\mathcal{O}(N)$

```
8f9fa8, 26 lines

typedef double T;
vector<T> tridiagonal(vector<T> diag, const vector<T>& super,
    const vector<T>& sub, vector<T> b) {
    int n = sz(b); vi tr(n);
    rep(i,0,n-1) {
        if (abs(diag[i]) < 1e-9 * abs(super[i])) { // diag[i] == 0
            b[i+1] -= b[i] * diag[i+1] / super[i];
            if (i+2 < n) b[i+2] -= b[i] * sub[i+1] / super[i];
            diag[i+1] = sub[i]; tr[++i] = 1;
        } else {
            diag[i+1] -= super[i]*sub[i]/diag[i];
            b[i+1] -= b[i]*sub[i]/diag[i];
        }
    }
    for (int i = n; i--;) {
        if (tr[i]) {
            swap(b[i], b[i-1]);
            diag[i-1] = diag[i];
            b[i] /= super[i-1];
        } else {
            b[i] /= diag[i];
            if (i) b[i-1] -= b[i]*super[i-1];
        }
    }
    return b;
}
```

4.4 Fourier transforms

FastFourierTransform.h

Description: $\text{fft}(a)$ computes $\hat{f}(k) = \sum_x a[x] \exp(2\pi i \cdot kx/N)$ for all k . *N* must be a power of 2. Useful for convolution: $\text{conv}(a, b) = c$, where $c[x] = \sum a[i]b[x-i]$. For convolution of complex numbers or more than two vectors: FFT, multiply pointwise, divide by *n*, reverse(start+1, end), FFT back. Rounding is safe if $(\sum a_i^2 + \sum b_i^2) \log_2 N < 9 \cdot 10^{14}$ (in practice 10^{16} ; higher for random inputs). Otherwise, use NTT/FFTMod.

Time: $\mathcal{O}(N \log N)$ with $N = |A| + |B|$ ($\sim 1s$ for $N = 2^{22}$)

```
00ced6, 35 lines

typedef complex<double> C;
typedef vector<double> vd;
void fft(vector<C>& a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1);
    static vector<C> rt(2, 1); // (^ 10% faster if double)
    for (static int k = 2; k < n; k *= 2) {
        R.resize(n); rt.resize(n);
        auto x = polar(1.0L, acos(-1.0L) / k);
        rep(i,k,2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
    }
    vi rev(n);
    rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
            C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-rolled)
            a[i + j + k] = a[i + j] - z;
            a[i + j] += z;
        }
    }
    vd conv(const vd& a, const vd& b) {
        if (a.empty() || b.empty()) return {};
        vd res(sz(a) + sz(b) - 1);
        int L = 32 - __builtin_clz(sz(res)), n = 1 << L;
        vector<C> in(n), out(n);
        copy(all(a), begin(in));
        rep(i,0,sz(b)) in[i].imag(b[i]);
        fft(in);
        for (C& x : in) x *= x;
        rep(i,0,n) out[i] = in[-i & (n - 1)] - conj(in[i]);
        fft(out);
        rep(i,0,sz(res)) res[i] = imag(out[i]) / (4 * n);
        return res;
    }
}
```

FastFourierTransformMod.h

Description: Higher precision FFT, can be used for convolutions modulo arbitrary integers as long as $N \log_2 N \cdot \text{mod} < 8.6 \cdot 10^{14}$ (in practice 10^{16} or higher). Inputs must be in $[0, \text{mod})$.

Time: $\mathcal{O}(N \log N)$, where $N = |A| + |B|$ (twice as slow as NTT or FFT)

```
"FastFourierTransform.h"
b82773, 22 lines

typedef vector<ll> vl;
template<int M> vl convMod(const vl &a, const vl &b) {
    if (a.empty() || b.empty()) return {};
    vl res(sz(a) + sz(b) - 1);
    int B=32-__builtin_clz(sz(res)), n=1<<B, cut=int(sqrt(M));
    vector<C> L(n), R(n), outs(n), outl(n);
    rep(i,0,sz(a)) L[i] = C((int)a[i] / cut, (int)a[i] % cut);
    rep(i,0,sz(b)) R[i] = C((int)b[i] / cut, (int)b[i] % cut);
    fft(L), fft(R);
    rep(i,0,n) {
        int j = -i & (n - 1);
        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
    }
    fft(outl), fft(outs);
    rep(i,0,sz(res)) {
        ll av = ll(real(outl[i])+.5), cv = ll(imag(outs[i])+.5);
        ll bv = ll(imag(outl[i])+.5) + ll(real(outs[i])+.5);
        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    }
    return res;
}
```

NumberTheoreticTransform.h

Description: $\text{ntt}(a)$ computes $\hat{f}(k) = \sum_x a[x]g^{xk}$ for all k , where $g = \text{root}^{(\text{mod}-1)/N}$. *N* must be a power of 2. Useful for convolution modulo specific nice primes of the form $2^a b + 1$, where the convolution result has size at most 2^a . For arbitrary modulo, see FFTMod. $\text{conv}(a, b) = c$, where $c[x] = \sum a[i]b[x-i]$. For manual convolution: NTT the inputs, multiply pointwise, divide by *n*, reverse(start+1, end), NTT back. Inputs must be in $[0, \text{mod})$.

Time: $\mathcal{O}(N \log N)$

```
"../number-theory/ModPow.h"
ced03d, 33 lines

const ll mod = (119 << 23) + 1, root = 62; // = 998244353
// For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 << 21
// and 483 << 21 (same root). The last two are > 10^9.
typedef vector<ll> vl;
void ntt(vl &a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vl rt(2, 1);
    for (static int k = 2, s = 2; k < n; k *= 2, s++) {
        rt.resize(n);
        ll z[] = {1, modpow(root, mod >> s)};
        rep(i,k,2*k) rt[i] = rt[i / 2] * z[i & 1] % mod;
    }
    vi rev(n);
    rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
            ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
            a[i + j + k] = ai - z + (z > ai ? mod : 0);
            ai += (ai + z >= mod ? z - mod : z);
        }
    }
    vl conv(const vl &a, const vl &b) {
        if (a.empty() || b.empty()) return {};
        int s = sz(a) + sz(b) - 1, B = 32 - __builtin_clz(s), n = 1 << B;
        int inv = modpow(n, mod - 2);
        vl L(a), R(b), out(n);
        L.resize(n), R.resize(n);
        ntt(L), ntt(R);
        rep(i,0,n) out[-i & (n - 1)] = (ll)L[i] * R[i] % mod * inv % mod;
        ntt(out);
        return {out.begin(), out.begin() + s};
    }
}
```

FastSubsetTransform.h

Description: Transform to a basis with fast convolutions of the form $c[z] = \sum_{z=x \oplus y} a[x] \cdot b[y]$, where $\oplus$ is one of AND, OR, XOR. The size of a must be a power of two.

Time: $\mathcal{O}(N \log N)$	464cf3, 16 lines
<pre>void FST(vi& a, bool inv) { for (int n = sz(a), step = 1; step < n; step *= 2) { for (int i = 0; i < n; i += 2 * step) rep(j,i,step) { int &u = a[j], &v = a[j + step]; tie(u, v) = inv ? pii(v - u, u) : pii(v, u + v); <i>// AND</i> inv ? pii(v, u - v) : pii(u + v, u); <i>// OR</i> pii(u + v, u - v); <i>// XOR</i> } } if (inv) for (int& x : a) x /= sz(a); <i>// XOR only</i> } vi conv(vi a, vi b) { FST(a, 0); FST(b, 0); rep(i,0,sz(a)) a[i] *= b[i]; FST(a, 1); return a; }</pre>	

Number theory (5)

5.1 Modular arithmetic

ModularArithmetic.h

Description: Operators for modular arithmetic. You need to set mod to some number first and then you can use the structure.

"euclid.h"	35bfea, 18 lines
<pre>const ll mod = 17; <i>// change to something else</i> struct Mod { ll x; Mod(ll xx) : x(xx) {} Mod operator+(Mod b) { return Mod((x + b.x) % mod); } Mod operator-(Mod b) { return Mod((x - b.x + mod) % mod); } Mod operator*(Mod b) { return Mod((x * b.x) % mod); } Mod operator/(Mod b) { return *this * invert(b); } Mod invert(Mod a) { ll x, y, g = euclid(a.x, mod, x, y); assert(g == 1); return Mod((x + mod) % mod); } Mod operator^(ll e) { if (!e) return Mod(1); Mod r = *this ^ (e / 2); r = r * r; return e&1 ? *this * r : r; } };</pre>	

ModInverse.h

Description: Pre-computation of modular inverses. Assumes $\text{LIM} \leq \text{mod}$ and that mod is a prime.

6f684f, 3 lines
<pre>const ll mod = 1000000007, LIM = 200000; ll* inv = new ll[LIM] - 1; inv[1] = 1; rep(i,2,LIM) inv[i] = mod - (mod / i) * inv[mod % i] % mod;</pre>

ModPow.h

b83e45, 8 lines
<pre>const ll mod = 1000000007; <i>// faster if const</i></pre>
<pre>ll modpow(ll b, ll e) { ll ans = 1; for (; e; b = b * b % mod, e /= 2) if (e & 1) ans = ans * b % mod; return ans; }</pre>

ModLog.h

Description: Returns the smallest $x > 0$ s.t. $a^x = b \pmod m$, or -1 if no such x exists. modLog(a,1,m) can be used to calculate the order of a .

$\mathcal{O}(\sqrt{m})$	c040b8, 11 lines
<pre>ll modLog(ll a, ll b, ll m) { ll n = (ll) sqrt(m) + 1, e = 1, f = 1, j = 1; unordered_map<ll, ll> A; while (j <= n && (e = f = e * a % m) != b % m) A[e * b % m] = j++; if (e == b % m) return j; if (__gcd(m, e) == __gcd(m, b))</pre>	

<pre> rep(i,2,n+2) if (A.count(e = e * f % m)) return n * i - A[e]; return -1; }</pre>

ModSum.h

Description: Sums of mod'ed arithmetic progressions.

modsum(to, c, k, m) = $\sum_{i=0}^{to-1} (ki + c) \% m$. divsum is similar but for floored division.

5c5bc5, 16 lines
Time: log(m), with a large constant.
<pre>typedef unsigned long long ull; ull sumsq(ull to) { return to / 2 * ((to-1) 1); }</pre>
<pre>ull divsum(ull to, ull c, ull k, ull m) { ull res = k / m * sumsq(to) + c / m * to; k %= m; c %= m; if (!k) return res; ull to2 = (to * k + c) / m; return res + (to - 1) * to2 - divsum(to2, m-1 - c, m, k); }</pre>
<pre>ll modsum(ull to, ll c, ll k, ll m) { c = ((c % m) + m) % m; k = ((k % m) + m) % m; return to * c + k * sumsq(to) - m * divsum(to, c, k, m); }</pre>

ModMulLL.h

Description: Calculate $a \cdot b \bmod c$ (or $a^b \bmod c$) for $0 \leq a, b \leq c \leq 7.2 \cdot 10^{18}$.

bbbd8f, 11 lines
Time: $\mathcal{O}(1)$ for modmul, $\mathcal{O}(\log b)$ for modpow
<pre>typedef unsigned long long ull; ull modmul(ull a, ull b, ull M) { ll ret = a * b - M * ull(1.L / M * a * b); return ret + M * (ret < 0) - M * (ret >= (ll)M); } ull modpow(ull b, ull e, ull mod) { ull ans = 1; for (; e; b = modmul(b, b, mod), e /= 2) if (e & 1) ans = modmul(ans, b, mod); return ans; }</pre>

ModSqrt.h

Description: Tonelli-Shanks algorithm for modular square roots. Finds x s.t. $x^2 = a \pmod p$ ($-x$ gives the other solution).

19a793, 24 lines
Time: $\mathcal{O}(\log^2 p)$ worst case, $\mathcal{O}(\log p)$ for most p
<pre>"ModPow.h" ll sqrt(ll a, ll p) { a %= p; if (a < 0) a += p; if (a == 0) return 0; assert(modpow(a, (p-1)/2, p) == 1); <i>// else no solution</i> if (p % 4 == 3) return modpow(a, (p+1)/4, p); <i>// a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5</i> ll s = p - 1, n = 2; int r = 0, m; while (s % 2 == 0) ++r, s /= 2; while (modpow(n, (p - 1) / 2, p) != p - 1) ++n; ll x = modpow(a, (s + 1) / 2, p); ll b = modpow(a, s, p), g = modpow(n, s, p); for (; r = m) { ll t = b; for (m = 0; m < r && t != 1; ++m) t = t * t % p; if (m == 0) return x; ll gs = modpow(g, 1LL << (r - m - 1), p); g = gs * gs % p; x = x * gs % p; b = b * g % p; } }</pre>

5.2 Primality

FastEratosthenes.h

Description: Prime sieve for generating all primes smaller than LIM.

Time: LIM=1e9 $\approx$ 1.5s	6b2912, 20 lines
<pre>const int LIM = 1e6; bitset<LIM> isPrime; vi eratosthenes() { const int S = (int)round(sqrt(LIM)), R = LIM / 2; vi pr = {2}, sieve(S+1); pr.reserve(int(LIM/log(LIM)*1.1)); vector<pii> cp; for (int i = 3; i <= S; i += 2) if (!sieve[i]) { cp.push_back({i, i * i / 2}); for (int j = i * i; j <= S; j += 2 * i) sieve[j] = 1; } for (int L = 1; L <= R; L += S) { array<bool, S> block{}; for (auto &[p, idx] : cp) for (int i=idx; i < S+L; idx = (i+=p)) block[i-L] = 1; rep(i,0,min(S, R - L)) if (!block[i]) pr.push_back((L + i) * 2 + 1); } for (int i : pr) isPrime[i] = 1; return pr; }</pre>	

MillerRabin.h

Description: Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to $7 \cdot 10^{18}$; for larger numbers, use Python and extend A randomly.

Time: 7 times the complexity of $a^b \bmod c$.

"ModMulLL.h"	60dcd1, 12 lines
<pre>bool isPrime(ull n) { if (n < 2 n % 6 % 4 != 1) return (n 1) == 3; ull A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022}, s = __builtin_ctzll(n-1), d = n >> s; for (ull a : A) { <i>// ^ count trailing zeroes</i> ull p = modpow(a%n, d, n), i = s; while (p != 1 && p != n - 1 && a % n && i--) p = modmul(p, p, n); if (p != n-1 && i != s) return 0; } return 1; }</pre>	

Factor.h

Description: Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).

Time: $\mathcal{O}\left(n^{1/4}\right)$, less for numbers with small factors.

"ModMulLL.h", "MillerRabin.h"	a33cf6, 18 lines
<pre>ull pollard(ull n) { auto f = [n](ull x) { return modmul(x, x, n) + 1; }; ull x = 0, y = 0, t = 30, prd = 2, i = 1, q; while (t++ % 40 __gcd(prd, n) == 1) { if (x == y) x = ++i, y = f(x); if ((q = modmul(prd, max(x,y) - min(x,y), n))) prd = q; x = f(x), y = f(f(y)); } return __gcd(prd, n); } vector<ull> factor(ull n) { if (n == 1) return {}; if (isPrime(n)) return {n}; ull x = pollard(n); auto l = factor(x), r = factor(n / x); l.insert(l.end(), all(r)); return l; }</pre>	

CountPrimes.h

Description: Count number of primes up to n.

Time: $\mathcal{O}(\sqrt{n})$	652db8, 43 lines
<pre>ll piSieve(const ll n) { if (n <= 1) return 0LL; if (n == 2) return 1LL; const int lim = int(sqrt(n)); int vsz = (lim + 1) >> 1; vector<int> smalls(vsz), roughs(vsz), larges(vsz);</pre>	

```
rep(i,0,vsz) smalls[i]=i, roughs[i]=(i<<1|1), larges[i]=(n/(i<<1|1)-1)>>1;
vector<bool> skips(lim+1,false);
int pCnt=0;
for(int p=3;p<=lim;p+=2){
    if (skips[p]) continue;
    int p2 = p * p;
    if (1LL * p2 * p2 > n) break;
    skips[p] = true;
    for(int cx = p2; cx <= lim; cx += (p << 1)) skips[cx] = true;
    int ns = 0;
    rep(cz, 0, vsz) {
        int cur = roughs[cx];
        if(skills[cur]) continue;
        ll d = 1LL * cur * p;
        larges[ns]=larges[cx]-(d<=lim?larges[smalls[d]>>1]-pCnt:smalls[(ll)((double(n)/d)-1))>>1]+pCnt;
        roughs[ns++]<=cur;
    }
    vsz=ns;
    for(int cx=(lim-1)>>1,cy=((lim/p)-1)|1;cy>=p;cy-=2){
        int cur=smalls[cy>>1]-pCnt;
        for(int cz=(cy*p)>>1;cz<=cx;--cx) smalls[cx]-=cur;
    }
    ++pCnt;
}
larges[0]+=1LL*(vsz+((pCnt-1)<<1))*(vsz-1)>>1;
for(int cx=1;cx<vsz;++cx) larges[0]-=larges[cx];
for(int cx=1;cx<vsz;++cx){
    int q=roughs[cx];
    ll m=n/q;
    int e=smalls[((m/q)-1)>>1]-pCnt;
    if (e<cx+1) break;
    ll t=0;
    for(int cy=cx+1;cy<=e;++cy) t+=smalls[ll((double(m)/roughs[cy])-1)>>1];
    larges[0]+=t-1LL*(e-cx)*(pCnt+cx-1);
}
return larges[0]+1;
}
```

5.3 Divisibility

euclid.h

Description: Finds two integers x and y , such that $ax + by = \gcd(a, b)$. If you just need gcd, use the built in `_gcd` instead. If a and b are coprime, then x is the inverse of $a \pmod b$.

```
ll euclid(ll a, ll b, ll &x, ll &y) {
    if (!b) return x = 1, y = 0, a;
    ll d = euclid(b, a % b, y, x);
    return y -= a/b * x, d;
}
```

CRT.h

Description: Chinese Remainder Theorem.

Usage: Iterate through all factors of M , call `add(x)` to multiply by x and `del(x)` to divide by x .

Time: $\log(n)$

```
ll px, py, pd;
void ExEuclid(ll a, ll b) {
    if (b == 0) {
        px = 1; py = 0; pd = a;
        return;
    }
    ExEuclid(b, a % b);
    ll x1 = px, y1 = py;
    px = y1, py = x1 - a / b * y1;
}
```

```
struct ChineseRemainderTheorem {
    ll ans, lcm;

    ll x, y, d;
    void ExtendedEuclidean(ll a, ll b) {
        if (b == 0) {
            x = 1; y = 0; d = a;
            return;
        }
    }
}
```

```
    }
    ExtendedEuclidean(b, a % b);
    ll x1 = x, y1 = y;
    x = y1, y = x1 - a / b * y1;
}

void add(ll a, ll n) {
    ExtendedEuclidean(lcm, n);
    ans = ans + x * (a - ans) / d % (n / d) * lcm; lcm = lcm * n / d;
    normalize(ans, lcm);
}
};
```

```
struct FactorData {
    int cnt_base = 0, non_zero = 1, mod, base, lim;

    int pw(const int &x, int y) {
        if (!y) return 1;
        int mid = pw(x, y / 2);
        if (y & 1) return 1LL * mid * mid % mod * x % mod;
        return 1LL * mid * mid % mod;
    }

    void add(int x) {
        while (x % base == 0) {
            ++cnt_base;
            x /= base;
        }
        non_zero = 1LL * non_zero * x % mod;
    }

    void del(int x) {
        while (x % base == 0) {
            --cnt_base;
            x /= base;
        }
        if (x == 1) return;
        ExEuclid(x, mod);
        non_zero = 1LL * non_zero * px % mod;
        if (non_zero < 0) non_zero += mod;
    }

    int get() {
        if (cnt_base >= lim) return 0;
        return 1LL * non_zero * pw(base, cnt_base) % mod;
    }
} factors[10];
```

```
void process(int M) {
    int x = M;
    for (int i = 2; i * i <= x && x != 1; ++i) {
        int cur = 1, cnt = 0;
        while (x % i == 0) {
            cur *= i;
            x /= i;
            ++cnt;
        }
        if (cnt) {
            factors[++it].mod = cur;
            factors[it].base = i;
            factors[it].lim = cnt;
        }
    }
    if (x != 1) {
        factors[++it].mod = x;
        factors[it].base = x;
        factors[it].lim = 1;
    }
}
```

```
void process(int M) {
    int x = M;
    for (int i = 2; i * i <= x && x != 1; ++i) {
        int cur = 1, cnt = 0;
        while (x % i == 0) {
            cur *= i;
            x /= i;
            ++cnt;
        }
        if (cnt) {
            factors[++it].mod = cur;
            factors[it].base = i;
            factors[it].lim = cnt;
        }
    }
    if (x != 1) {
        factors[++it].mod = x;
        factors[it].base = x;
        factors[it].lim = 1;
    }
}
```

phiFunction.h

Description: Euler's ϕ function is defined as $\phi(n) := \#$ of positive integers $\leq n$ that are coprime with n . $\phi(1) = 1$, p prime $\Rightarrow \phi(p^k) = (p - 1)p^{k-1}$, m, n coprime $\Rightarrow \phi(mn) = \phi(m)\phi(n)$. If $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$ then $\phi(n) = (p_1 - 1)p_1^{k_1-1} \dots (p_r - 1)p_r^{k_r-1}$. $\phi(n) = n \cdot \prod_{p|n} (1 - 1/p)$. $\sum_{d|n} \phi(d) = n$, $\sum_{1 \leq k \leq n, \gcd(k, n) = 1} k = n\phi(n)/2$, $n > 1$

Euler's thm: a, n coprime $\Rightarrow a^{\phi(n)} \equiv 1 \pmod n$.

Fermat's little thm: p prime $\Rightarrow a^{p-1} \equiv 1 \pmod p \ \forall a$. cf7d6d, 8 lines

```
const int LIM = 5000000;
int phi[LIM];
```

```
void calculatePhi() {
    rep(i,0,LIM) phi[i] = i&1 ? i : i/2;
    for (int i = 3; i < LIM; i += 2) if(phi[i] == i)
        for (int j = i; j < LIM; j += i) phi[j] -= phi[j] / i;
}
```

5.3.1 Bézout's identity

For $a \neq 0$, $b \neq 0$, then $d = \gcd(a, b)$ is the smallest positive integer for which there are integer solutions to

$$ax + by = d$$

If (x, y) is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a, b)}, y - \frac{ka}{\gcd(a, b)}\right), \quad k \in \mathbb{Z}$$

5.4 Fractions

ContinuedFractions.h

Description: Given N and a real number $x \geq 0$, finds the closest rational approximation p/q with $p, q \leq N$. It will obey $|p/q - x| \leq 1/qN$.

For consecutive convergents, $p_{k+1}q_k - q_{k+1}p_k = (-1)^k$. (p_k/q_k alternates between $> x$ and $< x$.) If x is rational, y eventually becomes ∞ ; if x is the root of a degree 2 polynomial the a 's eventually become cyclic.

Time: $\mathcal{O}(\log N)$

```
typedef double d; // for N ~ 1e7; long double for N ~ 1e9
pair<ll, ll> approximate(d x, ll N) {
    ll LP = 0, LQ = 1, P = 1, Q = 0, inf = LLONG_MAX; d y = x;
    for (;;) {
        ll lim = min(P ? (N-LP) / P : inf, Q ? (N-LQ) / Q : inf),
            a = (ll)floor(y), b = min(a, lim),
            NP = b*P + LP, NQ = b*Q + LQ;
        if (a > b) {
            // If b > a/2, we have a semi-convergent that gives us a
            // better approximation; if b = a/2, we *may* have one.
            // Return (P, Q) here for a more canonical approximation.
            return (abs(x - (d)NP / (d)NQ) < abs(x - (d)P / (d)Q)) ?
                make_pair(NP, NQ) : make_pair(P, Q);
        }
        if (abs(y = 1/(y - (d)a)) > 3*N) {
            return {NP, NQ};
        }
        LP = P; P = NP;
        LQ = Q; Q = NQ;
    }
}
```

FracBinarySearch.h

Description: Given f and N , finds the smallest fraction $p/q \in [0, 1]$ such that $f(p/q)$ is true, and $p, q \leq N$. You may want to throw an exception from f if it finds an exact solution, in which case N can be removed.

Usage: `fracBS([])(Frac f) { return f.p>=3*f.q; }, 10);` // {1,3}

Time: $\mathcal{O}(\log(N))$

```
struct Frac { ll p, q; };

template<class F>
Frac fracBS(F f, ll N) {
    bool dir = 1, A = 1, B = 1;
    Frac lo{0, 1}, hi{1, 1}; // Set hi to 1/0 to search (0, N]
    if (f(lo)) return lo;
    assert(f(hi));
    while (A || B) {
        ll adv = 0, step = 1; // move hi if dir, else lo
        for (int si = 0; step; (step *= 2) >= si) {
            adv += step;
            Frac mid{lo.p * adv + hi.p, lo.q * adv + hi.q};
            if (abs(mid.p) > N || mid.q > N || dir == !f(mid)) {
                adv -= step; si = 2;
            }
        }
    }
}
```

```
    }
}
hi.p += lo.p * adv;
hi.q += lo.q * adv;
dir = !dir;
swap(lo, hi);
A = B; B = !adv;
}
return dir ? hi : lo;
}
```

5.5 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$$

with $m > n > 0$, $k > 0$, $m \perp n$, and either m or n even.

5.6 Primes

$p = 962592769$ is such that $2^{21} \mid p - 1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a , except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

5.7 Estimates

$$\sum_{d \mid n} d = O(n \log \log n).$$

The number of divisors of n is at most around 100 for $n < 5e4$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

5.8 Mobius Function

$$\mu(n) = \begin{cases} 0 & n \text{ is not square free} \\ 1 & n \text{ has even number of prime factors} \\ -1 & n \text{ has odd number of prime factors} \end{cases}$$

Mobius Inversion:

$$g(n) = \sum_{d \mid n} f(d) \Leftrightarrow f(n) = \sum_{d \mid n} \mu(d) g(n/d)$$

Other useful formulas/forms:

$$\sum_{d \mid n} \mu(d) = [n = 1] \text{ (very useful)}$$

$$g(n) = \sum_{n \mid d} f(d) \Leftrightarrow f(n) = \sum_{n \mid d} \mu(d/n) g(d)$$

$$g(n) = \sum_{1 \leq m \leq n} f(\lfloor \frac{n}{m} \rfloor) \Leftrightarrow f(n) = \sum_{1 \leq m \leq n} \mu(m) g(\lfloor \frac{n}{m} \rfloor)$$

FlooredDivisors.h

```
Description: Returns a vector of maximum intervals [L, R] such that  $\lfloor \frac{n}{L} \rfloor = \lfloor \frac{n}{R} \rfloor$ .
Time:  $\mathcal{O}(2\sqrt{n})$ 
54ad0f, 8 lines

vector<pii> floors(int x) {
    vector<pii> ret;
    for (int L = 1, R; L <= x; L = R + 1) {
        R = x / (x / L); // every i in [L, R] has the same x / i
        ret.emplace_back(L, R);
    }
    return ret;
}
```

Combinatorial (6)

6.1 Permutations

6.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800
n	11	12	13	14	15	16	17			
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14			
n	20	25	30	40	50	100	150	171		
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

IntPerm.h

```
Description: Permutation -> integer conversion. (Not order preserving.) Integer
-> permutation can use a lookup table.
Time:  $\mathcal{O}(n)$ 
044568, 6 lines
```

```
int permToInt(vi& v) {
    int use = 0, i = 0, r = 0;
    for(int x:v) r = r * ++i + __builtin_popcount(use & ~(1<<x)),
        use |= 1 << x; // (note: minus, not ~!)
    return r;
}
```

6.1.2 Cycles

Let $g_S(n)$ be the number of n -permutations whose cycle lengths all belong to the set S . Then

$$\sum_{n=0}^\infty g_S(n) \frac{x^n}{n!} = \exp \left(\sum_{n \in S} \frac{x^n}{n} \right)$$

6.1.3 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n - 1)(D(n - 1) + D(n - 2)) = nD(n - 1) + (-1)^n = \left\lfloor \frac{n!}{e} \right\rfloor$$

6.1.4 Burnside’s lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k \mid n} f(k) \phi(n/k).$$

6.2 Partitions and subsets

6.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	$\sim 2e5$	$\sim 2e8$

6.2.2 Lucas’ Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + \dots + n_1 p + n_0$ and $m = m_k p^k + \dots + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod p$.

6.2.3 Stars and bars

Count nonnegative integer solutions to $x_1 + \dots + x_k = n$.

$$\#\{x_i \geq 0, \sum x_i = n\} = \binom{n + k - 1}{k - 1}, \quad \#\{x_i \geq 1, \sum x_i = n\} = \binom{n - 1}{k - 1}.$$

Bounded version via inclusion–exclusion: if $0 \leq x_i \leq U_i$ then

$$\sum_{J \subseteq [k]} (-1)^{|J|} \binom{n - \sum_{j \in J} (U_j + 1) + k - 1}{k - 1},$$

where terms with a negative top are taken as 0.

6.2.4 Inclusion–exclusion

For finite sets $A_1, \dots, A_m$,

$$\left| \bigcup_{i=1}^m A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots.$$

Equivalently, the number of elements with **none** of the properties A_i is

$$N - \sum_i |A_i| + \sum_{i < j} |A_i \cap A_j| - \dots.$$

Simple useful corollary: number of derangements

$$D(n) = n! \sum_{i=0}^n \frac{(-1)^i}{i!}.$$

6.2.5 Binomials

multinomial.h

```
Description: Computes  $\binom{k_1 + \dots + k_n}{k_1, k_2, \dots, k_n} = \frac{(\sum k_i)!}{k_1! k_2! \dots k_n!}$ .
a0a312, 6 lines

ll multinomial(vi& v) {
    ll c = 1, m = v.empty() ? 1 : v[0];
    rep(i, 1, sz(v)) rep(j, 0, v[i])
        c = c * ++m / (j+1);
    return c;
}
```

6.3 General purpose numbers

6.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able).

$$B[0, \dots] = [1, -\tfrac{1}{2}, \tfrac{1}{6}, 0, -\tfrac{1}{30}, 0, \tfrac{1}{42}, \dots]$$

Sums of powers:

$$\sum_{i=1}^n n^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\begin{aligned} \sum_{i=m}^\infty f(i) &= \int_m^\infty f(x) dx - \sum_{k=1}^\infty \frac{B_k}{k!} f^{(k-1)}(m) \\ &\approx \int_m^\infty f(x) dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m)) \end{aligned}$$

6.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n, k) = c(n - 1, k - 1) + (n - 1)c(n - 1, k), \quad c(0, 0) = 1$$
$$\sum_{k=0}^n c(n, k)x^k = x(x + 1) \dots (x + n - 1)$$

$$c(8, k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$$
$$c(n, 2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots$$

6.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. $k \ j$:s s.t. $\pi(j) > \pi(j + 1)$, $k + 1 \ j$:s s.t. $\pi(j) \geq j$, $k \ j$:s s.t. $\pi(j) > j$.

$$E(n, k) = (n - k)E(n - 1, k - 1) + (k + 1)E(n - 1, k)$$

$$E(n, 0) = E(n, n - 1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

6.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n, k) = S(n - 1, k - 1) + kS(n - 1, k)$$

$$S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

6.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$ For p prime,

$$B(p^m + n) \equiv mB(n) + B(n + 1) \pmod{p}$$

6.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i : $n_1 n_2 \dots n_k n^{k-2}$
with degrees d_i : $(n - 2)! / ((d_1 - 1)! \dots (d_n - 1)!)$

6.3.7 Catalan numbers

$$C_n = \frac{1}{n + 1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n + 1} = \frac{(2n)!}{(n + 1)!n!}$$

$$C_0 = 1, \quad C_{n+1} = \frac{2(2n + 1)}{n + 2} C_n, \quad C_{n+1} = \sum C_i C_{n-i}$$

$$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \dots$$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n + 1$ leaves (0 or 2 children).
- ordered trees with $n + 1$ vertices.
- ways a convex polygon with $n + 2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Graph (7)

7.1 Fundamentals

BellmanFord.h

Description: Calculates shortest paths from s in a graph that might have negative edge weights. Unreachable nodes get $\text{dist} = \text{inf}$; nodes reachable through negative-weight cycles get $\text{dist} = -\text{inf}$. Assumes $V^2 \max |w_i| < \sim 2^{63}$.

Time: $\mathcal{O}(VE)$ 830a8f, 23 lines

```
const ll inf = LLONG_MAX;
struct Ed { int a, b, w, s() { return a < b ? a : -a; }};
struct Node { ll dist = inf; int prev = -1; };

void bellmanFord(vector<Node>& nodes, vector<Ed>& eds, int s) {
    nodes[s].dist = 0;
    sort(all(eds), [](Ed a, Ed b) { return a.s() < b.s(); });

    int lim = sz(nodes) / 2 + 2; // /3+100 with shuffled vertices
    rep(i, 0, lim) for (Ed ed : eds) {
        Node cur = nodes[ed.a], &dest = nodes[ed.b];
        if (abs(cur.dist) == inf) continue;
        ll d = cur.dist + ed.w;
        if (d < dest.dist) {
            dest.prev = ed.a;
            dest.dist = (i < lim-1 ? d : -inf);
        }
    }
    rep(i, 0, lim) for (Ed e : eds) {
        if (nodes[e.a].dist == -inf)
            nodes[e.b].dist = -inf;
    }
}
```

TopoSort.h

Description: Topological sorting. Given is an oriented graph. Output is an ordering of vertices, such that there are edges only from left to right. If there are cycles, the returned list will have size smaller than n – nodes reachable from cycles will not be returned.

Time: $\mathcal{O}(|V| + |E|)$ 66a137, 14 lines

```
vi topoSort(const vector<vi>& gr) {
    vi indeg(sz(gr)), ret;
    for (auto& li : gr) for (int x : li) indeg[x]++;
    queue<int> q; // use priority-queue for lexic. largest ans.
    rep(i, 0, sz(gr)) if (indeg[i] == 0) q.push(i);
    while (!q.empty()) {
        int i = q.front(); // top() for priority queue
        ret.push_back(i);
        q.pop();
        for (int x : gr[i])
            if (--indeg[x] == 0) q.push(x);
    }
    return ret;
}
```

7.2 Network flow

PushRelabel.h

Description: Push-relabel using the highest label selection rule and the gap heuristic. Quite fast in practice. To obtain the actual flow, look at positive values only.

Time: $\mathcal{O}(V^2\sqrt{E})$ 0ae1d4, 48 lines

```
struct PushRelabel {
    struct Edge {
        int dest, back;
        ll f, c;
    };
    vector<vector<Edge>> g;
    vector<ll> ec;
    vector<Edge> cur;
    vector<vi> hs; vi H;
    PushRelabel(int n) : g(n), ec(n), cur(n), hs(2*n), H(n) {}

    void addEdge(int s, int t, ll cap, ll rcap=0) {
        if (s == t) return;
        g[s].push_back({t, sz(g[t]), 0, cap});
        g[t].push_back({s, sz(g[s])-1, 0, rcap});
    }
}
```

```
void addFlow(Edge& e, ll f) {
    Edge &back = g[e.dest][e.back];
    if (!ec[e.dest] && f) hs[H[e.dest]].push_back(e.dest);
    e.f += f; e.c -= f; ec[e.dest] += f;
    back.f -= f; back.c += f; ec[back.dest] -= f;
}

ll calc(int s, int t) {
    int v = sz(g); H[s] = v; ec[t] = 1;
    vi co(2*v); co[0] = v-1;
    rep(i, 0, v) cur[i] = g[i].data();
    for (Edge& e : g[s]) addFlow(e, e.c);

    for (int hi = 0;;) {
        while (hs[hi].empty()) if (!hi--) return -ec[s];
        int u = hs[hi].back(); hs[hi].pop_back();
        while (ec[u] > 0) // discharge u
            if (cur[u] == g[u].data() + sz(g[u])) {
                H[u] = le9;
                for (Edge& e : g[u]) if (e.c && H[u] > H[e.dest]+1)
                    H[u] = H[e.dest]+1, cur[u] = &e;
                if (++co[H[u]],!--co[hi] && hi < v)
                    rep(i, 0, v) if (hi < H[i] && H[i] < v)
                        --co[H[i]], H[i] = v + 1;
                hi = H[u];
            } else if (cur[u]->c && H[u] == H[cur[u]->dest]+1)
                addFlow(*cur[u], min(ec[u], cur[u]->c));
            else ++cur[u];
    }
}

bool leftOfMinCut(int a) { return H[a] >= sz(g); }
```

EdmondsKarp.h

Description: Flow algorithm with guaranteed complexity $\mathcal{O}(VE^2)$. To get edge flow values, compare capacities before and after, and take the positive values only.

Time: $\mathcal{O}(V^2E)$ 482f60, 35 lines

```
template<class T> T edmondsKarp(vector<unordered_map<int, T>& graph, int
    source, int sink) {
    assert(source != sink);
    T flow = 0;
    vi par(sz(graph)), q = par;

    for (;;) {
        fill(all(par), -1);
        par[source] = 0;
        int ptr = 1;
        q[0] = source;

        rep(i, 0, ptr) {
            int x = q[i];
            for (auto e : graph[x]) {
                if (par[e.first] == -1 && e.second > 0) {
                    par[e.first] = x;
                    q[ptr++] = e.first;
                    if (e.first == sink) goto out;
                }
            }
        }
        return flow;
    }
}

out:
T inc = numeric_limits<T>::max();
for (int y = sink; y != source; y = par[y])
    inc = min(inc, graph[par[y]][y]);

flow += inc;
for (int y = sink; y != source; y = par[y]) {
    int p = par[y];
    if ((graph[p][y] -= inc) <= 0) graph[p].erase(y);
    graph[y][p] += inc;
}
}
```

FlowWithDemands.h

Description: To solve flow with demands $d(e) \leq f(e) \leq c(e)$ for all edges e , make the following changes in the network: add a new source s' and a new sink t' , a new edge from the source s' to every other vertex, a new edge for every vertex to the sink t' , and one edge from t to s . Additionally we define the new capacity function c' as: $c'((s', v)) = \sum_{w \in V} d((u, v))$ for each edge (s', v) , $c'((v, t')) = \sum_{w \in V} d((v, w))$ for each edge (v, t') , $c'((u, v)) = c((u, v)) - d((u, v))$ for each edge (u, v) in the old network, $c'((t, s)) = \infty$. Solution exists if there's a saturating flow (every edge from s' or to t' is completely filled). To trace, for each edge e in the old network, its flow is the flow of the corresponding edge e' plus $d(e)$. To find minimum flow, binary search the minimum capacity of the edge (s, t) for saturating flow. For node demands, connect a new source to all supply nodes with capacity supply $s(u)$, and a new sink to all demand nodes with capacity demand $d(v)$. If there's both node demands and edge demands, set each edge to lower bound. For each node u , define $L(u) = in(u) - out(u)$. Subtract $L(u)$ from supply/demand of u , problem transforms to node demands.

Time: dependent on flow algorithm

MinCostMaxFlow.h

Description: Min-cost max-flow. $cap[i][j] \neq cap[j][i]$ is allowed; double edges are not. If costs can be negative, call `setpi` before `maxflow`, but note that negative cost cycles are not supported. To obtain the actual flow, look at positive values only.

Time: Approximately $\mathcal{O}(E^2)$

fe85cc, 81 lines

```
#include <bits/extc++.h>
```

```
const ll INF = numeric_limits<ll>::max() / 4;
typedef vector<ll> VL;
```

```
struct MCMF {
    int N;
    vector<vi> ed, red;
    vector<VL> cap, flow, cost;
    vi seen;
    VL dist, pi;
    vector<pi> par;
```

```
MCMF(int N) :
    N(N), ed(N), red(N), cap(N, VL(N)), flow(cap), cost(cap),
    seen(N), dist(N), pi(N), par(N) {}
```

```
void addEdge(int from, int to, ll cap, ll cost) {
    this->cap[from][to] = cap;
    this->cost[from][to] = cost;
    ed[from].push_back(to);
    red[to].push_back(from);
}
```

```
void path(int s) {
    fill(all(seen), 0);
    fill(all(dist), INF);
    dist[s] = 0; ll di;
```

```
    __gnu_pbds::priority_queue<pair<ll, int>> q;
    vector<decltype(q)::point_iterator> its(N);
    q.push({0, s});
```

```
    auto relax = [&](int i, ll cap, ll cost, int dir) {
        ll val = di - pi[i] + cost;
        if (cap && val < dist[i]) {
            dist[i] = val;
            par[i] = {s, dir};
            if (its[i] == q.end()) its[i] = q.push({-dist[i], i});
            else q.modify(its[i], {-dist[i], i});
        }
    };
```

```
while (!q.empty()) {
    s = q.top().second; q.pop();
    seen[s] = 1; di = dist[s] + pi[s];
    for (int i : ed[s]) if (!seen[i])
        relax(i, cap[s][i] - flow[s][i], cost[s][i], 1);
    for (int i : red[s]) if (!seen[i])
        relax(i, flow[i][s], -cost[i][s], 0);
}
rep(i,0,N) pi[i] = min(pi[i] + dist[i], INF);
}
```

```
pair<ll, ll> maxflow(int s, int t) {
    ll totflow = 0, totcost = 0;
    while (path(s), seen[t]) {
        ll fl = INF;
        for (int p,r,x = t; tie(p,r) = par[x], x != s; x = p)
            fl = min(fl, r ? cap[p][x] - flow[p][x] : flow[x][p]);
        totflow += fl;
        for (int p,r,x = t; tie(p,r) = par[x], x != s; x = p)
            if (r) flow[p][x] += fl;
            else flow[x][p] -= fl;
    }
    rep(i,0,N) rep(j,0,N) totcost += cost[i][j] * flow[i][j];
    return {totflow, totcost};
}
```

// If some costs can be negative, call this before maxflow:

```
void setpi(int s) { // (otherwise, leave this out)
    fill(all(pi), INF); pi[s] = 0;
    int it = N, ch = 1; ll v;
    while (ch-- && it--)
        rep(i,0,N) if (pi[i] != INF)
            for (int to : ed[i]) if (cap[i][to])
                if ((v = pi[i] + cost[i][to]) < pi[to])
                    pi[to] = v, ch = 1;
    assert(it >= 0); // negative cost cycle
}
```

};

MinCut.h

Description: After running max-flow, the left side of a min-cut from s to t is given by all vertices reachable from s , only traversing edges with positive residual capacity.

GlobalMinCut.h

Description: Find a global minimum cut in an undirected graph, as represented by an adjacency matrix.

Time: $\mathcal{O}(V^3)$

8b0e19, 21 lines

```
pair<int, vi> globalMinCut(vector<vi> mat) {
    pair<int, vi> best = {INT_MAX, {}};
    int n = sz(mat);
    vector<vi> co(n);
    rep(i,0,n) co[i] = {i};
    rep(ph,1,n) {
        vi w = mat[0];
        size_t s = 0, t = 0;
        rep(it,0,n-ph) { // O(V^2) -> O(E log V) with prio. queue
            w[t] = INT_MIN;
            s = t, t = max_element(all(w)) - w.begin();
            rep(i,0,n) w[i] += mat[t][i];
        }
        best = min(best, {w[t] - mat[t][t], co[t]});
        co[s].insert(co[s].end(), all(co[t]));
        rep(i,0,n) mat[s][i] += mat[t][i];
        rep(i,0,n) mat[i][s] = mat[s][i];
        mat[0][t] = INT_MIN;
    }
    return best;
}
```

GomoryHu.h

Description: Given a list of edges representing an undirected flow graph, returns edges of the Gomory-Hu tree. The max flow between any pair of vertices is given by minimum edge weight along the Gomory-Hu tree path.

Time: $\mathcal{O}(V)$ Flow Computations

"PushRelabel.h"

0418b3, 13 lines

```
typedef array<ll, 3> Edge;
vector<Edge> gomoryHu(int N, vector<Edge> ed) {
    vector<Edge> tree;
    vi par(N);
    rep(i,1,N) {
        PushRelabel D(N); // Dinic also works
        for (Edge t : ed) D.addEdge(t[0], t[1], t[2], t[2]);
        tree.push_back({i, par[i], D.calc(i, par[i])});
        rep(j,i+1,N)
            if (par[j] == par[i] && D.leftOfMinCut(j)) par[j] = i;
    }
}
```

```
    return tree;
}
```

7.3 Matching

hopcroftKarp.h

Description: Fast bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and $btoa$ should be a vector full of -1's of the same size as the right partition. Returns the size of the matching. $btoa[i]$ will be the match for vertex i on the right side, or -1 if it's not matched.

Usage: `vi btoa(m, -1); hopcroftKarp(g, btoa);`

Time: $\mathcal{O}(\sqrt{VE})$

f612e4, 42 lines

```
bool dfs(int a, int L, vector<vi>& g, vi& btoa, vi& A, vi& B) {
    if (A[a] != L) return 0;
    A[a] = -1;
    for (int b : g[a]) if (B[b] == L + 1) {
        B[b] = 0;
        if (btoa[b] == -1 || dfs(btoa[b], L + 1, g, btoa, A, B))
            return btoa[b] = a, 1;
    }
    return 0;
}
```

```
int hopcroftKarp(vector<vi>& g, vi& btoa) {
    int res = 0;
    vi A(g.size()), B(btoa.size()), cur, next;
    for (;;) {
        fill(all(A), 0);
        fill(all(B), 0);
        cur.clear();
        for (int a : btoa) if (a != -1) A[a] = -1;
        rep(a,0,sz(g)) if (A[a] == 0) cur.push_back(a);
        for (int lay = 1; lay++) {
            bool islast = 0;
            next.clear();
            for (int a : cur) for (int b : g[a]) {
                if (btoa[b] == -1) {
                    B[b] = lay;
                    islast = 1;
                }
                else if (btoa[b] != a && !B[b]) {
                    B[b] = lay;
                    next.push_back(btoa[b]);
                }
            }
            if (islast) break;
            if (next.empty()) return res;
            for (int a : next) A[a] = lay;
            cur.swap(next);
        }
        rep(a,0,sz(g))
            res += dfs(a, 0, g, btoa, A, B);
    }
}
```

DFSMatching.h

Description: Simple bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and $btoa$ should be a vector full of -1's of the same size as the right partition. Returns the size of the matching. $btoa[i]$ will be the match for vertex i on the right side, or -1 if it's not matched.

Usage: `vi btoa(m, -1); dfsMatching(g, btoa);`

Time: $\mathcal{O}(VE)$

522b98, 22 lines

```
bool find(int j, vector<vi>& g, vi& btoa, vi& vis) {
    if (btoa[j] == -1) return 1;
    vis[j] = 1; int di = btoa[j];
    for (int e : g[di])
        if (!vis[e] && find(e, g, btoa, vis)) {
            btoa[e] = di;
            return 1;
        }
    return 0;
}
int dfsMatching(vector<vi>& g, vi& btoa) {
    vi vis;
    rep(i,0,sz(g)) {
        vis.assign(sz(btoa), 0);
        for (int j : g[i])
```

```
    if (find(j, g, btoa, vis)) {
        btoa[j] = i;
        break;
    }
}
return sz(btoa) - (int)count(all(btoa), -1);
}
```

MinimumVertexCover.h

Description: Finds a minimum vertex cover in a bipartite graph. The size is the same as the size of a maximum matching, and the complement is a maximum independent set.

```
"DFSMatching.h"
da4196, 20 lines

vi cover(vector<vi>& g, int n, int m) {
    vi match(m, -1);
    int res = dfsMatching(g, match);
    vector<bool> lfound(n, true), seen(m);
    for (int it : match) if (it != -1) lfound[it] = false;
    vi q, cover;
    rep(i,0,n) if (lfound[i]) q.push_back(i);
    while (!q.empty()) {
        int i = q.back(); q.pop_back();
        lfound[i] = 1;
        for (int e : g[i]) if (!seen[e] && match[e] != -1) {
            seen[e] = true;
            q.push_back(match[e]);
        }
    }
    rep(i,0,n) if (!lfound[i]) cover.push_back(i);
    rep(i,0,m) if (seen[i]) cover.push_back(n+i);
    assert(sz(cover) == res);
    return cover;
}
```

WeightedMatching.h

Description: Given a weighted bipartite graph, matches every node on the left with a node on the right such that no nodes are in two matchings and the sum of the edge weights is minimal. Takes cost[N][M], where cost[i][j] = cost for L[i] to be matched with R[j] and returns (min cost, match), where L[i] is matched with R[match[i]]. Negate costs for max cost. Requires $N \leq M$.

```
"Time:  $\mathcal{O}(N^2M)$ "
1e0fe9, 31 lines
```

```
pair<int, vi> hungarian(const vector<vi> &a) {
    if (a.empty()) return {0, {}};
    int n = sz(a) + 1, m = sz(a[0]) + 1;
    vi u(n), v(m), p(m), ans(n - 1);
    rep(i,1,n) {
        p[0] = i;
        int j0 = 0; // add "dummy" worker 0
        vi dist(m, INT_MAX), pre(m, -1);
        vector<bool> done(m + 1);
        do { // dijkstra
            done[j0] = true;
            int i0 = p[j0], j1, delta = INT_MAX;
            rep(j,1,m) if (!done[j]) {
                auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
                if (dist[j] < delta) delta = dist[j], j1 = j;
            }
            rep(j,0,m) {
                if (done[j]) u[p[j]] += delta, v[j] -= delta;
                else dist[j] -= delta;
            }
            j0 = j1;
        } while (p[j0]);
        while (j0) { // update alternating path
            int j1 = pre[j0];
            p[j0] = p[j1], j0 = j1;
        }
    }
    rep(j,1,m) if (p[j]) ans[p[j] - 1] = j - 1;
    return {-v[0], ans}; // min cost
}
```

GeneralMatching.h

Description: Matching for general graphs. Fails with probability N/mod .

```
"Time:  $\mathcal{O}(N^3)$ "
"../numerical/MatrixInverse-mod.h"
cb1912, 40 lines
```

```
vector<pii> generalMatching(int N, vector<pii>& ed) {
    vector<vector<ll>> mat(N, vector<ll>(N)), A;
    for (pii pa : ed) {
        int a = pa.first, b = pa.second, r = rand() % mod;
        mat[a][b] = r, mat[b][a] = (mod - r) % mod;
    }

    int r = matInv(A = mat), M = 2*N - r, fi, fj;
    assert(r % 2 == 0);

    if (M != N) do {
        mat.resize(M, vector<ll>(M));
        rep(i,0,N) {
            mat[i].resize(M);
            rep(j,N,M) {
                int r = rand() % mod;
                mat[i][j] = r, mat[j][i] = (mod - r) % mod;
            }
        } while (matInv(A = mat) != M);

        vi has(M, 1); vector<pii> ret;
        rep(it,0,M/2) {
            rep(i,0,M) if (has[i])
                rep(j,i+1,M) if (A[i][j] && mat[i][j]) {
                    fi = i; fj = j; goto done;
                } assert(0); done:
            if (fj < N) ret.emplace_back(fi, fj);
            has[fi] = has[fj] = 0;
            rep(sw,0,2) {
                ll a = modpow(A[fi][fj], mod-2);
                rep(i,0,M) if (has[i] && A[i][fj]) {
                    ll b = A[i][fj] * a % mod;
                    rep(j,0,M) A[i][j] = (A[i][j] - A[fi][j] * b) % mod;
                }
                swap(fi,fj);
            }
        }
        return ret;
    }
}
```

7.4 DFS algorithms

BiconnectedComponents.h

Description: Finds all biconnected components in an undirected graph, and runs a callback for the edges in each. In a biconnected component there are at least two distinct paths between any two nodes. Note that a node can be in several components. An edge which is not in a component is a bridge, i.e., not part of any cycle.

Usage: int eid = 0; ed.resize(N);
for each edge (a,b) {
ed[a].emplace_back(b, eid);
ed[b].emplace_back(a, eid++); }
bicomps([&](const vi& edgelist) {...});
Time: $\mathcal{O}(E + V)$

```
2965e5, 33 lines
```

```
vi num, st;
vector<vector<pii>> ed;
int Time;
template<class F>
int dfs(int at, int par, F& f) {
    int me = num[at] = ++Time, e, y, top = me;
    for (auto pa : ed[at]) if (pa.second != par) {
        tie(y, e) = pa;
        if (num[y]) {
            top = min(top, num[y]);
            if (num[y] < me)
                st.push_back(e);
        } else {
            int si = sz(st);
            int up = dfs(y, e, f);
            top = min(top, up);
            if (up == me) {
                st.push_back(e);
                f(vi(st.begin() + si, st.end()));
                st.resize(si);
            }
            else if (up < me) st.push_back(e);
        }
    }
}
```

```
    else { /* e is a bridge */ }
    }
}
return top;
}
```

```
template<class F>
void bicomps(F f) {
    num.assign(sz(ed), 0);
    rep(i,0,sz(ed)) if (!num[i]) dfs(i, -1, f);
}
```

TwoConnectedComponents.h

Description: Lists out all edge biconnected components in a graph (components that if any edge is deleted is still connected).

```
"Time:  $\mathcal{O}(E + V)$ "
654bdc, 30 lines
```

```
vector<vector<pair<int, int>>> adj;
vector<int> num, low, color, id_twoconnected;
vector<vector<int>>> two_connected;
int cnt_twoconnected, Time;

stack<int> st;
void DFS(int u, int edge_dad) {
    num[u] = low[u] = ++Time;
    color[u] = 1;
    st.push(u);
    for(auto e : adj[u]) {
        int v = e.first, edge_id = e.second;
        if (edge_id != edge_dad) {
            if (color[v] == 0) {
                DFS(v, edge_id);
                low[u] = min(low[u], low[v]);
            } else low[u] = min(low[u], num[v]);
        }
    }
    if (low[u] == num[u]) {
        ++cnt_twoconnected;
        int v;
        do {
            v = st.top();
            st.pop();
            id_twoconnected[v] = cnt_twoconnected;
            two_connected[cnt_twoconnected].push_back(v);
        } while (v != u);
    }
}
```

2sat.h

Description: Calculates a valid assignment to boolean variables a, b, c,... to a 2-SAT problem, so that an expression of the type $(a \vee b) \wedge (\neg a \vee c) \wedge (d \vee \neg b) \wedge \dots$ becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions (~x).

Usage: TwoSat ts(number of boolean variables);
ts.either(0, ~3); // Var 0 is true or var 3 is false
ts.setValue(2); // Var 2 is true
ts.atMostOne({0,~1,2}); // <= 1 of vars 0, ~1 and 2 are true
ts.solve(); // Returns true iff it is solvable
ts.values[0..N-1] holds the assigned values to the vars

Time: $\mathcal{O}(N + E)$, where N is the number of boolean variables, and E is the number of clauses.

```
5f9706, 56 lines
```

```
struct TwoSat {
    int N;
    vector<vi> gr;
    vi values; // 0 = false, 1 = true

    TwoSat(int n = 0) : N(n), gr(2*n) {}

    int addVar() { // (optional)
        gr.emplace_back();
        gr.emplace_back();
        return N++;
    }

    void either(int f, int j) {
        f = max(2*f, -1-2*f);
        j = max(2*j, -1-2*j);
        gr[f].push_back(j^1);
    }
}
```

```
    gr[j].push_back(f^1);
}
void setValue(int x) { either(x, x); }

void atMostOne(const vi& li) { // (optional)
    if (sz(li) <= 1) return;
    int cur = ~li[0];
    rep(i,2,sz(li)) {
        int next = addVar();
        either(cur, ~li[i]);
        either(cur, next);
        either(~li[i], next);
        cur = ~next;
    }
    either(cur, ~li[1]);
}

vi val, comp, z; int time = 0;
int dfs(int i) {
    int low = val[i] = ++time, x; z.push_back(i);
    for(int e : gr[i]) if (!comp[e])
        low = min(low, val[e] ? dfs(e));
    if (low == val[i]) do {
        x = z.back(); z.pop_back();
        comp[x] = low;
        if (values[x>>1] == -1)
            values[x>>1] = x&1;
    } while (x != i);
    return val[i] = low;
}

bool solve() {
    values.assign(N, -1);
    val.assign(2*N, 0); comp = val;
    rep(i,0,2*N) if (!comp[i]) dfs(i);
    rep(i,0,N) if (comp[2*i] == comp[2*i+1]) return 0;
    return 1;
}
};
```

EulerWalk.h

Description: Eulerian undirected/directed path/cycle algorithm. Input should be a vector of (dest, global edge index), where for undirected graphs, forward/backward edges have the same index. Returns a list of nodes in the Eulerian path/cycle with src at both start and end, or empty list if no cycle/path exists. To get edge indices back, add .second to s and ret.
Time: $\mathcal{O}(V + E)$

```
vi eulerWalk(vector<vector<pii>>& gr, int nedges, int src=0) {
    int n = sz(gr);
    vi D(n), its(n), eu(nedges), ret, s = {src};
    D[src]++; // to allow Euler paths, not just cycles
    while (!s.empty()) {
        int x = s.back(), y, e, &it = its[x], end = sz(gr[x]);
        if (it == end){ ret.push_back(x); s.pop_back(); continue; }
        tie(y, e) = gr[x][it++];
        if (!eu[e]) {
            D[x]--, D[y]++;
            eu[e] = 1; s.push_back(y);
        }
    }
    for (int x : D) if (x < 0 || sz(ret) != nedges+1) return {};
    return {ret.rbegin(), ret.rend()};
}
```

7.5 Coloring
EdgeColoring.h

Description: Given a simple, undirected graph with max degree D , computes a $(D + 1)$ -coloring of the edges such that no neighboring edges share a color. (D -coloring is NP-hard, but can be done for bipartite graphs by repeated matchings of max-degree nodes.)
Time: $\mathcal{O}(NM)$

```
vi edgeColoring(int N, vector<pii> eds) {
    vi cc(N + 1), ret(sz(eds)), fan(N), free(N), loc;
    for (pii e : eds) ++cc[e.first], ++cc[e.second];
    int u, v, ncols = *max_element(all(cc)) + 1;
    vector<vi> adj(N, vi(ncols, -1));
    for (pii e : eds) {
```

```
        tie(u, v) = e;
        fan[0] = v;
        loc.assign(ncols, 0);
        int at = u, end = u, d, c = free[u], ind = 0, i = 0;
        while (d = free[v], !loc[d] && (v = adj[u][d]) != -1)
            loc[d] = ++ind, cc[ind] = d, fan[ind] = v;
        cc[loc[d]] = c;
        for (int cd = d; at != -1; cd ^= c ^ d, at = adj[at][cd])
            swap(adj[at][cd], adj[end = at][cd ^ c ^ d]);
        while (adj[fan[i]][d] != -1) {
            int left = fan[i], right = fan[++i], e = cc[i];
            adj[u][e] = left;
            adj[left][e] = u;
            adj[right][e] = -1;
            free[right] = e;
        }
        adj[u][d] = fan[i];
        adj[fan[i]][d] = u;
        for (int y : {fan[0], u, end})
            for (int& z = free[y] = 0; adj[y][z] != -1; z++);
    }
    rep(i,0,sz(eds))
        for (tie(u, v) = eds[i]; adj[u][ret[i]] != v;) ++ret[i];
    return ret;
}
```

7.6 Heuristics
MaximalCliques.h

Description: Runs a callback for all maximal cliques in a graph (given as a symmetric bitset matrix; self-edges not allowed). Callback is given a bitset representing the maximal clique.
Time: $\mathcal{O}\left(3^n/3\right)$, much faster for sparse graphs

```
typedef bitset<128> B;
template<class F>
void cliques(vector<B>& eds, F f, B P = ~B(), B X={}, B R={}) {
    if (!P.any()) { if (!X.any()) f(R); return; }
    auto q = (P | X)._find_first();
    auto cands = P & ~eds[q];
    rep(i,0,sz(eds)) if (cands[i]) {
        R[i] = 1;
        cliques(eds, f, P & eds[i], X & eds[i], R);
        R[i] = P[i] = 0; X[i] = 1;
    }
}
```

MaximumClique.h

Description: Quickly finds a maximum clique of a graph (given as symmetric bit-set matrix; self-edges not allowed). Can be used to find a maximum independent set by finding a clique of the complement graph.
Time: Runs in about 1s for n=155 and worst case random graphs (p=.90). Runs faster for sparse graphs.

```
typedef vector<bitset<200>> vb;
struct Maxclique {
    double limit=0.025, pk=0;
    struct Vertex { int i, d=0; };
    typedef vector<Vertex> vv;
    vb e;
    vv V;
    vector<vi> C;
    vi qmax, q, S, old;
    void init(vv& r) {
        for (auto& v : r) v.d = 0;
        for (auto& v : r) for (auto j : r) v.d += e[v.i][j.i];
        sort(all(r), [](auto a, auto b) { return a.d > b.d; });
        int mxD = r[0].d;
        rep(i,0,sz(r)) r[i].d = min(i, mxD) + 1;
    }
    void expand(vv& R, int lev = 1) {
        S[lev] += S[lev - 1] - old[lev];
        old[lev] = S[lev - 1];
        while (sz(R)) {
            if (sz(q) + R.back().d <= sz(qmax)) return;
            q.push_back(R.back().i);
            vv T;
            for(auto v:R) if (e[R.back().i][v.i]) T.push_back({v.i});
            if (sz(T)) {
```

```
                if (S[lev]++ / ++pk < limit) init(T);
                int j = 0, mxk = 1, mnk = max(sz(qmax) - sz(q) + 1, 1);
                C[1].clear(), C[2].clear();
                for (auto v : T) {
                    int k = 1;
                    auto f = [&](int i) { return e[v.i][i]; };
                    while (any_of(all(C[k]), f)) k++;
                    if (k > mxk) mxk = k, C[mxk + 1].clear();
                    if (k < mnk) T[j++] .i = v.i;
                    C[k].push_back(v.i);
                }
                if (j > 0) T[j - 1].d = 0;
                rep(k,mnk,mxk + 1) for (int i : C[k])
                    T[j].i = i, T[j++].d = k;
                expand(T, lev + 1);
            } else if (sz(q) > sz(qmax)) qmax = q;
            q.pop_back(), R.pop_back();
        }
    }
    vi maxClique() { init(V), expand(V); return qmax; }
    Maxclique(vb conn) : e(conn), C(sz(e)+1), S(sz(C)), old(S) {
        rep(i,0,sz(e)) V.push_back({i});
    }
};
```

MaximumIndependentSet.h

Description: To obtain a maximum independent set of a graph, find a max clique of the complement. If the graph is bipartite, see MinimumVertexCover.

7.7 Trees
CompressTree.h

Description: Given a rooted tree and a subset S of nodes, compute the minimal subtree that contains all the nodes by adding all (at most $|S| - 1$) pairwise LCA's and compressing edges. Returns a list of (par, orig_index) representing a tree rooted at 0. The root points to itself.
Time: $\mathcal{O}(|S| \log |S|)$

```
"LCA.h"
typedef vector<pair<int, int>> vpi;
vpi compressTree(LCA& lca, const vi& subset) {
    static vi rev; rev.resize(sz(lca.time));
    vi li = subset, &T = lca.time;
    auto cmp = [&](int a, int b) { return T[a] < T[b]; };
    sort(all(li), cmp);
    int m = sz(li)-1;
    rep(i,0,m) {
        int a = li[i], b = li[i+1];
        li.push_back(lca.lca(a, b));
    }
    sort(all(li), cmp);
    li.erase(unique(all(li)), li.end());
    rep(i,0,sz(li)) rev[li[i]] = i;
    vpi ret = {pii(0, li[0])};
    rep(i,0,sz(li)-1) {
        int a = li[i], b = li[i+1];
        ret.emplace_back(rev[lca.lca(a, b)], b);
    }
    return ret;
}
```

HLD.h

Description: Decomposes a tree into vertex disjoint heavy paths and light edges such that the path from any leaf to the root contains at most $\log(n)$ light edges. Code does additive modifications and max queries, but can support commutative segtree modifications/queries on paths and subtrees. Takes as input the full adjacency list. VALS_EDGES being true means that values are stored in the edges, as opposed to the nodes. All values initialized to the segtree default. Root must be 0.

```
Time:  $\mathcal{O}\left((\log N)^2\right)$ 
"../data-structures/LazySegmentTree.h"
template <bool VALS_EDGES> struct HLD {
    int N, tim = 0;
    vector<vi> adj;
    vi par, siz, depth, rt, pos;
    Node *tree;
    HLD(vector<vi> adj_)
        : N(sz(adj_)), adj(adj_), par(N, -1), siz(N, 1), depth(N),
```

```
rt(N),pos(N),tree(new Node(0, N)){ dfsSz(0); dfsHld(0); }
void dfsSz(int v) {
    if (par[v] != -1) adj[v].erase(find(all(adj[v]), par[v]));
    for (int& u : adj[v]) {
        par[u] = v, depth[u] = depth[v] + 1;
        dfsSz(u);
        siz[v] += siz[u];
        if (siz[u] > siz[adj[v][0]]) swap(u, adj[v][0]);
    }
}
void dfsHld(int v) {
    pos[v] = tim++;
    for (int u : adj[v]) {
        rt[u] = (u == adj[v][0] ? rt[v] : u);
        dfsHld(u);
    }
}
template <class B> void process(int u, int v, B op) {
    for (; rt[u] != rt[v]; v = par[rt[v]]) {
        if (depth[rt[u]] > depth[rt[v]]) swap(u, v);
        op(pos[rt[v]], pos[v] + 1);
    }
    if (depth[u] > depth[v]) swap(u, v);
    op(pos[u] + VALS_EDGES, pos[v] + 1);
}
void modifyPath(int u, int v, int val) {
    process(u, v, [&](int l, int r) { tree->add(l, r, val); });
}
int queryPath(int u, int v) { // Modify depending on problem
    int res = -1e9;
    process(u, v, [&](int l, int r) {
        res = max(res, tree->query(l, r));
    });
    return res;
}
int querySubtree(int v) { // modifySubtree is similar
    return tree->query(pos[v] + VALS_EDGES, pos[v] + siz[v]);
}
};
```

CentroidDecomposition.h

Description: Divide-and-conquer on a tree.
Time: $\mathcal{O}(n \cdot \log_2 n)$ (with high constant factor) 28cab3, 18 lines

```
const int N = 2e5 + 10;
int sz[N]; bool used[N];
vector<int> adj[N];
int calc(int x, int par) {
    sz[x] = 1;
    for (int &i : adj[x]) if (!used[i] && i != par) sz[x] += calc(i,x);
    return sz[x];
}
int getCentroid(int x, int par, int cnt) {
    for (int &i : adj[x]) if (!used[i] && i != par && sz[i] * 2 > cnt)
        return getCentroid(i, x, cnt);
    return x;
}
void centroid(int x, int par = 0) {
    int cnt = calc(x, x), node = getCentroid(x, x, cnt);
    // do work here
    used[node] = 1;
    for (int &i : adj[node]) if (!used[i]) centroid(i, node);
}
```

LinkCutTree.h

Description: Represents a forest of unrooted trees. You can add and remove edges (as long as the result is still a forest), and check whether two nodes are in the same tree.
Time: All operations take amortized $\mathcal{O}(\log N)$. f0f202, 206 lines

```
struct node {
    int p = 0, c[2] = {0, 0}, pp = 0;
    bool flip = 0;
    int sz = 0, ssz = 0, vsz = 0; // sz -> aux tree size, ssz = subtree size in rep tree, vsz = virtual tree size
    long long val = 0, sum = 0, lazy = 0, subsum = 0, vsum = 0;
    node() {}
    node(int x) {
        val = x; sum = x;
```

```
sz = 1; lazy = 0;
ssz = 1; vsz = 0;
subsum = x; vsum = 0;
}
};
struct LCT {
    vector<node> t;
    LCT() {}
    LCT(int n) : t(n + 1) {}

    // <independant splay tree code>
    int dir(int x, int y) { return t[x].c[1] == y; }
    void set(int x, int d, int y) {
        if (x) t[x].c[d] = y, pull(x);
        if (y) t[y].p = x;
    }
    void pull(int x) {
        if (!x) return;
        int &l = t[x].c[0], &r = t[x].c[1];
        push(l); push(r);
        t[x].sum = t[l].sum + t[r].sum + t[x].val;
        t[x].sz = t[l].sz + t[r].sz + 1;
        t[x].ssz = t[l].ssz + t[r].ssz + t[x].vsz + 1;
        t[x].subsum = t[l].subsum + t[r].subsum + t[x].vsum + t[x].val;
    }
    void push(int x) {
        if (!x) return;
        int &l = t[x].c[0], &r = t[x].c[1];
        if (t[x].flip) {
            swap(l, r);
            if (l) t[l].flip ^= 1;
            if (r) t[r].flip ^= 1;
            t[x].flip = 0;
        }
        if (t[x].lazy) {
            t[x].val += t[x].lazy;
            t[x].sum += t[x].lazy * t[x].sz;
            t[x].subsum += t[x].lazy * t[x].ssz;
            t[x].vsum += t[x].lazy * t[x].vsz;
            if (l) t[l].lazy += t[x].lazy;
            if (r) t[r].lazy += t[x].lazy;
            t[x].lazy = 0;
        }
    }
    void rotate(int x, int d) {
        int y = t[x].p, z = t[y].p, w = t[x].c[d];
        swap(t[x].pp, t[y].pp);
        set(y, !d, w);
        set(x, d, y);
        set(z, dir(z, y), x);
    }
    void splay(int x) {
        for (push(x); t[x].p;) {
            int y = t[x].p, z = t[y].p;
            push(z); push(y); push(x);
            int dx = dir(y, x), dy = dir(z, y);
            if (!z) rotate(x, !dx);
            else if (dx == dy) rotate(y, !dx), rotate(x, !dx);
            else rotate(x, dy), rotate(x, dx);
        }
    }
    // </independant splay tree code>

    // making it a root in the rep. tree
    void make_root(int u) {
        access(u);
        int l = t[u].c[0];
        t[l].flip ^= 1;
        swap(t[l].p, t[l].pp);
        t[u].vsz += t[l].ssz;
        t[u].vsum += t[l].subsum;
        set(u, 0, 0);
    }
    // make the path from root to u a preferred path
    // returns last path-parent of a node as it moves up the tree
    int access(int _u) {
        int last = _u;
        for (int v = 0, u = _u; u; u = t[v = u].pp) {
            splay(u); splay(v);
```

```
t[u].vsz -= t[v].ssz;
t[u].vsum -= t[v].subsum;
int r = t[u].c[1];
t[u].vsz += t[r].ssz;
t[u].vsum += t[r].subsum;
t[v].pp = 0;
swap(t[r].p, t[r].pp);
set(u, 1, v);
last = u;
}
splay(_u);
return last;
}
void link(int u, int v) { // u -> v
    // assert(!connected(u, v));
    make_root(v);
    access(u); splay(u);
    t[v].pp = u;
    t[u].vsz += t[v].ssz;
    t[u].vsum += t[v].subsum;
}
void cut(int u) { // cut par[u] -> u, u is non root vertex
    access(u);
    assert(t[u].c[0] != 0);
    t[t[u].c[0]].p = 0;
    t[u].c[0] = 0;
    pull(u);
}
// parent of u in the rep. tree
int get_parent(int u) {
    access(u); splay(u); push(u);
    u = t[u].c[0]; push(u);
    while (t[u].c[1]) {
        u = t[u].c[1]; push(u);
    }
    splay(u);
    return u;
}
// root of the rep. tree containing this node
int find_root(int u) {
    access(u); splay(u); push(u);
    while (t[u].c[0]) {
        u = t[u].c[0]; push(u);
    }
    splay(u);
    return u;
}
bool connected(int u, int v) {
    return find_root(u) == find_root(v);
}
// depth in the rep. tree
int depth(int u) {
    access(u); splay(u);
    return t[u].sz;
}
int lca(int u, int v) {
    // assert(connected(u, v));
    if (u == v) return u;
    if (depth(u) > depth(v)) swap(u, v);
    access(v);
    return access(u);
}
int is_root(int u) {
    return get_parent(u) == 0;
}
int component_size(int u) {
    return t[find_root(u)].ssz;
}
int subtree_size(int u) {
    int p = get_parent(u);
    if (p == 0) {
        return component_size(u);
    }
    cut(u);
    int ans = component_size(u);
    link(p, u);
    return ans;
}
long long component_sum(int u) {
```

```
        return t[find_root(u)].subsum;
    }
    long long subtree_sum(int u) {
        int p = get_parent(u);
        if (p == 0) {
            return component_sum(u);
        }
        cut(u);
        long long ans = component_sum(u);
        link(p, u);
        return ans;
    }
    // sum of the subtree of u when root is specified
    long long subtree_query(int u, int root) {
        int cur = find_root(u);
        make_root(root);
        long long ans = subtree_sum(u);
        make_root(cur);
        return ans;
    }
    // path sum
    long long query(int u, int v) {
        int cur = find_root(u);
        make_root(u); access(v);
        long long ans = t[v].sum;
        make_root(cur);
        return ans;
    }
    void upd(int u, int x) {
        access(u); splay(u);
        t[u].val += x;
    }
    // add x to the nodes on the path from u to v
    void upd(int u, int v, int x) {
        int cur = find_root(u);
        make_root(u); access(v);
        t[v].lazy += x;
        make_root(cur);
    }
};
```

DirectedMST.h

Description: Finds a minimum spanning tree/arborescence of a directed graph, given a root node. If no MST exists, returns -1.
Time: $\mathcal{O}(E \log V)$

../data-structures/UnionFindRollback.h 39e620, 60 lines

```
struct Edge { int a, b; ll w; };
struct Node {
    Edge key;
    Node *l, *r;
    ll delta;
    void prop() {
        key.w += delta;
        if (l) l->delta += delta;
        if (r) r->delta += delta;
        delta = 0;
    }
    Edge top() { prop(); return key; }
};
Node *merge(Node *a, Node *b) {
    if (!a || !b) return a ? b : a->prop(), b->prop();
    if (a->key.w > b->key.w) swap(a, b);
    swap(a->l, (a->r = merge(b, a->r)));
    return a;
}
void pop(Node& a) { a->prop(); a = merge(a->l, a->r); }

pair<ll, vi> dmst(int n, int r, vector<Edge>& g) {
    RollbackUF uf(n);
    vector<Node*> heap(n);
    for (Edge e : g) heap[e.b] = merge(heap[e.b], new Node(e));
    ll res = 0;
    vi seen(n, -1), path(n), par(n);
    seen[r] = r;
    vector<Edge> Q(n), in(n, {-1,-1}), comp;
    deque<tuple<int, int, vector<Edge>>> cys;
    rep(s,0,n) {
```

```
int u = s, qi = 0, w;
while (seen[u] < 0) {
    if (!heap[u]) return {-1,{};};
    Edge e = heap[u]->top();
    heap[u]->delta -= e.w, pop(heap[u]);
    Q[qi] = e, path[qi++] = u, seen[u] = s;
    res += e.w, u = uf.find(e.a);
    if (seen[u] == s) {
        Node* cyc = 0;
        int end = qi, time = uf.time();
        do cyc = merge(cyc, heap[w = path[--qi]]);
        while (uf.join(u, w));
        u = uf.find(u), heap[u] = cyc, seen[u] = -1;
        cys.push_front({u, time, {Q[qi], Q[end]}));
    }
    rep(i,0,qi) in[uf.find(Q[i].b)] = Q[i];
}

for (auto& [u,t,comp] : cys) { // restore sol (optional)
    uf.rollback(t);
    Edge inEdge = in[u];
    for (auto& e : comp) in[uf.find(e.b)] = e;
    in[uf.find(inEdge.b)] = inEdge;
}
rep(i,0,n) par[i] = in[i].a;
return {res, par};
}
```

7.8 Math

7.8.1 Number of Spanning Trees

Create an $N \times N$ matrix mat , and for each edge $a \rightarrow b \in G$, do $mat[a][b]--$, $mat[b][b]++$ (and $mat[b][a]--$, $mat[a][a]++$ if G is undirected). Remove the i th row and column and take the determinant; this yields the number of directed spanning trees rooted at i (if G is undirected, remove any row/column).

7.8.2 Erdős–Gallai theorem

A simple graph with node degrees $d_1 \geq \dots \geq d_n$ exists iff $d_1 + \dots + d_n$ is even and for every $k = 1 \dots n$,

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k).$$

Geometry (8)

8.1 Geometric primitives

Point.h

Description: Class to handle points in the plane. T can be e.g. double or long long. (Avoid int.)

template <class T> int sgn(T x) { return (x > 0) - (x < 0); }
template<class T> 47ec0a, 28 lines

```
template <class T> int sgn(T x) { return (x > 0) - (x < 0); }
template<class T>
struct Point {
    typedef Point P;
    T x, y;
    explicit Point(T x=0, T y=0) : x(x), y(y) {}
    bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
    bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
    P operator+(P p) const { return P(x+p.x, y+p.y); }
    P operator-(P p) const { return P(x-p.x, y-p.y); }
    P operator*(T d) const { return P(x*d, y*d); }
    P operator/(T d) const { return P(x/d, y/d); }
    T dot(P p) const { return x*p.x + y*p.y; }
    T cross(P p) const { return x*p.y - y*p.x; }
    T cross(P a, P b) const { return (a->this).cross(b->this); }
    T dist2() const { return x*x + y*y; }
    double dist() const { return sqrt((double)dist2()); }
    // angle to x-axis in interval [-pi, pi]
```

```
double angle() const { return atan2(y, x); }
P unit() const { return *this/dist(); } // makes dist()==1
P perp() const { return P(-y, x); } // rotates +90 degrees
P normal() const { return perp().unit(); }
// returns point rotated 'a' radians ccw around the origin
P rotate(double a) const {
    return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); }
friend ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.x << ", " << p.y << ")"; }
};
```

lineDistance.h

Description: Returns the signed distance between point p and the line containing points a and b. Positive value on left side and negative on right as seen from a towards b. $a=b$ gives nan. P is supposed to be $\text{Point}<T>$ or $\text{Point3D}<T>$ where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long. Using Point3D will always give a non-negative distance. For Point3D , call $.dist$ on the result of the cross product.
"Point.h" f6bf6b, 4 lines

```
template<class P>
double lineDist(const P& a, const P& b, const P& p) {
    return (double) (b-a).cross(p-a) / (b-a).dist();
}
```

SegmentDistance.h

Description: Returns the shortest distance between point p and the line segment from point s to e.
Usage: $\text{Point}<\text{double}> a, b(2,2), p(1,1);$
 $\text{bool onSegment} = \text{segDist}(a,b,p) < 1e-10;$
"Point.h" 5c88f4, 6 lines

```
typedef Point<double> P;
double segDist(P& s, P& e, P& p) {
    if (s==e) return (p-s).dist();
    auto d = (e-s).dist2(), t = min(d,max(.0, (p-s).dot(e-s)));
    return ((p-s)*d-(e-s)*t).dist()/d;
}
```

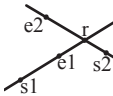
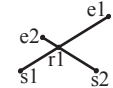
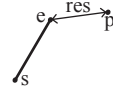
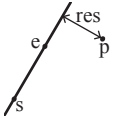
SegmentIntersection.h

Description: If a unique intersection point between the line segments going from s_1 to e_1 and from s_2 to e_2 exists then it is returned. If no intersection point exists an empty vector is returned. If infinitely many exist a vector with 2 elements is returned, containing the endpoints of the common line segment. The wrong position will be returned if P is $\text{Point}<\text{ll}>$ and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or long long.
Usage: $\text{vector}<P> \text{inter} = \text{segInter}(s_1,e_1,s_2,e_2);$
if ($\text{sz}(\text{inter})==1$)
cout << "segments intersect at " << inter[0] << endl;
"Point.h", "OnSegment.h" 9d5ff2, 13 lines

```
template<class P> vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b),
        oc = a.cross(b, c), od = a.cross(b, d);
    // Checks if intersection is single non-endpoint point.
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa)};
    set<P> s;
    if (onSegment(c, d, a)) s.insert(a);
    if (onSegment(c, d, b)) s.insert(b);
    if (onSegment(a, b, c)) s.insert(c);
    if (onSegment(a, b, d)) s.insert(d);
    return {all(s)};
}
```

lineIntersection.h

Description: If a unique intersection point of the lines going through s_1,e_1 and s_2,e_2 exists $\{1, \text{point}\}$ is returned. If no intersection point exists $\{0, (0,0)\}$ is returned and if infinitely many exists $\{-1, (0,0)\}$ is returned. The wrong position will be returned if P is $\text{Point}<\text{ll}>$ and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or ll.



```
Usage: auto res = lineInter(s1,e1,s2,e2);
if (res.first == 1)
cout << "intersection point at " << res.second << endl;
"Point.h" a01f81, 8 lines

template<class P>
pair<int, P> lineInter(P s1, P e1, P s2, P e2) {
    auto d = (e1 - s1).cross(e2 - s2);
    if (d == 0) // if parallel
        return {-(s1.cross(e1, s2) == 0), P(0, 0)};
    auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
    return {1, (s1 * p + e1 * q) / d};
}
```

sideOf.h

Description: Returns where p is as seen from s towards e . $1/0/-1 \Leftrightarrow$ left/on line/right. If the optional argument eps is given 0 is returned if p is within distance eps from the line. P is supposed to be $\text{Point}<T>$ where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long.

Usage: bool left = sideOf(p1,p2,q)==1;

```
"Point.h" 3af81c, 9 lines

template<class P>
int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }
```

```
template<class P>
int sideOf(const P& s, const P& e, const P& p, double eps) {
    auto a = (e-s).cross(p-s);
    double l = (e-s).dist()*eps;
    return (a > l) - (a < -l);
}
```

OnSegment.h

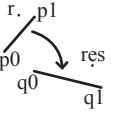
Description: Returns true iff p lies on the line segment from s to e . Use (segDist(s,e,p)<=epsilon) instead when using $\text{Point}<\text{double}>$.

```
"Point.h" c597e8, 3 lines

template<class P> bool onSegment(P s, P e, P p) {
    return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
}
```

linearTransformation.h

Description:



Apply the linear transformation (translation, rotation and scaling) which takes line p_0-p_1 to line q_0-q_1 to point r .

```
"Point.h" 03a306, 6 lines

typedef Point<double> P;
P linearTransformation(const P& p0, const P& p1,
    const P& q0, const P& q1, const P& r) {
    P dp = p1-p0, dq = q1-q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + P((r-p0).cross(num), (r-p0).dot(num))/dp.dist2();
}
```

LineProjectionReflection.h

Description: Projects point p onto line ab . Set refl=true to get reflection of point p across line ab insted. The wrong point will be returned if P is an integer point and the desired point doesn't have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow.

```
"Point.h" b5562d, 5 lines

template<class P>
P lineProj(P a, P b, P p, bool refl=false) {
    P v = b - a;
    return p - v.perp()*(1+refl)*v.cross(p-a)/v.dist2();
}
```

Angle.h

Description: A class for ordering angles (as represented by int points and a number of rotations around the origin). Useful for rotational sweeping. Sometimes also represents points or vectors.

Usage: vector<Angle> v = {w[0], w[0].t360() ...}; // sorted

```
int j = 0; rep(i,0,n) { while (v[j] < v[i].t180()) ++j; }
// sweeps j such that [j-i) represents the number of positively oriented
triangles with vertices at 0 and i
```

```
struct Angle {
    int x, y;
    int t;
};
```

```
Angle(int x, int y, int t=0) : x(x), y(y), t(t) {}
Angle operator-(Angle b) const { return {x-b.x, y-b.y, t}; }
int half() const {
    assert(x || y);
    return y < 0 || (y == 0 && x < 0);
}
Angle t90() const { return {-y, x, t + (half() && x >= 0)}; }
Angle t180() const { return {-x, -y, t + half()}; }
Angle t360() const { return {x, y, t + 1}; }
};

bool operator<(Angle a, Angle b) {
    // add a.dist2() and b.dist2() to also compare distances
    return make_tuple(a.t, a.half(), a.y * (1l)b.x) <
        make_tuple(b.t, b.half(), a.x * (1l)b.y);
}

// Given two points, this calculates the smallest angle between
// them, i.e., the angle that covers the defined line segment.
pair<Angle, Angle> segmentAngles(Angle a, Angle b) {
    if (b < a) swap(a, b);
    return (b < a.t180() ?
        make_pair(a, b) : make_pair(b, a.t360()));
}

Angle operator+(Angle a, Angle b) { // point a + vector b
    Angle r(a.x + b.x, a.y + b.y, a.t);
    if (a.t180() < r) r.t--;
    return r.t180() < a ? r.t360() : r;
}

Angle angleDiff(Angle a, Angle b) { // angle b - angle a
    int tu = b.t - a.t; a.t = b.t;
    return {a.x*b.x + a.y*b.y, a.x*b.y - a.y*b.x, tu - (b < a)};
}
```

8.2 Circles

CircleIntersection.h

Description: Computes the pair of points at which two circles intersect. Returns false in case of no intersection.

```
"Point.h" 84d6d3, 11 lines

typedef Point<double> P;
bool circleInter(P a,P b,double r1,double r2,pair<P, P>* out) {
    if (a == b) { assert(r1 != r2); return false; }
    P vec = b - a;
    double d2 = vec.dist2(), sum = r1+r2, dif = r1-r2,
        p = (d2 + r1*r1 - r2*r2)/(d2*2), h2 = r1*r1 - p*p*d2;
    if (sum*sum < d2 || dif*dif > d2) return false;
    P mid = a + vec*p, per = vec.perp() * sqrt(fmax(0, h2) / d2);
    *out = {mid + per, mid - per};
    return true;
}
```

CircleTangents.h

Description: Finds the external tangents of two circles, or internal if r_2 is negated. Can return 0, 1, or 2 tangents – 0 if one circle contains the other (or overlaps it, in the internal case, or if the circles are the same); 1 if the circles are tangent to each other (in which case .first = .second and the tangent line is perpendicular to the line between the centers). .first and .second give the tangency points at circle 1 and 2 respectively. To find the tangents of a circle with a point set r_2 to 0.

```
"Point.h" b0153d, 13 lines

template<class P>
vector<pair<P, P>> tangents(P c1, double r1, P c2, double r2) {
    P d = c2 - c1;
    double dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 == 0 || h2 < 0) return {};
    vector<pair<P, P>> out;
    for (double sign : {-1, 1}) {
        P v = (d * dr + d.perp() * sqrt(h2) * sign) / d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    if (h2 == 0) out.pop_back();
    return out;
}
```

CircleLine.h

Description: Finds the intersection between a circle and a line. Returns a vector of either 0, 1, or 2 intersection points. P is intended to be $\text{Point}<\text{double}>$.

```
"Point.h" e0cfba, 9 lines
```

```
template<class P>
vector<P> circleLine(P c, double r, P a, P b) {
    P ab = b - a, p = a + ab * (c-a).dot(ab) / ab.dist2();
    double s = a.cross(b, c), h2 = r*r - s*s / ab.dist2();
    if (h2 < 0) return {};
    if (h2 == 0) return {p};
    P h = ab.unit() * sqrt(h2);
    return {p - h, p + h};
}
```

CirclePolygonIntersection.h

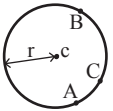
Description: Returns the area of the intersection of a circle with a ccw polygon. Time: $\mathcal{O}(n)$

```
".../content/geometry/Point.h" aleec63, 19 lines

typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
    auto tri = [&](P p, P q) {
        auto r2 = r * r / 2;
        P d = q - p;
        auto a = d.dot(p)/d.dist2(), b = (p.dist2()-r*r)/d.dist2();
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a-sqrt(det)), t = min(1., -a+sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = p + d * t;
        return arg(p,u) * r2 + u.cross(v)/2 + arg(v,q) * r2;
    };
    auto sum = 0.0;
    rep(i,0,sz(ps))
        sum += tri(ps[i] - c, ps[(i + 1) % sz(ps)] - c);
    return sum;
}
```

circumcircle.h

Description:



The circumcircle of a triangle is the circle intersecting all three vertices. ccRadius returns the radius of the circle going through points A , B and C and ccCenter returns the center of the same circle.

```
"Point.h" 1caa3a, 9 lines

typedef Point<double> P;
double ccRadius(const P& A, const P& B, const P& C) {
    return (B-A).dist()*(C-B).dist()*(A-C).dist()/
        abs((B-A).cross(C-A))/2;
}

P ccCenter(const P& A, const P& B, const P& C) {
    P b = C-A, c = B-A;
    return A + (b*c.dist2()-c*b.dist2()).perp()/b.cross(c)/2;
}
```

MinimumEnclosingCircle.h

Description: Computes the minimum circle that encloses a set of points. Time: expected $\mathcal{O}(n)$

```
"circumcircle.h" 09dd0a, 17 lines

pair<P, double> mec(vector<P> ps) {
    shuffle(all(ps), mt19937(time(0)));
    P o = ps[0];
    double r = 0, EPS = 1 + 1e-8;
    rep(i,0,sz(ps)) if ((o - ps[i]).dist() > r * EPS) {
        o = ps[i], r = 0;
        rep(j,0,i) if ((o - ps[j]).dist() > r * EPS) {
            o = (ps[i] + ps[j]) / 2;
            r = (o - ps[i]).dist();
            rep(k,0,j) if ((o - ps[k]).dist() > r * EPS) {
                o = ccCenter(ps[i], ps[j], ps[k]);
                r = (o - ps[i]).dist();
            }
        }
    }
    return {o, r};
}
```

8.3 Polygons

InsidePolygon.h

Description: Returns true if p lies within the polygon. If strict is true, it returns false for points on the boundary. The algorithm uses products in intermediate steps so watch out for overflow.

Usage: vector<P> v = {P{4,4}, P{1,2}, P{2,1}};

bool in = inPolygon(v, P{3, 3}, false);

Time: $\mathcal{O}(n)$

"Point.h", "OnSegment.h", "SegmentDistance.h"2bf504, 11 lines

```
template<class P>
bool inPolygon(vector<P> &p, P a, bool strict = true) {
    int cnt = 0, n = sz(p);
    rep(i,0,n) {
        P q = p[(i + 1) % n];
        if (onSegment(p[i], q, a)) return !strict;
        //or: if (segDist(p[i], q, a) <= eps) return !strict;
        cnt ^= ((a.y<p[i].y) - (a.y<q.y)) * a.cross(p[i], q) > 0;
    }
    return cnt;
}
```

PolygonArea.h

Description: Returns twice the signed area of a polygon. Clockwise enumeration gives negative area. Watch out for overflow if using int as T!

"Point.h"fl2300, 6 lines

```
template<class T>
T polygonArea2(vector<Point<T>&& v) {
    T a = v.back().cross(v[0]);
    rep(i,0,sz(v)-1) a += v[i].cross(v[i+1]);
    return a;
}
```

PolygonCenter.h

Description: Returns the center of mass for a polygon.

Time: $\mathcal{O}(n)$

"Point.h"9706dc, 9 lines

```
typedef Point<double> P;
P polygonCenter(const vector<P>& v) {
    P res(0, 0); double A = 0;
    for (int i = 0, j = sz(v) - 1; i < sz(v); j = i++) {
        res = res + (v[i] + v[j]) * v[j].cross(v[i]);
        A += v[j].cross(v[i]);
    }
    return res / A / 3;
}
```

PolygonCut.h

Description:

Returns a vector with the vertices of a polygon with everything to the left of the line going from s to e cut away.

Usage: vector<P> p = ...;

p = polygonCut(p, P(0,0), P(1,0));

"Point.h", "LineIntersection.h"f2b7d4, 13 lines

```
typedef Point<double> P;
vector<P> polygonCut(const vector<P>& poly, P s, P e) {
    vector<P> res;
    rep(i,0,sz(poly)) {
        P cur = poly[i], prev = i ? poly[i-1] : poly.back();
        bool side = s.cross(e, cur) < 0;
        if (side != (s.cross(e, prev) < 0))
            res.push_back(lineInter(s, e, cur, prev).second);
        if (side)
            res.push_back(cur);
    }
    return res;
}
```

ConvexHull.h

Description:

Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull.

Time: $\mathcal{O}(n \log n)$

"Point.h"310954, 13 lines

```
typedef Point<ll> P;
```

```
vector<P> convexHull(vector<P> pts) {
    if (sz(pts) <= 1) return pts;
    sort(all(pts));
    vector<P> h(sz(pts)+1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(pts)))
        for (P p : pts) {
            while (t >= s + 2 && h[t-2].cross(h[t-1], p) <= 0) t--;
            h[t++] = p;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}
```

HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).

Time: $\mathcal{O}(n)$

"Point.h"c571b8, 12 lines

```
typedef Point<ll> P;
array<P, 2> hullDiameter(vector<P> S) {
    int n = sz(S), j = n < 2 ? 0 : 1;
    pair<ll, array<P, 2>> res({0, {S[0], S[0]}});
    rep(i,0,j)
        for (; j = (j + 1) % n) {
            res = max(res, {(S[i] - S[j]).dist2(), {S[i], S[j]}});
            if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)
                break;
        }
    return res.second;
}
```

PointInsideHull.h

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included.

Time: $\mathcal{O}(\log N)$

"Point.h", "sideOf.h", "OnSegment.h"71446b, 14 lines

```
typedef Point<ll> P;

bool inHull(const vector<P>& l, P p, bool strict = true) {
    int a = 1, b = sz(l) - 1, r = !strict;
    if (sz(l) < 3) return r && onSegment(l[0], l.back(), p);
    if (sideOf(l[0], l[a], l[b]) > 0) swap(a, b);
    if (sideOf(l[0], l[a], p) >= r || sideOf(l[0], l[b], p) <= -r)
        return false;
    while (abs(a - b) > 1) {
        int c = (a + b) / 2;
        (sideOf(l[0], l[c], p) > 0 ? b : a) = c;
    }
    return sgn(l[a].cross(l[b], p)) < r;
}
```

LineHullIntersection.h

Description: Line-convex polygon intersection. The polygon must be ccw and have no collinear points. lineHull(line, poly) returns a pair describing the intersection of a line with the polygon: $\bullet (-1, -1)$ if no collision, $\bullet (i, -1)$ if touching the corner i , $\bullet (i, i)$ if along side $(i, i + 1)$, $\bullet (i, j)$ if crossing sides $(i, i + 1)$ and $(j, j + 1)$. In the last case, if a corner i is crossed, this is treated as happening on side $(i, i + 1)$. The points are returned in the same order as the line hits the polygon. extrVertex returns the point of a hull with the max projection onto a line.

Time: $\mathcal{O}(\log n)$

"Point.h"7cf45b, 39 lines

```
#define cmp(i,j) sgn(dir.perp().cross(poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i + 1, i) >= 0 && cmp(i, i - 1 + n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (extr(m)) return m;
        int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
    }
    return lo;
}
```

```
#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
    int endB = extrVertex(poly, (b - a).perp());
    if (cmpL(endA) < 0 || cmpL(endB) > 0)
        return {-1, -1};
    array<int, 2> res;
    rep(i,0,2) {
        int lo = endB, hi = endA, n = sz(poly);
        while ((lo + 1) % n != hi) {
            int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
            (cmpL(m) == cmpL(endB) ? lo : hi) = m;
        }
        res[i] = (lo + !cmpL(hi)) % n;
        swap(endA, endB);
    }
    if (res[0] == res[1]) return {res[0], -1};
    if (!cmpL(res[0]) && !cmpL(res[1]))
        switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
            case 0: return {res[0], res[0]};
            case 2: return {res[1], res[1]};
        }
    return res;
}
```

8.4 Misc. Point Set Problems

ClosestPair.h

Description: Finds the closest pair of points.

Time: $\mathcal{O}(n \log n)$

"Point.h"ac41a6, 17 lines

```
typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
    assert(sz(v) > 1);
    set<P> S;
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d(1 + (ll)sqrt(ret.first), 0);
        while (v[j].y <= p.y - d.x) S.erase(v[j++]);
        auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
        for (; lo != hi; ++lo)
            ret = min(ret, {(lo - p).dist2(), {lo, p}});
        S.insert(p);
    }
    return ret.second;
}
```

ManhattanMST.h

Description: Given N points, returns up to 4*N edges, which are guaranteed to contain a minimum spanning tree for the graph with edge weights $w(p, q) = -p.x - q.x + -p.y - q.y$. Edges are in the form (distance, src, dst). Use a standard MST algorithm on the result to find the final MST.

Time: $\mathcal{O}(N \log N)$

"Point.h"df6f59, 23 lines

```
typedef Point<int> P;
vector<array<int, 3>> manhattanMST(vector<P> ps) {
    vi id(sz(ps));
    iota(all(id), 0);
    vector<array<int, 3>> edges;
    rep(k,0,4) {
        sort(all(id), [&](int i, int j) {
            return (ps[i]-ps[j]).x < (ps[j]-ps[i]).y;});
        map<int, int> sweep;
        for (int i : id) {
            for (auto it = sweep.lower_bound(-ps[i].y);
                it != sweep.end(); sweep.erase(it++)) {
                int j = it->second;
                P d = ps[i] - ps[j];
                if (d.y > d.x) break;
                edges.push_back({d.y + d.x, i, j});
            }
            sweep[-ps[i].y] = i;
        }
        for (P& p : ps) if (k & 1) p.x = -p.x; else swap(p.x, p.y);
    }
}
```

```
    return edges;
}

kdTree.h
Description: KD-tree (2d, can be extended to 3d)
"Point.h" bac5b0, 63 lines

typedef long long T;
typedef Point<T> P;
const T INF = numeric_limits<T>::max();

bool on_x(const P& a, const P& b) { return a.x < b.x; }
bool on_y(const P& a, const P& b) { return a.y < b.y; }

struct Node {
    P pt; // if this is a leaf, the single point in it
    T x0 = INF, x1 = -INF, y0 = INF, y1 = -INF; // bounds
    Node *first = 0, *second = 0;

    T distance(const P& p) { // min squared distance to a point
        T x = (p.x < x0 ? x0 : p.x > x1 ? x1 : p.x);
        T y = (p.y < y0 ? y0 : p.y > y1 ? y1 : p.y);
        return (P(x,y) - p).dist2();
    }

    Node(vector<P>&& vp) : pt(vp[0]) {
        for (P p : vp) {
            x0 = min(x0, p.x); x1 = max(x1, p.x);
            y0 = min(y0, p.y); y1 = max(y1, p.y);
        }
        if (vp.size() > 1) {
            // split on x if width>= height (not ideal...)
            sort(all(vp), x1 - x0 >= y1 - y0 ? on_x : on_y);
            // divide by taking half the array for each child (not
            // best performance with many duplicates in the middle)
            int half = sz(vp)/2;
            first = new Node({vp.begin(), vp.begin() + half});
            second = new Node({vp.begin() + half, vp.end()});
        }
    };

struct KDTree {
    Node* root;
    KDTree(const vector<P>& vp) : root(new Node({all(vp)})) {}

    pair<T, P> search(Node *node, const P& p) {
        if (!node->first) {
            // uncomment if we should not find the point itself:
            // if (p == node->pt) return {INF, P()};
            return make_pair((p - node->pt).dist2(), node->pt);
        }

        Node *f = node->first, *s = node->second;
        T bfirst = f->distance(p), bsec = s->distance(p);
        if (bfirst > bsec) swap(bsec, bfirst), swap(f, s);

        // search closest side first, other side if needed
        auto best = search(f, p);
        if (bsec < best.first)
            best = min(best, search(s, p));
    }

    // find nearest point to a point, and its squared distance
    // (requires an arbitrary operator< for Point)
    pair<T, P> nearest(const P& p) {
        return search(root, p);
    }
};
```

FastDelaunay.h

Description: Fast Delaunay triangulation. Each circumcircle contains none of the input points. There must be no duplicate points. If all points are on a line, no triangles will be returned. Should work for doubles as well, though there may be precision issues in 'circ'. Returns triangles in order {t[0][0], t[0][1], t[0][2], t[1][0], ...}, all counter-clockwise.
Time: $\mathcal{O}(n \log n)$
"Point.h" eefd5, 88 lines

```
typedef Point<ll> P;
typedef struct Quad* Q;
typedef __int128_t lll; // (can be ll if coords are < 2e4)
P arb(LLONG_MAX, LLONG_MAX); // not equal to any other point

struct Quad {
    Q rot, o; P p = arb; bool mark;
    P& F() { return r()->p; }
    Q& r() { return rot->rot; }
    Q prev() { return rot->o->rot; }
    Q next() { return r()->prev(); }
} *H;

bool circ(P p, P a, P b, P c) { // is p in the circumcircle?
    lll p2 = p.dist2(), A = a.dist2()-p2,
        B = b.dist2()-p2, C = c.dist2()-p2;
    return p.cross(a,b)*C + p.cross(b,c)*A + p.cross(c,a)*B > 0;
}

Q makeEdge(P orig, P dest) {
    Q r = H ? H : new Quad(new Quad(new Quad{0}));
    H = r->o; r->r()->r() = r;
    rep(i,0,4) r = r->rot, r->p = arb, r->o = i & 1 ? r : r->r();
    r->p = orig; r->F() = dest;
    return r;
}

void splice(Q a, Q b) {
    swap(a->o->rot->o, b->o->rot->o); swap(a->o, b->o);
}

Q connect(Q a, Q b) {
    Q q = makeEdge(a->F(), b->p);
    splice(q, a->next());
    splice(q->r(), b);
    return q;
}

pair<Q,Q> rec(const vector<P>& s) {
    if (sz(s) <= 3) {
        Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.back());
        if (sz(s) == 2) return { a, a->r() };
        splice(a->r(), b);
        auto side = s[0].cross(s[1], s[2]);
        Q c = side ? connect(b, a) : 0;
        return {side < 0 ? c->r() : a, side < 0 ? c : b->r() };
    }

#define H(e) e->F(), e->p
#define valid(e) (e->F().cross(H(base)) > 0)
    Q A, B, ra, rb;
    int half = sz(s) / 2;
    tie(ra, A) = rec({all(s) - half});
    tie(B, rb) = rec({sz(s) - half + all(s)});
    while ((B->p.cross(H(A)) < 0 && (A = A->next()) ||
        (A->p.cross(H(B)) > 0 && (B = B->r()->o)));
    Q base = connect(B->r(), A);
    if (A->p == ra->p) ra = base->r();
    if (B->p == rb->p) rb = base;

#define DEL(e, init, dir) Q e = init->dir; if (valid(e)) \
    while (circ(e->dir->F(), H(base), e->F())) { \
        Q t = e->dir; \
        splice(e, e->prev()); \
        splice(e->r(), e->r()->prev()); \
        e->o = H; H = e; e = t; \
    }
    for (;;) {
        DEL(LC, base->r(), o); DEL(RC, base, prev());
        if (!valid(LC) && !valid(RC)) break;
        if (!valid(LC) || (valid(RC) && circ(H(RC), H(LC))))
            base = connect(RC, base->r());
        else
            base = connect(base->r(), LC->r());
    }
    return { ra, rb };
}

vector<P> triangulate(vector<P> pts) {
    sort(all(pts)); assert(unique(all(pts)) == pts.end());
    if (sz(pts) < 2) return {};
```

```
    Q e = rec(pts).first;
    vector<Q> q = {e};
    int qi = 0;
    while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.push_back(c->p); \
    q.push_back(c->r()); c = c->next(); } while (c != e); }
    ADD; pts.clear();
    while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
    return pts;
}
```

8.5 3D

PolyhedronVolume.h

Description: Magic formula for the volume of a polyhedron. Faces should point outwards.
3058c3, 6 lines

```
template<class V, class L>
double signedPolyVolume(const V& p, const L& trilst) {
    double v = 0;
    for (auto i : trilst) v += p[i.a].cross(p[i.b]).dot(p[i.c]);
    return v / 6;
}
```

Point3D.h

Description: Class to handle points in 3D space. T can be e.g. double or long long.
8058ae, 32 lines

```
template<class T> struct Point3D {
    typedef Point3D P;
    typedef const P& R;
    T x, y, z;
    explicit Point3D(T x=0, T y=0, T z=0) : x(x), y(y), z(z) {}
    bool operator<(R p) const {
        return tie(x, y, z) < tie(p.x, p.y, p.z); }
    bool operator==(R p) const {
        return tie(x, y, z) == tie(p.x, p.y, p.z); }
    P operator+(R p) const { return P(x+p.x, y+p.y, z+p.z); }
    P operator-(R p) const { return P(x-p.x, y-p.y, z-p.z); }
    P operator*(T d) const { return P(x*d, y*d, z*d); }
    P operator/(T d) const { return P(x/d, y/d, z/d); }
    T dot(R p) const { return x*p.x + y*p.y + z*p.z; }
    P cross(R p) const {
        return P(y*p.z - z*p.y, z*p.x - x*p.z, x*p.y - y*p.x);
    }
    T dist2() const { return x*x + y*y + z*z; }
    double dist() const { return sqrt((double)dist2()); }
    //Azimuthal angle (longitude) to x-axis in interval [-pi, pi]
    double phi() const { return atan2(y, x); }
    //Zenith angle (latitude) to the z-axis in interval [0, pi]
    double theta() const { return atan2(sqrt(x*x+y*y),z); }
    P unit() const { return *this/(T)dist(); } //makes dist()==1
    //returns unit vector normal to *this and p
    P normal(P p) const { return cross(p).unit(); }
    //returns point rotated 'angle' radians ccw around axis
    P rotate(double angle, P axis) const {
        double s = sin(angle), c = cos(angle); P u = axis.unit();
        return u*dot(u)*1-c + (*this)*c - cross(u)*s;
    }
};
```

3dHull.h

Description: Computes all faces of the 3-dimension hull of a point set. *No four points must be coplanar*, or else random results will be returned. All faces will point outwards.

Time: $\mathcal{O}(n^2)$

```
"Point3D.h" 5b45fc, 49 lines

typedef Point3D<double> P3;

struct PR {
    void ins(int x) { (a == -1 ? a : b) = x; }
    void rem(int x) { (a == x ? a : b) = -1; }
    int cnt() { return (a != -1) + (b != -1); }
    int a, b;
};

struct F { P3 q; int a, b, c; };
```

```
vector<F> hull3d(const vector<P3>& A) {
    assert(sz(A) >= 4);
    vector<vector<PR>> E(sz(A), vector<PR>(sz(A), {-1, -1}));
#define E(x,y) E[f.x][f.y]
    vector<F> FS;
    auto mf = [&](int i, int j, int k, int l) {
        P3 q = (A[j] - A[i]).cross((A[k] - A[i]));
        if (q.dot(A[l]) > q.dot(A[i]))
            q = q * -1;
        F f(q, i, j, k);
        E(a,b).ins(k); E(a,c).ins(j); E(b,c).ins(i);
        FS.push_back(f);
    };
    rep(i,0,4) rep(j,i+1,4) rep(k,j+1,4)
        mf(i, j, k, 6 - i - j - k);

    rep(i,4,sz(A)) {
        rep(j,0,sz(FS)) {
            F f = FS[j];
            if(f.q.dot(A[i]) > f.q.dot(A[f.a])) {
                E(a,b).rem(f.c);
                E(a,c).rem(f.b);
                E(b,c).rem(f.a);
                swap(FS[j-], FS.back());
                FS.pop_back();
            }
        }
        int nw = sz(FS);
        rep(j,0,nw) {
            F f = FS[j];
#define C(a, b, c) if (E(a,b).cnt() != 2) mf(f.a, f.b, i, f.c);
            C(a, b, c); C(a, c, b); C(b, c, a);
        }
        for (F& it : FS) if ((A[it.b] - A[it.a]).cross(
            A[it.c] - A[it.a]).dot(it.q) <= 0) swap(it.c, it.b);
        return FS;
    };
};
```

sphericalDistance.h

Description: Returns the shortest distance on the sphere with radius radius between the points with azimuthal angles (longitude) f_1 (ϕ_1) and f_2 (ϕ_2) from x axis and zenith angles (latitude) t_1 (θ_1) and t_2 (θ_2) from z axis (0 = north pole). All angles measured in radians. The algorithm starts by converting the spherical coordinates to cartesian coordinates so if that is what you have you can use only the two last rows. $dx*$ radius is then the difference between the two points in the x direction and $d*$ radius is the total distance between the points.

```
double sphericalDistance(double f1, double t1,
    double f2, double t2, double radius) {
    double dx = sin(t2)*cos(f2) - sin(t1)*cos(f1);
    double dy = sin(t2)*sin(f2) - sin(t1)*sin(f1);
    double dz = cos(t2) - cos(t1);
    double d = sqrt(dx*dx + dy*dy + dz*dz);
    return radius*2*asin(d/2);
}
```

Strings (9)

KMP.h

Description: $pi[x]$ computes the length of the longest prefix of s that ends at x , other than $s[0...x]$ itself (abacaba -> 0010123). Can be used to find all occurrences of a string.

```
Time:  $\mathcal{O}(n)$ 
d4375c, 16 lines

vi pi(const string& s) {
    vi p(sz(s));
    rep(i,1,sz(s)) {
        int g = p[i-1];
        while (g && s[i] != s[g]) g = p[g-1];
        p[i] = g + (s[i] == s[g]);
    }
    return p;
}
```

```
vi match(const string& s, const string& pat) {
    vi p = pi(pat + '\0' + s), res;
```

```
    rep(i,sz(p)-sz(s),sz(p))
        if (p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
    return res;
}
```

Zfunc.h

Description: $z[x]$ computes the length of the longest common prefix of $s[i]$ and s , except $z[0] = 0$. (abacaba -> 0010301)

```
Time:  $\mathcal{O}(n)$ 
ee09e2, 12 lines

vi Z(const string& S) {
    vi z(sz(S));
    int l = -1, r = -1;
    rep(i,1,sz(S)) {
        z[i] = i >= r ? 0 : min(r - i, z[i - l]);
        while (i + z[i] < sz(S) && S[i + z[i]] == S[z[i]])
            z[i]++;
        if (i + z[i] > r)
            l = i, r = i + z[i];
    }
    return z;
}
```

Manacher.h

Description: For each position in a string, computes $p[0][i] =$ half length of longest even palindrome around pos i , $p[1][i] =$ longest odd (half rounded down).

```
Time:  $\mathcal{O}(N)$ 
e7ad79, 13 lines

array<vi, 2> manacher(const string& s) {
    int n = sz(s);
    array<vi,2> p = {vi(n+1), vi(n)};
    rep(z,0,2) for (int i=0,l=0,r=0; i < n; i++) {
        int t = r-i+!z;
        if (<r) p[z][i] = min(t, p[z][l+t]);
        int L = i-p[z][i], R = i+p[z][i]-!z;
        while (L>=1 && R+1<n && s[L-1] == s[R+1])
            p[z][i]++, L--, R++;
        if (R>r) l=L, r=R;
    }
    return p;
}
```

MinRotation.h

Description: Finds the lexicographically smallest rotation of a string.

Usage: rotate(v.begin(), v.begin()+minRotation(v), v.end());

```
Time:  $\mathcal{O}(N)$ 
d07a42, 8 lines

int minRotation(string s) {
    int a=0, N=sz(s); s += s;
    rep(b,0,N) rep(k,0,N) {
        if (a+k == b || s[a+k] < s[b+k]) {b += max(0, k-1); break;}
        if (s[a+k] > s[b+k]) {a = b; break;}
    }
    return a;
}
```

SuffixArray.h

Description: Builds suffix array for a string. $sa[i]$ is the starting index of the suffix which is i 'th in the sorted suffix array. The returned vector is of size $n+1$, and $sa[0] = n$. The lcp array contains longest common prefixes for neighbouring strings in the suffix array: $lcp[i] = lcp(sa[i], sa[i-1])$, $lcp[0] = 0$. The input string must not contain any zero bytes.

```
Time:  $\mathcal{O}(n \log n)$ 
38db9f, 23 lines

struct SuffixArray {
    vi sa, lcp;
    SuffixArray(string& s, int lim=256) { // or basic_string<int>
        int n = sz(s) + 1, k = 0, a, b;
        vi x(all(s)+1), y(n), ws(max(n, lim)), rank(n);
        sa = lcp = y, iota(all(sa), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
            p = j, iota(all(y), n - j);
            rep(i,0,n) if (sa[i] >= j) y[p++] = sa[i] - j;
            fill(all(ws), 0);
            rep(i,0,n) ws[x[i]]++;
            rep(i,1,lim) ws[i] += ws[i - 1];
            for (int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
            swap(x, y), p = 1, x[sa[0]] = 0;
            rep(i,1,n) a = sa[i - 1], b = sa[i], x[b] =
```

```
            (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
        }
        rep(i,1,n) rank[sa[i]] = i;
        for (int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
            for (k && k--, j = sa[rank[i] - 1];
                s[i + k] == s[j + k]; k++);
    }
};
```

SuffixTree.h

Description: Ukkonen's algorithm for online suffix tree construction. Each node contains indices $[l, r]$ into the string, and a list of child nodes. Suffixes are given by traversals of this tree, joining $[l, r]$ substrings. The root is 0 (has $l = -1, r = 0$), non-existent children are -1. To get a complete tree, append a dummy symbol – otherwise it may contain an incomplete path (still useful for substring matching, though).

```
Time:  $\mathcal{O}(26N)$ 
aae0b8, 50 lines

struct SuffixTree {
    enum { N = 200010, ALPHA = 26 }; // N ~ 2*maxlen+10
    int toi(char c) { return c - 'a'; }
    string a; // v = cur node, q = cur position
    int t[N][ALPHA], l[N], r[N], p[N], s[N], v=0, q=0, m=2;

    void ukkadd(int i, int c) { suff:
        if (r[v]<=q) {
            if (t[v][c]==-1) { t[v][c]=m; l[m]=i;
                p[m++]=v; v=s[v]; q=r[v]; goto suff; }
            v=t[v][c]; q=l[v];
        }
        if (q==-1 || c==toi(a[q])) q++; else {
            l[m+1]=i; p[m+1]=m; l[m]=l[v]; r[m]=q;
            p[m]=p[v]; t[m][c]=m+1; t[m][toi(a[q])]=v;
            l[v]=q; p[v]=m; t[p[m]][toi(a[l[m]])]=m;
            v=s[p[m]]; q=l[m];
            while (q<r[m]) { v=t[v][toi(a[q])]; q+=r[v]-l[v]; }
            if (q==r[m]) s[m]=v; else s[m]=m+2;
            q=r[v]-(q-r[m]); m+=2; goto suff;
        }
    }

    SuffixTree(string a) : a(a) {
        fill(r,r+N,sz(a));
        memset(s, 0, sizeof s);
        memset(t, -1, sizeof t);
        fill(t[1],t[1]+ALPHA,0);
        s[0] = 1; l[0] = l[1] = -1; r[0] = r[1] = p[0] = p[1] = 0;
        rep(i,0,sz(a)) ukkadd(i, toi(a[i]));
    }
};
```

```

SuffixTree(string a) : a(a) {
    fill(r,r+N,sz(a));
    memset(s, 0, sizeof s);
    memset(t, -1, sizeof t);
    fill(t[1],t[1]+ALPHA,0);
    s[0] = 1; l[0] = l[1] = -1; r[0] = r[1] = p[0] = p[1] = 0;
    rep(i,0,sz(a)) ukkadd(i, toi(a[i]));
}

// example: find longest common substring (uses ALPHA = 28)
pii best;
int lcs(int node, int i1, int i2, int olen) {
    if (l[node] <= i1 && i1 < r[node]) return 1;
    if (l[node] <= i2 && i2 < r[node]) return 2;
    int mask = 0, len = node ? olen + (r[node] - l[node]) : 0;
    rep(c,0,ALPHA) if (t[node][c] != -1)
        mask |= lcs(t[node][c], i1, i2, len);
    if (mask == 3)
        best = max(best, {len, r[node] - len});
    return mask;
}

static pii LCS(string s, string t) {
    SuffixTree st(s + (char)('z' + 1) + t + (char)('z' + 2));
    st.lcs(0, sz(s), sz(s) + 1 + sz(t), 0);
    return st.best;
}
};
```

Hashing.h

```
Description: Self-explanatory methods for string hashing.
2d2a67, 44 lines

// Arithmetic mod 2^64-1. 2x slower than mod 2^64 and more
// code, but works on evil test data (e.g. Thue-Morse, where
// ABBA... and BAAB... of length 2^10 hash the same mod 2^64).
// "typedef ull H;" instead if you think test data is random,
// or work mod 10^9+7 if the Birthday paradox is not a problem.
typedef uint64_t ull;
struct H {
```

```
ull x; H(ull x=0) : x(x) {}
H operator+(H o) { return x + o.x + (x + o.x < x); }
H operator-(H o) { return *this + ~o.x; }
H operator*(H o) { auto m = (__uint128_t)x * o.x;
return H((ull)m) + (ull)(m >> 64); }
ull get() const { return x + !~x; }
bool operator==(H o) const { return get() == o.get(); }
bool operator<(H o) const { return get() < o.get(); }
};

static const H C = (1l)1e11+3; // (order ~ 3e9; random also ok)

struct HashInterval {
vector<H> ha, pw;
HashInterval(string& str) : ha(sz(str)+1), pw(ha) {
pw[0] = 1;
rep(i,0,sz(str))
ha[i+1] = ha[i] * C + str[i],
pw[i+1] = pw[i] * C;
}
H hashInterval(int a, int b) { // hash [a, b)
return ha[b] - ha[a] * pw[b - a];
}
};

vector<H> getHashes(string& str, int length) {
if (sz(str) < length) return {};
H h = 0, pw = 1;
rep(i,0,length)
h = h * C + str[i], pw = pw * C;
vector<H> ret = {h};
rep(i,length,sz(str)) {
ret.push_back(h = h * C + str[i] - pw * str[i-length]);
}
return ret;
}

H hashString(string& s){H h{}; for(char c:s) h=h*C+c;return h;}
```

AhoCorasick.h

Description: Aho-Corasick automaton, used for multiple pattern matching. Initialize with AhoCorasick ac(patterns); the automaton start node will be at index 0. find(word) returns for each position the index of the longest word that ends there, or -1 if none. findAll(−, word) finds all words (up to $N\sqrt{N}$ many if no duplicate patterns) that start at each position (shortest first). Duplicate patterns are allowed; empty patterns are not. To find the longest words that start at each position, reverse all input. For large alphabets, split each symbol into chunks, with sentinel bits for symbol boundaries. Important property: The suffix links form a tree, can be exploited using DFS traversal.
Time: construction takes $\mathcal{O}(26N)$, where N = sum of length of patterns. find(x) is $\mathcal{O}(N)$, where N = length of x. findAll is $\mathcal{O}(NM)$.

f35677, 66 lines

```
struct AhoCorasick {
enum {alpha = 26, first = 'A'}; // change this!
struct Node {
// (nmatches is optional)
int back, next[alpha], start = -1, end = -1, nmatches = 0;
Node(int v) { memset(next, v, sizeof(next)); }
};
vector<Node> N;
vi backp;
void insert(string& s, int j) {
assert(!s.empty());
int n = 0;
for (char c : s) {
int& m = N[n].next[c - first];
if (m == -1) { n = m = sz(N); N.emplace_back(-1); }
else n = m;
}
if (N[n].end == -1) N[n].start = j;
backp.push_back(N[n].end);
N[n].end = j;
N[n].nmatches++;
}
AhoCorasick(vector<string>& pat) : N(1, -1) {
rep(i,0,sz(pat)) insert(pat[i], i);
N[0].back = sz(N);
N.emplace_back(0);

queue<int> q;
```

```
for (q.push(0); !q.empty(); q.pop()) {
int n = q.front(), prev = N[n].back;
rep(i,0,alpha) {
int &ed = N[n].next[i], y = N[prev].next[i];
if (ed == -1) ed = y;
else {
N[ed].back = y;
(N[ed].end == -1 ? N[ed].end : backp[N[ed].start])
= N[y].end;
N[ed].nmatches += N[y].nmatches;
q.push(ed);
}
}
}
vi find(string word) {
int n = 0;
vi res; // ll count = 0;
for (char c : word) {
n = N[n].next[c - first];
res.push_back(N[n].end);
// count += N[n].nmatches;
}
return res;
}
vector<vi> findAll(vector<string>& pat, string word) {
vi r = find(word);
vector<vi> res(sz(word));
rep(i,0,sz(word)) {
int ind = r[i];
while (ind != -1) {
res[i - sz(pat[ind]) + 1].push_back(ind);
ind = backp[ind];
}
}
return res;
}
}
```

Various (10)

10.1 Intervals

IntervalContainer.h

Description: Add and remove intervals from a set of disjoint intervals. Will merge the added interval with any overlapping intervals in the set when adding. Intervals are [inclusive, exclusive).
Time: $\mathcal{O}(\log N)$

edce47, 23 lines

```
set<pii>::iterator addInterval(set<pii>& is, int L, int R) {
if (L == R) return is.end();
auto it = is.lower_bound({L, R}), before = it;
while (it != is.end() && it->first <= R) {
R = max(R, it->second);
before = it = is.erase(it);
}
if (it != is.begin() && (--it)->second >= L) {
L = min(L, it->first);
R = max(R, it->second);
is.erase(it);
}
return is.insert(before, {L,R});
}

void removeInterval(set<pii>& is, int L, int R) {
if (L == R) return;
auto it = addInterval(is, L, R);
auto r2 = it->second;
if (it->first == L) is.erase(it);
else (int&)it->second = L;
if (R != r2) is.emplace(R, r2);
}
```

IntervalCover.h

Description: Compute indices of smallest set of intervals covering another interval. Intervals should be [inclusive, exclusive). To support [inclusive, inclusive], change (A) to add || R.empty(). Returns empty set on failure (or if G is empty).

Time: $\mathcal{O}(N \log N)$ 9e9d8d, 19 lines

```
template<class T>
vi cover(pair<T, T> G, vector<pair<T, T>> I) {
vi S(sz(I)), R;
iota(all(S), 0);
sort(all(S), [&](int a, int b) { return I[a] < I[b]; });
T cur = G.first;
int at = 0;
while (cur < G.second) { // (A)
pair<T, int> mx = make_pair(cur, -1);
while (at < sz(I) && I[S[at]].first <= cur) {
mx = max(mx, make_pair(I[S[at]].second, S[at]));
at++;
}
if (mx.second == -1) return {};
cur = mx.first;
R.push_back(mx.second);
}
return R;
}
```

ConstantIntervals.h

Description: Split a monotone function on [from, to) into a minimal set of half-open intervals on which it has the same value. Runs a callback g for each such interval.
Usage: constantIntervals(0, sz(v), [&](int x){return v[x];}, [&](int lo, int hi, T val){...});
Time: $\mathcal{O}\left(k \log \frac{n}{k}\right)$ 753a4c, 19 lines

```
template<class F, class G, class T>
void rec(int from, int to, F& f, G& g, int& i, T& p, T q) {
if (p == q) return;
if (from == to) {
g(i, to, p);
i = to; p = q;
} else {
int mid = (from + to) >> 1;
rec(from, mid, f, g, i, p, f(mid));
rec(mid+1, to, f, g, i, p, q);
}
}

template<class F, class G>
void constantIntervals(int from, int to, F f, G g) {
if (to <= from) return;
int i = from; auto p = f(i), q = f(to-1);
rec(from, to-1, f, g, i, p, q);
g(i, to, q);
}
```

10.2 Misc. algorithms

TernarySearch.h

Description: Find the smallest i in [a,b] that maximizes $f(i)$, assuming that $f(a) < \dots < f(i) \geq \dots \geq f(b)$. To reverse which of the sides allows non-strict inequalities, change the < marked with (A) to <=, and reverse the loop at (B). To minimize f, change it to >, also at (B).
Usage: int ind = ternSearch(0,n-1,&[](int i){return a[i];});
Time: $\mathcal{O}(\log(b - a))$

9155b4, 11 lines

```
template<class F>
int ternSearch(int a, int b, F f) {
assert(a <= b);
while (b - a >= 5) {
int mid = (a + b) / 2;
if (f(mid) < f(mid+1)) a = mid; // (A)
else b = mid+1;
}
rep(i,a+1,b+1) if (f(a) < f(i)) a = i; // (B)
return a;
}
```

LIS.h

Description: Compute indices for the longest increasing subsequence.
Time: $\mathcal{O}(N \log N)$ 2932a0, 17 lines

```
template<class I> vi lis(const vector<I>& S) {
if (S.empty()) return {};
vi prev(sz(S));
```

```
typedef pair<I, int> p;
vector<p> res;
rep(i,0,sz(S)) {
    // change 0 -> i for longest non-decreasing subsequence
    auto it = lower_bound(all(res), p{S[i], 0});
    if (it == res.end()) res.emplace_back(), it = res.end()-1;
    *it = {S[i], i};
    prev[i] = it == res.begin() ? 0 : (it-1)->second;
}
int L = sz(res), cur = res.back().second;
vi ans(L);
while (L--) ans[L] = cur, cur = prev[cur];
return ans;
}
```

FastKnapsack.h

Description: Given N non-negative integer weights w and a non-negative target t, computes the maximum S <= t such that S is the sum of some subset of the weights.

Time: $\mathcal{O}(N \max(w_i))$ b20ccc, 16 lines

```
int knapsack(vi w, int t) {
    int a = 0, b = 0, x;
    while (b < sz(w) && a + w[b] <= t) a += w[b++];
    if (b == sz(w)) return a;
    int m = *max_element(all(w));
    vi u, v(2*m, -1);
    v[a+m-t] = b;
    rep(i,b,sz(w)) {
        u = v;
        rep(x,0,m) v[x+w[i]] = max(v[x+w[i]], u[x]);
        for (x = 2*m; --x > m;) rep(j, max(0,u[x]), v[x])
            v[x-w[j]] = max(v[x-w[j]], j);
    }
    for (a = t; v[a+m-t] < 0; a--);
    return a;
}
```

Mo.h

Description: Mo's algorithm. Includes 3d and 4d expansion.

Time: $\mathcal{O}\left(n \cdot \sqrt{n}\right)$ for 2d. 3d and 4d time in comments. 4cddcc, 186 lines

```
struct query {
    int l, r, id;
    friend bool operator < (query a, query b) {
        if (a.l / L != b.l / L) return a.l / L < b.l / L;
        if ((a.l / L) & 1) return a.r > b.r;
        else return a.r < b.r;
    }
};
void work() {
    repn(i, 1, m) {
        while (r < a[i].r) inc(++r);
        while (l > a[i].l) inc(--l);
        while (r > a[i].r) dec(r--);
        while (l < a[i].l) dec(l++);
    }
}
// n^{5/3}
const int L = pow(N, 2.0 / 3.0) + 10;
struct operation {
    int p, x, y;
}ops[N];
int ans[N];
struct query {
    int l, r, t, id;
    friend bool operator < (query a, query b) {
        if (a.l / L != b.l / L) return a.l / L < b.l / L;
        if (a.r / L != b.r / L) return a.r / L < b.r / L;
        if ((a.r / L) & 1) return a.t > b.t;
        else return a.t < b.t;
    }
};
vector<query> qry;
void MoAlgo() {
    int l = 1, r = 0, t = 0;
    for (auto p: qry) {
        int pl = p.l, pr = p.r, pt = p.t, id = p.id;
        while (t < pt) tinc(++t, l, r);
    }
}
```

```
while (t > pt) tdec(t--, l, r);
while (r < pr) inc(++r);
while (l > pl) inc(--l);
while (r > pr) dec(r--);
while (l < pl) dec(l++);
ans[id] = getans();
}
// n^{7/4}
const int B = 10000;
struct query {
    int l1, r1, l2, r2, id;
    friend bool operator < (query a, query b) {
        if (a.l1 / B != b.l1 / B) return a.l1 / B < b.l1 / B;
        if (a.r1 / B != b.r1 / B) return a.r1 / B < b.r1 / B;
        if (a.l2 / B != b.l2 / B) return a.l2 / B < b.l2 / B;
        return a.r2 < b.r2;
    }
}a[N];
void work() {
    repn(i, 1, q) {
        while (l1 > a[i].l1) inc(--l1);
        while (r1 < a[i].r1) inc(++r1);
        while (l2 > a[i].l2) inc(--l2);
        while (r2 < a[i].r2) inc(++r2);
        while (l1 < a[i].l1) dec(l1++);
        while (r1 > a[i].r1) dec(r1--);
        while (l2 < a[i].l2) dec(l2++);
        while (r2 > a[i].r2) dec(r2--);
    }
}
// roll back
#define mo RollBackMo
namespace RollBackMo {
    LL sum;
    struct record {
        int idx;
        LL dp[2][2], sum;
    };
    stack<record> st;
    void copy(LL dp1[2][2], LL dp2[2][2]) {
        dp1[0][0] = dp2[0][0];
        dp1[0][1] = dp2[0][1];
        dp1[1][0] = dp2[1][0];
        dp1[1][1] = dp2[1][1];
    }
    void add(int x, int dir, int l, int r) {
        int idx = x % k;
        record tmp;
        tmp.idx = idx, tmp.sum = sum;
        copy(tmp.dp, dp[idx]);
        st.push(tmp);
        int cnt = 1;
        if (x + k <= r) cnt++;
        if (x + k + k <= r) cnt++;
        if (x - k >= l) cnt++;
        if (x - k - k >= l) cnt++;
        if (cnt == 1) {
            dp[idx][0][0] = 0;
            dp[idx][1][0] = dp[idx][0][1] = dp[idx][1][1] = -INF;
        }
        else if (cnt == 2) {
            dp[idx][0][0] = 0;
            dp[idx][1][0] = dp[idx][0][1] = -INF;
            dp[idx][1][1] = a[x];
            if (x - k >= l) dp[idx][1][1] += a[x - k];
            if (x + k <= r) dp[idx][1][1] += a[x + k];
        }
        else if (dir == 1) {
            sum -= max(max(dp[idx][0][0], dp[idx][0][1]), max(dp[idx][1][0], dp[idx][1][1]));
            if (x - k >= l) {
                dp[idx][0][1] = tmp.dp[0][0] + a[x] + a[x - k];
                dp[idx][1][1] = tmp.dp[1][0] + a[x] + a[x - k];
            }
            else dp[idx][0][1] = dp[idx][1][1] = -INF;
            dp[idx][0][0] = max(tmp.dp[0][0], tmp.dp[0][1]);
            dp[idx][1][0] = max(tmp.dp[1][0], tmp.dp[1][1]);
        }
    }
}
```

```
else {
    sum -= max(max(dp[idx][0][0], dp[idx][0][1]), max(dp[idx][1][0], dp[idx][1][1]));
    if (x + k <= r) {
        dp[idx][1][0] = tmp.dp[0][0] + a[x] + a[x + k];
        dp[idx][1][1] = tmp.dp[0][1] + a[x] + a[x + k];
    }
    else dp[idx][1][0] = dp[idx][1][1] = -INF;
    dp[idx][0][0] = max(tmp.dp[0][0], tmp.dp[1][0]);
    dp[idx][0][1] = max(tmp.dp[0][1], tmp.dp[1][1]);
}
sum += max(max(dp[idx][0][0], dp[idx][0][1]), max(dp[idx][1][0], dp[idx][1][1]));
}
LL getans() {
    return sum;
}
void clear() {
    while (!st.empty()) st.pop();
}
void back() {
    while (!st.empty()) {
        auto p = st.top();
        st.pop();
        copy(dp[p.idx], p.dp);
        sum = p.sum;
    }
}
const int L = 320;
struct query {
    int l, r, id;
    friend bool operator < (const query &a, const query &b) {
        if (a.l / L != b.l / L) return a.l / L < b.l / L;
        return a.r < b.r;
    }
};
LL bwork(int cl, int cr) {
    int l = cl, r = cl - 1;
    sum = 0;
    while (r < cr) r++, add(r, 1, l, r);
    LL res = getans();
    back();
    return res;
}
vector<LL> work(vector<PII> &qry) {
    int q = qry.size();
    vector<LL> ans(q);
    vector<query> seq;
    rep(i, 0, q) {
        if (qry[i].se - qry[i].fi <= L) ans[i] = bwork(qry[i].fi, qry[i].se);
        else seq.pb({qry[i].fi, qry[i].se, i});
    }
    sort(all(seq));
    int ql = 0, qr;
    q = seq.size();
    while (ql < q) {
        qr = ql;
        while (qr < q && seq[qr + 1].l / L == seq[ql].l / L) qr++;
        sum = 0;
        int l = seq[ql].l / L * L + L, r = l - 1;
        repn(i, ql, qr) {
            l = seq[ql].l / L * L + L;
            while (r < seq[i].r) ++r, add(r, 1, l, r);
            clear();
            while (l > seq[i].l) --l, add(l, 0, l, r);
            ans[seq[i].id] = getans();
            back();
        }
        ql = qr + 1;
    }
    return ans;
}
```

MoOnTree.h

Description: Do DFS traversal to get in and out times (ed[x] = ++it, not ed[x] = it). If x = lca(x, y), query = [st[x], st[y]], else query = [ed[x], st[y]] + st[lca(x, y)]. Nodes that appear an even number of times should not be counted.

Time: $\mathcal{O}\left(n \cdot \sqrt{n}\right)$.

10.3 Dynamic programming

KnuthDP.h

Description: When doing DP on intervals: $a[i][j] = \min_{i < k < j} (a[i][k] + a[k][j]) + f(i, j)$, where the (minimal) optimal k increases with both i and j , one can solve intervals in increasing order of length, and search $k = p[i][j]$ for $a[i][j]$ only between $p[i][j-1]$ and $p[i+1][j]$. This is known as Knuth DP. Sufficient criteria for this are if $f(b, c) \leq f(a, d)$ and $f(a, c) + f(b, d) \leq f(a, d) + f(b, c)$ for all $a \leq b \leq c \leq d$. Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.

Time: $\mathcal{O}\left(N^2\right)$

```
int knuth() {
    int N; // read N and input
    int dp[N][N], opt[N][N];
    auto C = [&](int i, int j) {
        return i + j; // Implement cost function C.
    };
    for (int i = 0; i < N; i++) {
        opt[i][i] = i;
        // Initialize dp[i][i] according to the problem
    }
    for (int i = N-2; i >= 0; i--) {
        for (int j = i+1; j < N; j++) {
            int mn = INT_MAX;
            int cost = C(i, j);
            for (int k = opt[i][j-1]; k <= min(j-1, opt[i+1][j]); k++) {
                if (mn >= dp[i][k] + dp[k+1][j] + cost) {
                    opt[i][j] = k;
                    mn = dp[i][k] + dp[k+1][j] + cost;
                }
            }
            dp[i][j] = mn;
        }
    }
    return dp[0][N-1];
}
```

DivideAndConquerDP.h

Description: Given $a[i] = \min_{lo(i) \leq k < hi(i)} (f(i, k))$ where the (minimal) optimal k increases with i , computes $a[i]$ for $i = L..R-1$.

Time: $\mathcal{O}\left((N + (hi - lo)) \log N\right)$

```
struct DP { // Modify at will:
    int lo(int ind) { return 0; }
    int hi(int ind) { return ind; }
    ll f(int ind, int k) { return dp[ind][k]; }
    void store(int ind, int k, ll v) { res[ind] = pii(k, v); }

    void rec(int L, int R, int LO, int HI) {
        if (L >= R) return;
        int mid = (L + R) >> 1;
        pair<ll, int> best(LLONG_MAX, LO);
        rep(k, max(LO, lo(mid)), min(HI, hi(mid)))
            best = min(best, make_pair(f(mid, k), k));
        store(mid, best.second, best.first);
        rec(L, mid, LO, best.second+1);
        rec(mid+1, R, best.second, HI);
    }
    void solve(int L, int R) { rec(L, R, INT_MIN, INT_MAX); }
};
```

10.4 Debugging tricks

- `signal(SIGSEGV, [](int) { _Exit(0); });` converts segfaults into Wrong Answers. Similarly one can catch SIGABRT (assertion failures) and SIGFPE (zero divisions). `_GLIBCXX_DEBUG` failures generate SIGABRT (or SIGSEGV on gcc 5.4.0 apparently).
- `feenableexcept(29);` kills the program on NaNs (1), 0-divs (4), infinities (8) and denormals (16).

10.5 Optimization tricks

`__builtin_ia32_ldmxcsr(40896);` disables denormals (which make floats 20x slower near their minimum value).

10.5.1 Bit hacks

- `x & -x` is the least bit in `x`.
- `for (int x = m; x; ) { --x &= m; ... }` loops over all subset masks of `m` (except `m` itself).
- `c = x&-x, r = x+c; ((r^x) >> 2)/c | r` is the next number after `x` with the same number of bits set.
- `rep(b, 0, K) rep(i, 0, (1 << K))`
if `(i & 1 << b) D[i] += D[i^(1 << b)];` computes all sums of subsets.

10.5.2 Pragmas

- `#pragma GCC optimize ("Ofast")` will make GCC auto-vectorize loops and optimizes floating points better.
- `#pragma GCC target ("avx2")` can double performance of vectorized code, but causes crashes on old machines.
- `#pragma GCC optimize ("trapv")` kills the program on integer overflows (but is really slow).

FastMod.h

Description: Compute $a \% b$ about 5 times faster than usual, where b is constant but not known at compile time. Returns a value congruent to $a \pmod b$ in the range $[0, 2b)$.

```
#define FASTMOD 751a02, 8 lines
typedef unsigned long long ull;
struct FastMod {
    ull b, m;
    FastMod(ull b) : b(b), m(-1ULL / b) {}
    ull reduce(ull a) { // a % b + (0 or b)
        return a - (ull)((__uint128_t(m) * a) >> 64) * b;
    }
};
```

FastInput.h

Description: Read an integer from stdin. Usage requires your program to pipe in input from file.

Usage: `./a.out < input.txt`

Time: About 5x as fast as `cin/scanf`.

```
inline char gc() { // like getchar()
    static char buf[1 << 16];
    static size_t bc, be;
    if (bc >= be) {
        buf[0] = 0, bc = 0;
        be = fread(buf, 1, sizeof(buf), stdin);
    }
    return buf[bc++]; // returns 0 on EOF
}

int readInt() {
    int a, c;
    while ((a = gc()) < 40);
    if (a == '-') return -readInt();
    while ((c = gc()) >= 48) a = a * 10 + c - 48;
    return a - 48;
}
```

BumpAllocator.h

Description: When you need to dynamically allocate many objects and don't care about freeing them. "new X" otherwise has an overhead of something like 0.05us + 16 bytes per allocation.

```
#define BUMP 745db2, 8 lines
// Either globally or in a single class:
static char buf[450 << 20];
void* operator new(size_t s) {
    static size_t i = sizeof buf;
```

```
    assert(s < i);
    return (void*)&buf[i -= s];
}

void operator delete(void*) {}

SmallPtr.h
Description: A 32-bit pointer that points into BumpAllocator memory.
"BumpAllocator.h" 2dd6c9, 10 lines
template<class T> struct ptr {
    unsigned ind;
    ptr(T* p = 0) : ind(p ? unsigned((char*)p - buf) : 0) {
        assert(ind < sizeof buf);
    }
    T& operator*() const { return *(T*)(buf + ind); }
    T* operator->() const { return &*this; }
    T& operator[](int a) const { return (&*this)[a]; }
    explicit operator bool() const { return ind; }
};
```

BumpAllocatorSTL.h

Description: BumpAllocator for STL containers.

Usage: `vector<vector<int, small<int>>> ed(N);`

```
#define BUMP 666d4, 14 lines
char buf[450 << 20] alignas(16);
size_t buf_ind = sizeof buf;

template<class T> struct small {
    typedef T value_type;
    small() {}
    template<class U> small(const U&) {}
    T* allocate(size_t n) {
        buf_ind -= n * sizeof(T);
        buf_ind &= 0 - alignof(T);
        return (T*)(buf + buf_ind);
    }
    void deallocate(T*, size_t) {}
};
```